

5-5b

1. Let f be a function that is even and continuous on the closed interval $[-3, 3]$. The function f and its derivatives have the properties indicated in the table below.

x	0	$0 < x < 1$	1	$1 < x < 2$	2	$2 < x < 3$
$f(x)$	1	Positive	0	Negative	-1	Negative
$f'(x)$	Undefined	Negative	0	Negative	Undefined	Positive
$f''(x)$	Undefined	Positive	0	Negative	Undefined	Negative

Find the x -coordinate of each point at which f attains an absolute maximum value or an absolute minimum value. For each x -coordinate you give, state whether f attains an absolute maximum or an absolute minimum.

Part A

1 point is earned for first point

1 point is earned for second point

1 point is earned for third point

1 point is **deducted** for each additional point or incorrect point

Absolute maximum at $x = 0$

Absolute minimum at $x = \pm 2$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for first point

1 point is earned for second point

1 point is earned for third point

1 point is **deducted** for each additional point or incorrect point

Absolute maximum at $x = 0$

Absolute minimum at $x = \pm 2$

5-5b

2. Consider the graph of the function h given by $h(x) = e^{-x^2}$ for $0 \leq x \leq \infty$.

Let $A(w)$ be the area of the shaded rectangle shown in the figure above. Show that $A(w)$ has its maximum value when w is the x -coordinate of the point of inflection of the graph of h .

Part B

1 point is earned for the correct $A(w)$

Maximum:

$$\begin{aligned} A(w) &= we^{-w^2}, \\ A'(w) &= e^{-w^2} - 2w^2e^{-w^2} \\ &= e^{-w^2}(1 - 2w^2). \end{aligned}$$

1 point is earned for correctly setting $A'(w) = 0$

$$\begin{aligned} A'(w) &> 0 \text{ when } w < \frac{1}{\sqrt{2}}, \\ A'(w) &= 0 \text{ when } w = \frac{1}{\sqrt{2}}, \\ A'(w) &< 0 \text{ when } w > \frac{1}{\sqrt{2}}. \end{aligned}$$

Therefore, max occurs when $w = \frac{1}{\sqrt{2}}$

1 point is earned for correctly setting $h''(x) = 0$

Inflection:

$$\begin{aligned} h(x) &= e^{-x^2}, h'(x) = -2xe^{-x^2}, \\ h''(x) &= -2e^{-x^2} - 2x(-2x)e^{-x^2} \\ &= 2e^{-x^2}(-1 + 2x^2). \end{aligned}$$

$$\begin{aligned} h''(x) &< 0 \text{ when } x < \frac{1}{\sqrt{2}}, \\ h''(x) &= 0 \text{ when } x = \frac{1}{\sqrt{2}}, \\ h''(x) &> 0 \text{ when } x > \frac{1}{\sqrt{2}}, \end{aligned}$$

Therefore, inflection point when $x = \frac{1}{\sqrt{2}}$.

Therefore, the maximum value of $A(w)$ and the inflection point of $h(x)$ occur when x and w are $\frac{1}{\sqrt{2}}$.

1 point is earned for correctly indicates that the equation has the same solution

5-5b



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for the correct $A(w)$

Maximum:

$$\begin{aligned} A(w) &= we^{-w^2}, \\ A'(w) &= e^{-w^2} - 2w^2e^{-w^2} \\ &= e^{-w^2}(1 - 2w^2). \end{aligned}$$

1 point is earned for correctly setting $A'(w) = 0$

$$\begin{aligned} A'(w) &> 0 \text{ when } w < \frac{1}{\sqrt{2}}, \\ A'(w) &= 0 \text{ when } w = \frac{1}{\sqrt{2}}, \\ A'(w) &< 0 \text{ when } w > \frac{1}{\sqrt{2}}. \end{aligned}$$

Therefore, max occurs when $w = \frac{1}{\sqrt{2}}$

1 point is earned for correctly setting $h''(x) = 0$

Inflection:

$$\begin{aligned} h(x) &= e^{-x^2}, h'(x) = -2xe^{-x^2}, \\ h''(x) &= -2e^{-x^2} - 2x(-2x)e^{-x^2} \\ &= 2e^{-x^2}(-1 + 2x^2). \end{aligned}$$

$$\begin{aligned} h''(x) &< 0 \text{ when } x < \frac{1}{\sqrt{2}}, \\ h''(x) &= 0 \text{ when } x = \frac{1}{\sqrt{2}}, \\ h''(x) &> 0 \text{ when } x > \frac{1}{\sqrt{2}}, \end{aligned}$$

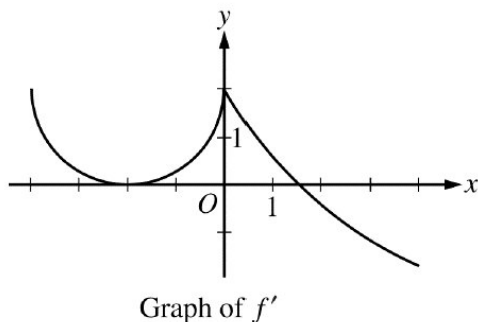
Therefore, inflection point when $x = \frac{1}{\sqrt{2}}$.

Therefore, the maximum value of $A(w)$ and the inflection point of $h(x)$ occur when x and w are $\frac{1}{\sqrt{2}}$.

1 point is earned for correctly indicates that the equation has the same solution

5-5b

3.



The derivative of a function f is defined by $f'(x) = \begin{cases} g(x) & \text{for } -4 \leq x \leq 0 \\ 5e^{-x/3} - 3 & \text{for } 0 < x \leq 4 \end{cases}$.

The graph of the continuous function f , shown in the figure above, has x -intercepts at $x = -2$ and $x = 3 \ln\left(\frac{5}{3}\right)$. The graph of g on $-4 \leq x \leq 0$ is a semicircle, and $f(0) = 5$.

For $-4 \leq x \leq 4$, find the value of x at which f has an absolute maximum. Justify your answer.

Part C

1 point is earned for the answer

1 point is earned for the justification

Since $f'(x) > 0$ on the intervals $-4 < x < -2$ and $-2 < x < 3 \ln\left(\frac{5}{3}\right)$, f is increasing on the interval $-4 \leq x \leq 3 \ln\left(\frac{5}{3}\right)$.

Since $f'(x) < 0$ on the interval $3 \ln\left(\frac{5}{3}\right) < x < 4$, f is decreasing on the interval $3 \ln\left(\frac{5}{3}\right) \leq x \leq 4$.

Therefore, f has an absolute maximum at $x = 3 \ln\left(\frac{5}{3}\right)$.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the answer

1 point is earned for the justification

Since $f'(x) > 0$ on the intervals $-4 < x < -2$ and $-2 < x < 3 \ln\left(\frac{5}{3}\right)$, f is increasing on the interval $-4 \leq x \leq 3 \ln\left(\frac{5}{3}\right)$.

Since $f'(x) < 0$ on the interval $3 \ln\left(\frac{5}{3}\right) < x < 4$, f is decreasing on the interval $3 \ln\left(\frac{5}{3}\right) \leq x \leq 4$.

Therefore, f has an absolute maximum at $x = 3 \ln\left(\frac{5}{3}\right)$.

5-5b

4. Let f be the function defined by $f(x) = \sin^2 x - \sin x$ for $0 \leq x \leq 3\pi/2$. Find the absolute maximum value and the absolute minimum value of f . Justify your answer.

Part C

1 point is earned for max value

1 point is earned for min value

1 point is earned for justification

Maximum value: 2
Minimum value: $-1/4$

x	$f(x)$
0	0
$\frac{\pi}{6}$	$-\frac{1}{4}$
$\frac{\pi}{2}$	0
$\frac{5\pi}{6}$	$-\frac{1}{4}$
$\frac{3\pi}{2}$	2



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for max value

1 point is earned for min value

1 point is earned for justification

Maximum value: 2
Minimum value: $-1/4$

5-5b

x	$f(x)$
0	0
$\frac{\pi}{6}$	$-\frac{1}{4}$
$\frac{\pi}{2}$	0
$\frac{5\pi}{6}$	$-\frac{1}{4}$
$\frac{3\pi}{2}$	2

5-5b

5. Consider the function f defined by $f(x) = e^x \cos x$ with domain $[0, 2\pi]$.
Find the absolute maximum and minimum values of $f(x)$.

Part A

1 point is earned for derivative

1 point is earned for finding critical values

1 point is earned for evaluating at critical points and ends

1 point is earned for choosing absolute max

1 point is earned for absolute min

$$\begin{aligned} f'(x) &= -e^x \sin x + e^x \cos x \\ &= e^x [\cos x - \sin x] \end{aligned}$$

$$f'(x) = 0 \text{ when } \sin x = \cos x, x = \frac{\pi}{4}, \frac{5\pi}{4}$$

x	$f(x)$
0	1
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2} e^{\pi/4}$
$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2} e^{5\pi/4}$
2π	$e^{2\pi}$

$$\text{Max : } e^{2\pi}; \text{Min : } -\frac{\sqrt{2}}{2} e^{5\pi/4}$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns five of the following points:

1 point is earned for derivative

1 point is earned for finding critical values

1 point is earned for evaluating at critical points and ends

1 point is earned for choosing absolute max

1 point is earned for absolute min

5-5b

$$\begin{aligned}f'(x) &= -e^x \sin x + e^x \cos x \\&= e^x [\cos x - \sin x]\end{aligned}$$

$$f'(x) = 0 \text{ when } \sin x = \cos x, x = \frac{\pi}{4}, \frac{5\pi}{4}$$

x	$f(x)$
0	1
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}e^{\pi/4}$
$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2}e^{5\pi/4}$
2π	$e^{2\pi}$

$$\text{Max : } e^{2\pi}; \text{Min : } -\frac{\sqrt{2}}{2}e^{5\pi/4}$$

5-5b

6. Let f be the function defined by $f(x) = \ln(2 + \sin x)$ for $\pi \leq x \leq 2\pi$.

Find the absolute maximum value and the absolute minimum value of f . Show the analysis that leads to your conclusion.

Part A

1 point is earned for $f(x)$

1 point is earned for solutions for $f'(x)=0$ which includes at least one solution

1 point is earned for evaluation f at π and 2π

1 point is earned for evaluation of f at student's critical point(s)

1 point is earned for answer

$$f'(x) = \frac{1}{2 + \sin x} \cos x;$$

$$\text{In } [\pi, 2\pi], \cos x = 0 \text{ when } x = \frac{3\pi}{2}$$

x	$f(x)$
π	$\ln(2) = 0.693$
2π	$\ln(2)$
$\frac{3\pi}{2}$	$\ln(1) = 0$

absolute maximum value is $\ln 2$

absolute minimum value is 0

5-5b



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns five of the following points:

1 point is earned for $f'(x)$

1 point is earned for solutions for $f'(x)=0$ which includes at least one solution

1 point is earned for evaluation f at π and 2π

1 point is earned for evaluation of f at student's critical point(s)

1 point is earned for answer

$$f'(x) = \frac{1}{2+\sin x} \cos x;$$

$$\text{In } [\pi, 2\pi], \cos x = 0 \text{ when } x = \frac{3\pi}{2}$$

x	$f(x)$
π	$\ln(2) = 0.693$
2π	$\ln(2)$
$\frac{3\pi}{2}$	$\ln(1) = 0$

absolute maximum value is $\ln 2$

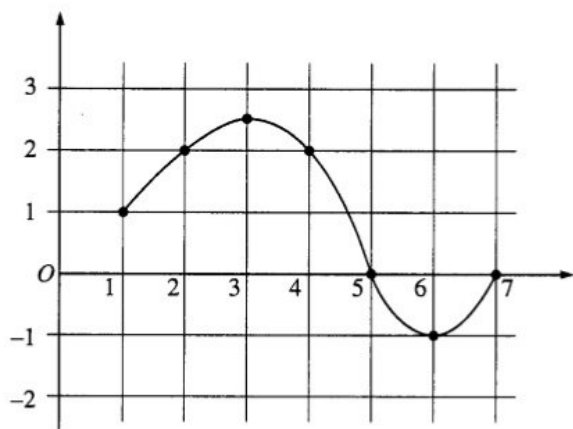
absolute minimum value is 0

5-5b

7. The absolute maximum values of $f(x) = x^3 - 3x^2 + 12$ on the closed interval $[-2, 4]$ occurs at $x =$
- (A) 4 ✓
- (B) 2
- (C) 1
- (D) 0
- (E) -2
8. The point on the curve $x^2 + 2y = 0$ that is nearest the point $(0, -1/2)$ occurs where y is
- (A) $1/2$
- (B) 0 ✓
- (C) $-1/2$
- (D) -1
- (E) none of the above
9. The maximum acceleration attained on the interval $0 \leq t \leq 3$ by the particle whose velocity is given by $v(t) = t^3 - 3t^2 + 12t + 4$ is
- (A) 9
- (B) 12
- (C) 14
- (D) 21 ✓
- (E) 40
10. If $f(x) = \frac{1}{3}x^3 - 4x^2 + 12x - 5$ and the domain is the set of all x such that $0 \leq x \leq 9$, then the absolute maximum value of the function f occurs when x is
- (A) 0
- (B) 2
- (C) 4
- (D) 6
- (E) 9 ✓

5-5b

11.



The graph of a differentiable function f on the closed interval $[1, 7]$ is shown above. Let $h(x) = \int_1^x f(t) dt$ for $1 \leq x \leq 7$.

Find the value of x at which h has its minimum on the closed interval $[1, 7]$. Justify your answer.

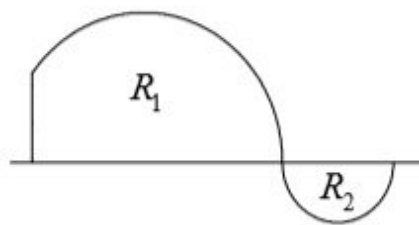
Part D

1 point is earned for the correct answer

1 point is earned for the correct justification where one point is earned for eliminating interior points and one point is earned for eliminating 7

minimum at $x = 1$ because:

h increases on $[1, 5]$ and decreases on $[5, 7]$, so minimum is at an endpoint



$$h(7) = \text{area } R_1 - \text{area } R_2 > 0 \text{ and } h(1) = 0$$



0	1	2	3
---	---	---	---

The student response earns one of the following points:

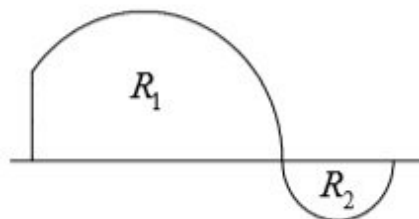
1 point is earned for the correct answer

5-5b

2 points are earned for the correct justification where one point is earned for eliminating interior points and one point is earned for eliminating 7

minimum at $x = 1$ because:

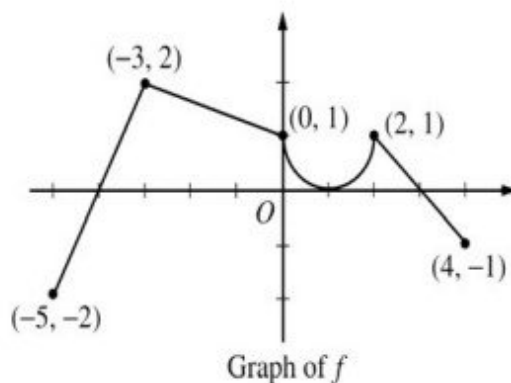
h increases on $[1, 5]$ and decreases on $[5, 7]$, so minimum is at an endpoint



$$h(7) = \text{area } R_1 - \text{area } R_2 > 0 \text{ and } h(1) = 0$$

5-5b

12.



The graph of the function f shown above consists of a semicircle and three line segments. Let g be the function given by $g(x) = \int_{-3}^x f(t) dt$.

Find the absolute minimum value of g on the closed interval $[-5, 4]$. Justify your answer.

Part C

1 point is earned for identifies $x = -4$ as a candidate

1 point is earned for: $g(-4) = -1$

1 point is earned for justification and answer

The only x -value where f changes from negative to positive is $x = -4$. The other candidates for the location of the absolute minimum value are the endpoints.

$$g(-5) = 0$$

$$g(-4) = \int_{-3}^{-4} f(t) dt = -1$$

$$g(4) = \frac{9}{2} + \left(2 - \frac{\pi}{2}\right) = \frac{13 - \pi}{2}$$

So the absolute minimum value of g is -1 .



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for identifies $x = -4$ as a candidate

1 point is earned for: $g(-4) = -1$

5-5b

1 point is earned for justification and answer

The only x -value where f changes from negative to positive is $x = -4$. The other candidates for the location of the absolute minimum value are the endpoints.

$$g(-5) = 0$$

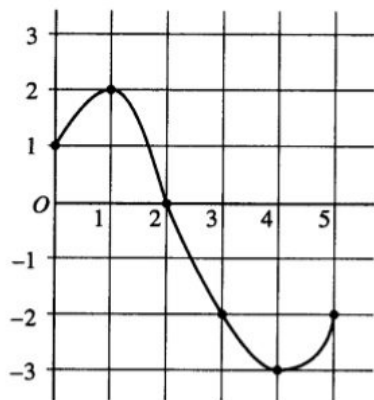
$$g(-4) = \int_{-3}^{-4} f(t) dt = -1$$

$$g(4) = \frac{9}{2} + \left(2 - \frac{\pi}{2}\right) = \frac{13 - \pi}{2}$$

So the absolute minimum value of g is -1 .

5-5b

13.

Graph of f

Let f be a function whose domain is the closed interval $[0, 5]$. The graph of f is shown above.

Let $h(x) = \int_0^{\frac{x}{2}+3} f(t) dt$.

At what x is $h(x)$ a minimum? Show the analysis that leads to your conclusion.

Part C

1 point is earned for correctly identifying the minimum is not in interior of domain

h' is positive, then negative, so minimum is at an endpoint

1 point is earned for correctly stating $h(-6) = 0$

$$h(-6) = \int_0^0 f(t) dt = 0$$

1 point is earned for correctly stating $h(4)$ is negative

$$h(4) = \int_0^5 f(t) dt < 0$$

1 point is earned for the correct answer, $x = 4$

Since the area below the axis is greater than the area above the axis therefore minimum at $x = 4$

Note: Domain $[0, 5]$, max $1/4$, $1 - 0 - 0 - 0$

No interior critical point in student's domain, max $1/4$, $0 - 0 - 0 - 1$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

5-5b

1 point is earned for correctly identifying the minimum is not in interior of domain

h' is positive, then negative, so minimum is at an endpoint

1 point is earned for correctly stating $h(-6) = 0$

$$h(-6) = \int_0^0 f(t)dt = 0$$

1 point is earned for correctly stating $h(4)$ is negative

$$h(4) = \int_0^5 f(t)dt < 0$$

1 point is earned for the correct answer, $x = 4$

Since the area below the axis is greater than the area above the axis therefore minimum at $x = 4$

Note: Domain $[0,5]$, max $1/4$, 1 - 0 - 0 - 0

No interior critical point in student's domain, max $1/4$, 0 - 0 - 0 - 1

5-5b

14. Let f be the function given by $f(x) = 3x^4 + x^3 - 21x^2$.

Find the absolute minimum value of f . Show the analysis that leads to your conclusion.

Part B

1 point is earned for the correct zeros of $f'(x)$

0/1: if f' is quadratic

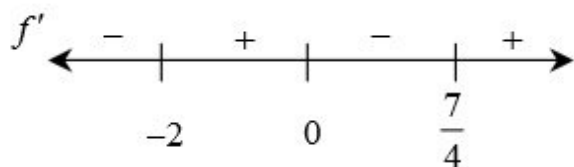
$$12x^3 + 3x^2 - 42x = 0$$

$$3x(4x^2 + x - 14) = 0$$

$$3x(4x - 7)(x + 2) = 0$$

$$x = 0, x = \frac{7}{4}, x = -2$$

1 point is earned for the correct analysis -- must indicate behavior of function over entire real line



Min must be at -2 or $\frac{7}{4}$

$$f(-2) = -44, f\left(\frac{7}{4}\right) = -30.816$$

1 point is earned for the correct answer

Absolute min is -44



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for the correct zeros of $f'(x)$

0/1: if f' is quadratic

$$12x^3 + 3x^2 - 42x = 0$$

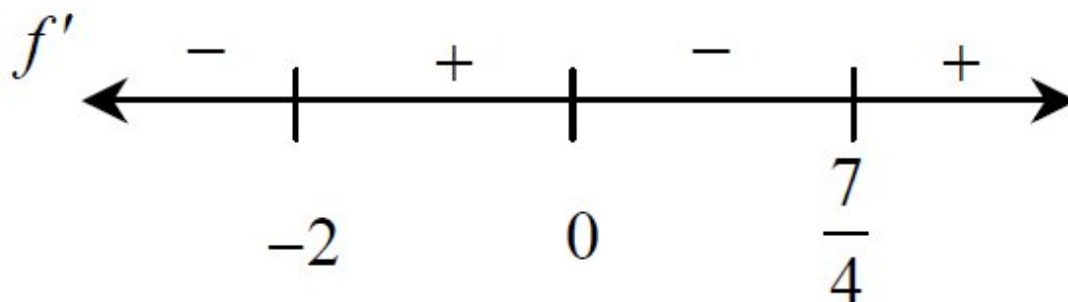
$$3x(4x^2 + x - 14) = 0$$

$$3x(4x - 7)(x + 2) = 0$$

$$x = 0, x = \frac{7}{4}, x = -2$$

1 point is earned for the correct analysis -- must indicate behavior of function over entire real line

5-5b



Min must be at -2 or $\frac{7}{4}$

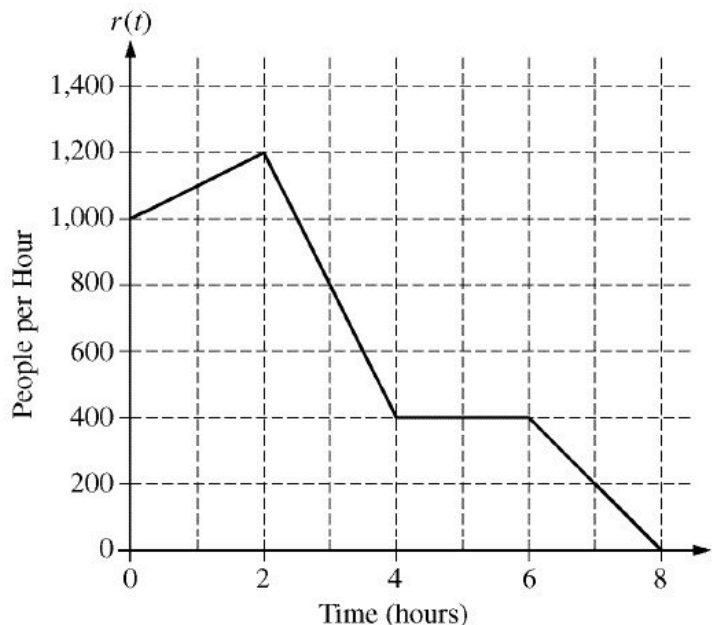
$$f(-2) = -44, f\left(\frac{7}{4}\right) = -30.816$$

1 point is earned for the correct answer

Absolute min is -44

5-5b

15.



There are 700 people in line for a popular amusement-park ride when the ride begins operation in the morning. Once it begins operation, the ride accepts passengers until the park closes 8 hours later. While there is a line, people move onto the ride at a rate of 800 people per hour. The graph above shows the rate, $r(t)$, at which people arrive at the ride throughout the day. Time t is measured in hours from the time the ride begins operation.

- How many people arrive at the ride between $t = 0$ and $t = 3$? Show the computations that lead to your answer.
- Is the number of people waiting in line to get on the ride increasing or decreasing between $t = 2$ and $t = 3$? Justify your answer.
- At what time t is the line for the ride the longest? How many people are in line at that time? Justify your answers.
- Write, but do not solve, an equation involving an integral expression of r whose solution gives the earliest time t at which there is no longer a line for the ride.

Part A

1 point is earned for integral

1 point is earned for the answer

$$\int_0^3 r(t) dt = 2 \cdot \frac{1000 + 1200}{2} + \frac{1200 + 800}{2} = 3200 \text{ people}$$



0	1	2
---	---	---

5-5b

The student response earns two of the following points:

1 point is earned for integral

1 point is earned for the answer

$$\int_0^3 r(t)dt = 2 \cdot \frac{1000 + 1200}{2} + \frac{1200 + 800}{2} = 3200 \text{ people}$$

Part B

1 point is earned for answer with reason

The number of people waiting in line is increasing because people move onto the ride at a rate of 800 people per hour and for $2 < t < 3$, $r(t) > 800$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer with reason

The number of people waiting in line is increasing because people move onto the ride at a rate of 800 people per hour and for $2 < t < 3$, $r(t) > 800$.

Part C

1 point is earned for identifying $t = 3$

1 point is earned for the number of people in line

1 point is earned for justification

$$r(t) = 800 \text{ only at } t = 3$$

For $0 \leq t < 3$, $r(t) > 800$. For $3 < t \leq 8$, $r(t) < 800$.

Therefore, the line is longest at time $t = 3$.

There are $700 + 3200 - 800 \cdot 3 = 1500$ people waiting in line at time $t = 3$.

5-5b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for identifying $t = 3$

1 point is earned for the number of people in line

1 point is earned for justification

$r(t) = 800$ only at $t = 3$

For $0 \leq t < 3$, $r(t) > 800$. For $3 < t \leq 8$, $r(t) < 800$.

Therefore, the line is longest at time $t = 3$.

There are $700 + 3200 - 800 \cdot 3 = 1500$ people waiting in line at time $t = 3$.

Part D

1 point is earned for $800t$

1 point is earned for integral

1 point is earned for the answer

$$0 = 700 + \int_0^t r(s)ds - 800t$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $800t$

1 point is earned for integral

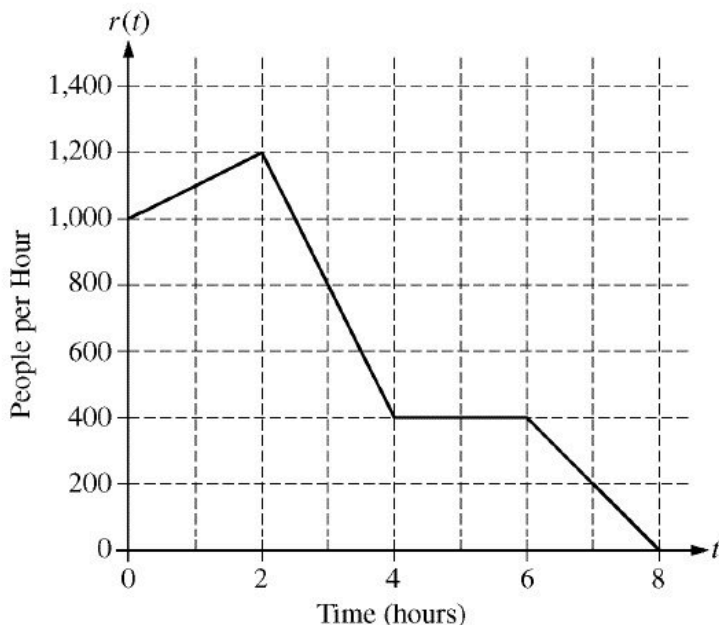
1 point is earned for the answer

5-5b

$$0 = 700 + \int_0^t r(s) ds - 800t$$

5-5b

16.



There are 700 people in line for a popular amusement-park ride when the ride begins operation in the morning. Once it begins operation, the ride accepts passengers until the park closes 8 hours later. While there is a line, people move onto the ride at a rate of 800 people per hour. The graph above shows the rate, $r(t)$, at which people arrive at the ride throughout the day. Time t is measured in hours from the time the ride begins operation.

At what time t is the line for the ride the longest? How many people are in line at that time? Justify your answers.

Part C

1 point is earned for identifying $t=3$

1 point is earned for the number of people in line

1 point is earned for justification

$r(t)=800$ only at $t=3$

For $0 \leq t < 3$, $r(t) > 800$. For $3 < t \leq 8$, $r(t) < 800$.

Therefore, the line is longest at time $t=3$.

There are $700+3200-800 \cdot 3=1500$ people waiting in line at time $t=3$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

5-5b

1 point is earned for identifying $t=3$

1 point is earned for the number of people in line

1 point is earned for justification

$r(t)=800$ only at $t=3$

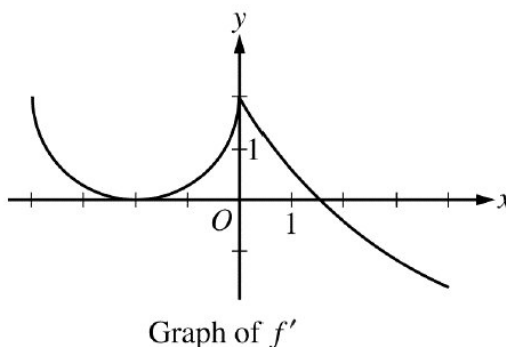
For $0 \leq t < 3$, $r(t) > 800$. For $3 < t \leq 8$, $r(t) < 800$.

Therefore, the line is longest at time $t=3$.

There are $700+3200-800 \cdot 3=1500$ people waiting in line at time $t=3$.

5-5b

17.



The derivative of a function f is defined by $f'(x) = \begin{cases} g(x) & \text{for } -4 \leq x \leq 0 \\ 5e^{-x/3} - 3 & \text{for } 0 < x \leq 4 \end{cases}$.

The graph of the continuous function f , shown in the figure above, has x -intercepts at $x = -2$ and $x = 3 \ln\left(\frac{5}{3}\right)$. The graph of g on $-4 \leq x \leq 0$ is a semicircle, and $f(0) = 5$.

- (a) For $-4 < x < 4$, find all values of x at which the graph of f has a point of inflection. Justify your answer
 (b) Find $f(-4)$ and $f(4)$.
 (c) For $-4 \leq x \leq 4$, find the value of x at which f has an absolute maximum. Justify your answer.

Part A

1 point is earned if identifies $x = -2$ or $x = 0$

1 point is earned for answer with justification

f changes from decreasing to increasing at $x = -2$ and from increasing to decreasing at $x = 0$. Therefore, the graph of f has points of inflections at $x = -2$ and $x = 0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if identifies $x = -2$ or $x = 0$

1 point is earned for answer with justification

f changes from decreasing to increasing at $x = -2$ and from increasing to decreasing at $x = 0$. Therefore, the graph of f has points of inflections at $x = -2$ and $x = 0$.

Part B

1 point is earned for integral of $f(-4)$

5-5b

1 point is earned for value of $f(-4)$

1 point is earned for integral of $f(4)$

1 point is earned for antiderivative of $f(4)$

1 point is earned for value of $f(4)$

$$\begin{aligned} f(-4) &= 5 + \int_0^{-4} g(x) dx \\ &= 5 - (8 - 2\pi) = 2\pi - 3 \end{aligned}$$

$$\begin{aligned} f(4) &= 5 + \int_0^4 (5e^{-x/3} - 3) dx \\ &= 5 + (-15e^{-x/3} - 3x) \Big|_{x=0}^{x=4} \\ &= 8 - 15e^{-4/3} \end{aligned}$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for integral of $f(-4)$

1 point is earned for value of $f(-4)$

1 point is earned for integral of $f(4)$

1 point is earned for antiderivative of $f(4)$

1 point is earned for value of $f(4)$

$$\begin{aligned} f(-4) &= 5 + \int_0^{-4} g(x) dx \\ &= 5 - (8 - 2\pi) = 2\pi - 3 \end{aligned}$$

$$\begin{aligned} f(4) &= 5 + \int_0^4 (5e^{-x/3} - 3) dx \\ &= 5 + (-15e^{-x/3} - 3x) \Big|_{x=0}^{x=4} \\ &= 8 - 15e^{-4/3} \end{aligned}$$

5-5b

Part C

1 point is earned for the answer

1 point is earned for the justification

Since $f'(x) > 0$ on the intervals $-4 < x < -2$ and $-2 < x < 3 \ln\left(\frac{5}{3}\right)$, f is increasing on the interval $-4 \leq x \leq 3 \ln\left(\frac{5}{3}\right)$.

Since $f'(x) < 0$ on the interval $3 \ln\left(\frac{5}{3}\right) < x < 4$, f is decreasing on the interval $3 \ln\left(\frac{5}{3}\right) \leq x \leq 4$.

Therefore, f has an absolute maximum at $x = 3 \ln\left(\frac{5}{3}\right)$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

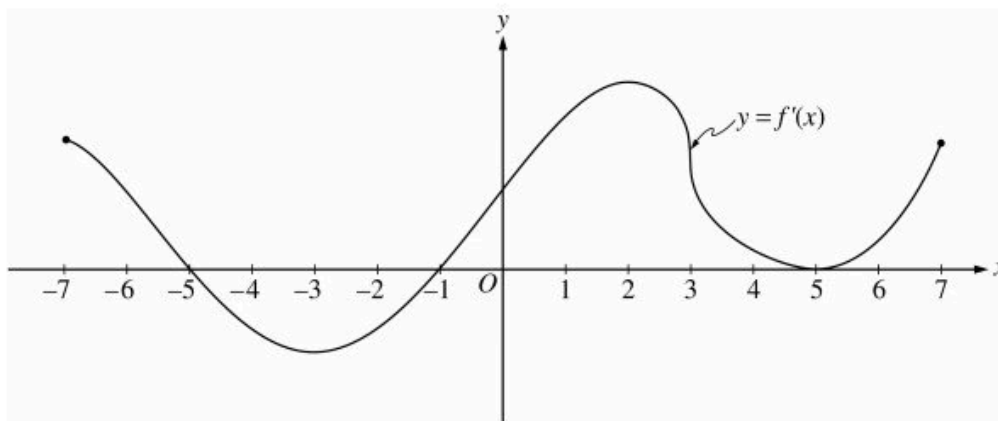
Since $f'(x) > 0$ on the intervals $-4 < x < -2$ and $-2 < x < 3 \ln\left(\frac{5}{3}\right)$, f is increasing on the interval $-4 \leq x \leq 3 \ln\left(\frac{5}{3}\right)$.

Since $f'(x) < 0$ on the interval $3 \ln\left(\frac{5}{3}\right) < x < 4$, f is decreasing on the interval $3 \ln\left(\frac{5}{3}\right) \leq x \leq 4$.

Therefore, f has an absolute maximum at $x = 3 \ln\left(\frac{5}{3}\right)$.

5-5b

18.



The figure above shows the graph of f' , the derivative of the function f , for $-7 \leq x \leq 7$. The graph of f has horizontal tangent lines at $x = -3$, $x = 2$, and $x = 5$, and a vertical tangent line at $x = 3$.

- Find all values of x , for $-7 < x < 7$, at which f attains a relative minimum. Justify your answer.
- Find all values of x , for $-7 < x < 7$, at which f attains a relative maximum. Justify your answer.
- Find all values of x , for $-7 < x < 7$, at which $f''(x) < 0$.
- At what value of x , for $-7 \leq x \leq 7$, does f attain its absolute maximum? Justify your answer.

Part A

1 point is earned for the answer

1 point is earned for the justification

$x = -1$

$f'(x)$ changes from negative to positive at $x = -1$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$x = -1$

$f'(x)$ changes from negative to positive at $x = -1$

Part B

1 point is earned for the answer

5-5b

1 point is earned for the justification

$$x = -5$$

$f'(x)$ changes from positive to negative at $x = -5$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$$x = -5$$

$f'(x)$ changes from positive to negative at $x = -5$

Part C

1 point is earned for $(-7, -3)$

1 point is earned for $(2, 3) \cup (3, 5)$

$f''(x)$ exists and f' is decreasing on the intervals $(-7, -3)$, $(2, 3)$, and $(3, 5)$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $(-7, -3)$

1 point is earned for $(2, 3) \cup (3, 5)$

$f''(x)$ exists and f' is decreasing on the intervals $(-7, -3)$, $(2, 3)$, and $(3, 5)$

Part D

1 point is earned for the answer

1 point is earned for identifying $x = -5$ and $x = 7$ as candidates

- or -

indicating that the graph of f increases, decreases, then increases

5-5b

1 point is earned for justifying $f(7) > f(-5)$

$$x = 7$$

The absolute maximum must occur at $x = -5$ or at an endpoint.

$f(-5) > f(-7)$ because f is increasing on $(-7, -5)$

The graph of f' shows that the magnitude of the negative change in f from $x = -5$ to $x = -1$ is smaller than the positive change in f from $x = -1$ to $x = 7$. Therefore the net change in f is positive from $x = -5$ to $x = 7$, and $f(7) > f(-5)$. So $f(7)$ is the absolute maximum.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for identifying $x = -5$ and $x = 7$ as candidates

- or -

indicating that the graph of f increases, decreases, then increases

1 point is earned for justifying $f(7) > f(-5)$

$$x = 7$$

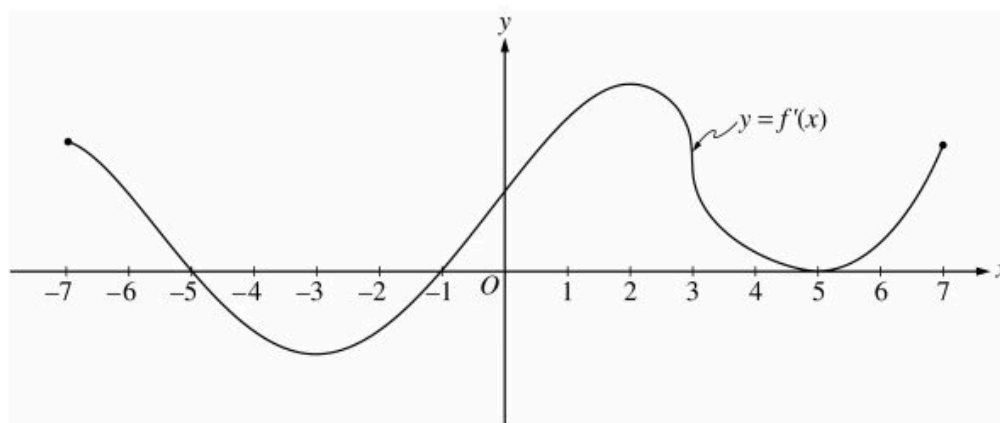
The absolute maximum must occur at $x = -5$ or at an endpoint.

$f(-5) > f(-7)$ because f is increasing on $(-7, -5)$

The graph of f' shows that the magnitude of the negative change in f from $x = -5$ to $x = -1$ is smaller than the positive change in f from $x = -1$ to $x = 7$. Therefore the net change in f is positive from $x = -5$ to $x = 7$, and $f(7) > f(-5)$. So $f(7)$ is the absolute maximum.

5-5b

19.



The figure above shows the graph of f' , the derivative of the function f , for $-7 \leq x \leq 7$. The graph of f has horizontal tangent lines at $x = -3$, $x = 2$, and $x = 5$, and a vertical tangent line at $x = 3$.

At what value of x , for $-7 \leq x \leq 7$, does f attain its absolute maximum? Justify your answer.

Part D

1 point is earned for the answer

1 point is earned for identifies $x = -5$ and $x = 7$ as candidates

- or -

indicates that the graph of f increase, decreases, then increases

1 point is earned for justifies $f(7) > f(-5)$

$x = 7$

The absolute maximum must occur at $x = -5$ or at an endpoint.

$f(-5) > f(-7)$ because f is increasing on $(-7, -5)$

The graph of f' shows that the magnitude of the negative change in f from $x = -5$ to $x = -1$ is smaller than the positive change in f from $x = -1$ to $x = 7$. Therefore the net change in f is positive from $x = -5$ to $x = 7$, and $f(7) > f(-5)$. So $f(7)$ is the absolute maximum.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for identifies $x = -5$ and $x = 7$ as candidates

5-5b

- or -

indicates that the graph of f increase, decreases, then increases

1 point is earned for justifies $f(7) > f(-5)$

 $x = 7$

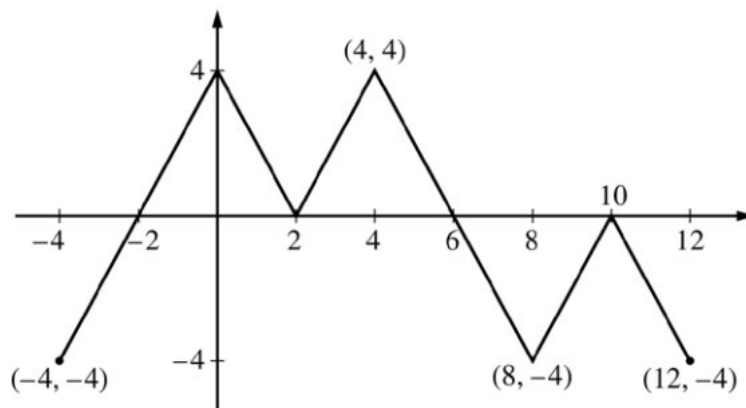
The absolute maximum must occur at $x = -5$ or at an endpoint.

$f(-5) > f(-7)$ because f is increasing on $(-7, -5)$

The graph of f' shows that the magnitude of the negative change in f from $x = -5$ to $x = -1$ is smaller than the positive change in f from $x = -1$ to $x = 7$. Therefore the net change in f is positive from $x = -5$ to $x = 7$, and $f(7) > f(-5)$. So $f(7)$ is the absolute maximum.

5-5b

20.

Graph of f

The figure above shows the graph of the piecewise-linear function f . For $-4 \leq x \leq 12$, the function g is defined by $g(x) = \int_2^x f(t) dt$.

- Does g have a relative minimum, a relative maximum, or neither at $x = 10$? Justify your answer.
- Does the graph of g have a point of inflection at $x = 4$? Justify your answer.
- Find the absolute minimum value and the absolute maximum value of g on the interval $-4 \leq x \leq 12$. Justify your answers.
- For $-4 \leq x \leq 12$, find all intervals for which $g(x) \leq 0$.

Part A

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for answer with justification

The function g has neither a relative minimum nor a relative maximum at $x = 10$ since $g'(x) = f(x)$ and $f(x) \leq 0$ for $8 \leq x \leq 12$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for answer with justification

The function g has neither a relative minimum nor a relative maximum at $x = 10$ since $g'(x) = f(x)$ and $f(x) \leq 0$ for $8 \leq x \leq 12$.

Part B

5-5b

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for answer with justification

The graph of g has a point of inflection at $x = 4$ since $g'(x) = f(x)$ is increasing for $2 \leq x \leq 4$ and decreasing for $4 \leq x \leq 8$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for answer with justification

The graph of g has a point of inflection at $x = 4$ since $g'(x) = f(x)$ is increasing for $2 \leq x \leq 4$ and decreasing for $4 \leq x \leq 8$.

Part C

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for considering $x = -2$ and $x = 6$ as candidates

$g'(x) = f(x)$ changes sign only at $x = -2$ and $x = 6$

1 point is earned for considering $x = -4$ and $x = 12$

x	$g(x)$
-4	-4
-2	-8
6	8
12	-4

2 points are earned for answers with justification

On the interval $-4 \leq x \leq 12$, the absolute minimum value is $g(-2) = -8$ and the absolute maximum value is $g(6) = 8$.



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

5-5b

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for considering $x = -2$ and $x = 6$ as candidates

$g'(x) = f(x)$ changes sign only at $x = -2$ and $x = 6$

1 point is earned for considering $x = -4$ and $x = 12$

x	$g(x)$
-4	-4
-2	-8
6	8
12	-4

2 points are earned for answers with justification

On the interval $-4 \leq x \leq 12$, the absolute minimum value is $g(-2) = -8$ and the absolute maximum value is $g(6) = 8$.

Part D

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

2 point are earned for intervals

$g(x) \leq 0$ for $-4 \leq x \leq 2$ and $10 \leq x \leq 12$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

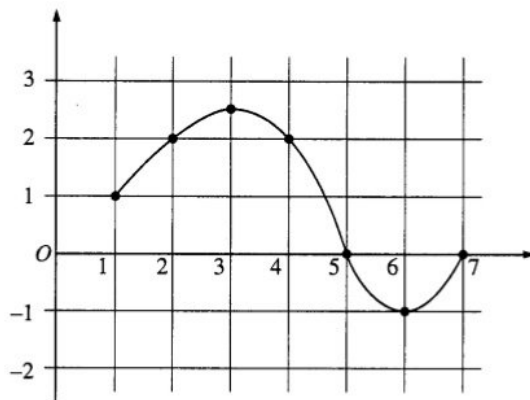
1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

2 point are earned for intervals

$g(x) \leq 0$ for $-4 \leq x \leq 2$ and $10 \leq x \leq 12$.

5-5b

21.



The graph of a differentiable function f on the closed interval $[1, 7]$ is shown above. Let $h(x) = \int_1^x f(t) dt$

for $1 \leq x \leq 7$.

(a) Find $h(1)$.

(b) Find $h'(4)$.

(c) On what interval or intervals is the graph of h concave upward? Justify your answer.

(d) Find the value of x at which h has its minimum on the closed interval $[1, 7]$. Justify your answer.

Part A

1 point is earned for the correct answer

$$h(1) = \int_1^1 f(t) dt = 0$$



0

1

The student response earns all of the following points:

1 point is earned for the correct answer

$$h(1) = \int_1^1 f(t) dt = 0$$

Part B

1 point is earned for correctly identifying $h'(x) = f(x)$

1 point is earned for the correct answer

$$h'(4) = f(4) =$$

5-5b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly identifying $h'(x) = f(x)$

1 point is earned for the correct answer

$$h'(4) = f(4) =$$

Part C

2 points are earned for the correct intervals where one point is earned for the first interval and one point is earned for the second interval

$$1 < x < 3 \text{ and } 6 < x < 7$$

1 point is earned for the correct justification

h is concave up when:

- h' is increasing, or
- f is increasing, or
- $h''(x) > 0$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct intervals where one point is earned for the first interval and one point is earned for the second interval

$$1 < x < 3 \text{ and } 6 < x < 7$$

1 point is earned for the correct justification

h is concave up when:

- h' is increasing, or
- f is increasing, or

5-5b

- $h''(x) > 0$

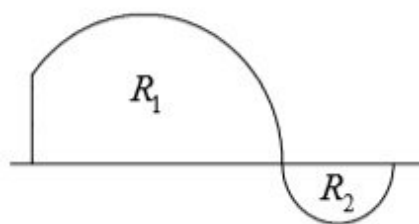
Part D

1 point is earned for the correct answer

1 point is earned for the correct justification where one point is earned for eliminating interior points and one point is earned for eliminating 7

minimum at $x = 1$ because:

h increases on $[1, 5]$ and decreases on $[5, 7]$, so minimum is at an endpoint



$$h(7) = \text{area } R_1 - \text{area } R_2 > 0 \text{ and } h(1) = 0$$



0	1	2	3
---	---	---	---

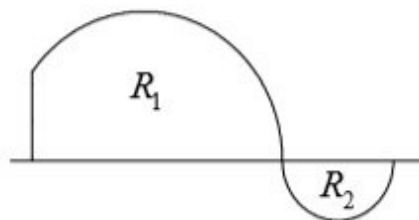
The student response earns all of the following points:

1 point is earned for the correct answer

1 point is earned for the correct justification where one point is earned for eliminating interior points and one point is earned for eliminating 7

minimum at $x = 1$ because:

h increases on $[1, 5]$ and decreases on $[5, 7]$, so minimum is at an endpoint



5-5b

$$h(7) = \text{area } R_1 - \text{area } R_2 > 0 \text{ and } h(1) = 0$$

5-5b

22. Let f be the function defined by $f(x) = \sin^2 x - \sin x$ for $0 \leq x \leq 3\pi/2$.

- (a) Find the x -intercepts of the graph of f .
- (b) Find the intervals on which f is increasing.
- (c) Find the absolute maximum value and the absolute minimum value of f . Justify your answer.

Part A

1 point is earned for $\sin^2 x - \sin x = 0$

1 point is earned for three appropriate roots

$$\sin^2 x - \sin x = 0$$

$$\therefore \sin x = 0 \text{ or } \sin x = 1$$

$$x = 0, \frac{\pi}{2}, \pi$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\sin^2 x - \sin x = 0$

1 point is earned for three appropriate roots

$$\sin^2 x - \sin x = 0$$

$$\therefore \sin x = 0 \text{ or } \sin x = 1$$

$$x = 0, \frac{\pi}{2}, \pi$$

Part B

1 point is earned for $f'(x)$

5-5b

1 point is earned for finding 2 interior critical points

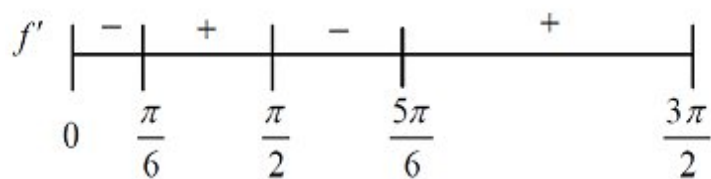
1 point is earned for analysis

1 point is earned for answer

$$f'(x) = 2 \sin x \cos x - \cos x$$

$$= \cos x(2 \sin x - 1)$$

$$f'(x) = 0 \text{ where } x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$$



$$\frac{\pi}{6} \leq x \leq \frac{\pi}{2} \text{ and } \frac{5\pi}{6} \leq x \leq \frac{3\pi}{2}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for finding 2 interior critical points

1 point is earned for analysis

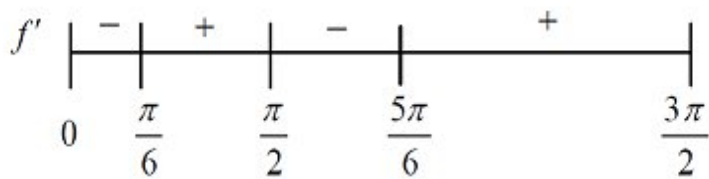
1 point is earned for answer

$$f'(x) = 2 \sin x \cos x - \cos x$$

$$= \cos x(2 \sin x - 1)$$

$$f'(x) = 0 \text{ where } x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$$

5-5b



$$\frac{\pi}{6} \leq x \leq \frac{\pi}{2} \text{ and } \frac{5\pi}{6} \leq x \leq \frac{3\pi}{2}$$

Part C

1 point is earned for max value

1 point is earned for min value

1 point is earned for justification

Maximum value: 2

Minimum value : -1/4

x	$f(x)$
0	0
$\frac{\pi}{6}$	$-\frac{1}{4}$
$\frac{\pi}{2}$	0
$\frac{5\pi}{6}$	$-\frac{1}{4}$
$\frac{3\pi}{2}$	2



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for max value

5-5b

1 point is earned for min value

1 point is earned for justification

Maximum value: 2

Minimum value : $-\frac{1}{4}$

x	$f(x)$
0	0
$\frac{\pi}{6}$	$-\frac{1}{4}$
$\frac{\pi}{2}$	0
$\frac{5\pi}{6}$	$-\frac{1}{4}$
$\frac{3\pi}{2}$	2

5-5b

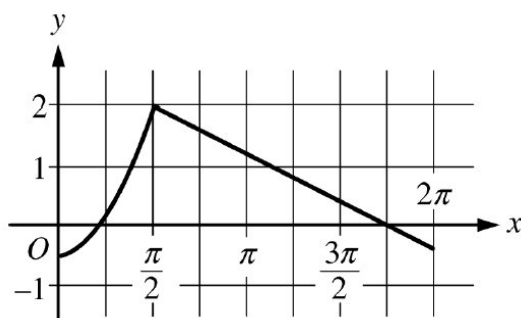
23. Let f be the function defined by $f(x) = e^x \cos(x)$.

(a) Find the average rate of change of f on the interval $0 \leq x \leq \pi$.

(b) What is the slope of the line tangent to the graph of f at $x = \frac{3\pi}{2}$?

(c) Find the absolute minimum value of f on the interval $0 \leq x \leq 2\pi$. Justify your answer.

(d) Let g be a differentiable function such that $g\left(\frac{\pi}{2}\right) = 0$. The graph of g' , the derivative of g , is shown below. Find the value of $\lim_{x \rightarrow \pi/2} \frac{f(x)}{g(x)}$ or state that it does not exist. Justify your answer.



Graph of g'

Part A

1 point is earned for answer

The average rate of change of f on the interval $0 \leq x \leq \pi$ is $\frac{f(\pi) - f(0)}{\pi - 0} = \frac{-e^\pi - 1}{\pi}$.



0

1

The student response earns one of the following points:

1 point is earned for answer

The average rate of change of f on the interval $0 \leq x \leq \pi$ is $\frac{f(\pi) - f(0)}{\pi - 0} = \frac{-e^\pi - 1}{\pi}$.

Part B

1 point is earned for $f'(x)$

1 point is earned for slope

$$f'(x) = e^x \cos x - e^x \sin x$$

$$f'\left(\frac{3\pi}{2}\right) = e^{3\pi/2} \cos\left(\frac{3\pi}{2}\right) - e^{3\pi/2} \sin\left(\frac{3\pi}{2}\right) = e^{3\pi/2}$$

5-5b

The slope of the line tangent to the graph of f at $x = \frac{3\pi}{2}$ is $e^{3\pi/2}$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for slope

$$f'(x) = e^x \cos x - e^x \sin x$$

$$f'\left(\frac{3\pi}{2}\right) = e^{3\pi/2} \cos\left(\frac{3\pi}{2}\right) - e^{3\pi/2} \sin\left(\frac{3\pi}{2}\right) = e^{3\pi/2}$$

The slope of the line tangent to the graph of f at $x = \frac{3\pi}{2}$ is $e^{3\pi/2}$.

Part C

1 point is earned for sets $f'(x) = 0$

1 point is earned for identifies $x = \frac{\pi}{4}$, $x = \frac{5\pi}{4}$ as candidates

1 point is earned for answer with justification

$$f'(x) = 0 \Rightarrow \cos x - \sin x = 0 \Rightarrow x = \frac{\pi}{4}, x = \frac{5\pi}{4}$$

5-5b

x	$f(x)$
0	1
$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}e^{\pi/4}$
$\frac{5\pi}{4}$	$-\frac{1}{\sqrt{2}}e^{5\pi/4}$
2π	$e^{2\pi}$

The absolute minimum value of f on $0 \leq x \leq 2\pi$ is $-\frac{1}{\sqrt{2}}e^{5\pi/4}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for sets $f'(x) = 0$

1 point is earned for identifies as $x = \frac{\pi}{4}$, $x = \frac{5\pi}{4}$ candidates

1 point is earned for answer with justification

$$f'(x) = 0 \Rightarrow \cos x - \sin x = 0 \Rightarrow x = \frac{\pi}{4}, x = \frac{5\pi}{4}$$

5-5b

x	$f(x)$
0	1
$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}e^{\pi/4}$
$\frac{5\pi}{4}$	$-\frac{1}{\sqrt{2}}e^{5\pi/4}$
2π	$e^{2\pi}$

The absolute minimum value of f on $0 \leq x \leq 2\pi$ is $-\frac{1}{\sqrt{2}}e^{5\pi/4}$.

Part D

1 point is earned for g is continuous at $x = \pi/2$ and limits equal 0

1 point is earned for applies L'Hospital's Rule

1 point is earned for answer

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives

$$\lim_{x \rightarrow \pi/2} f(x) = 0$$

Because g is differentiable, g is continuous.

$$\lim_{x \rightarrow \pi/2} g(x) = g\left(\frac{\pi}{2}\right) = 0$$

By L'Hospital's Rule,

$$\lim_{x \rightarrow \pi/2} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \pi/2} \frac{f'(x)}{g'(x)} = \frac{-e^{\pi/2}}{2}.$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for g is continuous at $x = \pi/2$ and limits equal 0

5-5b

1 point is earned for applies L'Hospital's Rule

1 point is earned for answer

Note: max 1/3 [1-0-0] if no limit notation attached to a ratio of derivatives

$$\lim_{x \rightarrow \pi/2} f(x) = 0$$

Because g is differentiable, g is continuous.

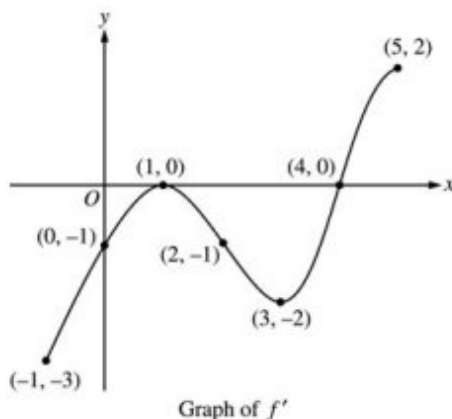
$$\lim_{x \rightarrow \pi/2} g(x) = g\left(\frac{\pi}{2}\right) = 0$$

By L'Hospital's Rule,

$$\lim_{x \rightarrow \pi/2} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \pi/2} \frac{f'(x)}{g'(x)} = \frac{-e^{\pi/2}}{2}.$$

5-5b

24.



The figure above shows the graph of f' , the derivative of the function f , on the closed interval $-1 \leq x \leq 5$. The graph of f' has horizontal tangent lines at $x = 1$ and $x = 3$. The function f is twice differentiable with $f(2) = 6$.

- (a) Find the x -coordinate of each of the points of inflection of the graph of f . Give a reason for your answer.
- (b) At what value of x does f attain its absolute minimum value on the closed interval $-1 \leq x \leq 5$? At what value of x does f attain its absolute maximum value on the closed interval $-1 \leq x \leq 5$? Show the analysis that leads to your answers.
- (c) Let g be the function defined by $g(x) = xf(x)$. Find an equation for the line tangent to the graph of g at $x = 2$.

Part A

1 point is earned for $x = 1, x = 3$

1 point is earned for the reason

$x = 1$ and $x = 3$ because the graph of f' changes from increasing to decreasing at $x = 1$, and changes from decreasing to increasing at $x = 3$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $x = 1, x = 3$

1 point is earned for the reason

$x = 1$ and $x = 3$ because the graph of f' changes from increasing to decreasing at $x = 1$, and changes from decreasing to increasing at $x = 3$.

Part B

5-5b

1 point is earned for indicating f decreases then increases

1 point is earned for eliminating $x = 5$ for maximum

1 point is earned for absolute minimum at $x = 4$

1 point is earned for absolute maximum at $x = -1$

The function f decreases from $x = -1$ to $x = 4$, then increases from $x = 4$ to $x = 5$.

Therefore, the absolute minimum value for f is at $x = 4$. The absolute maximum value must occur at $x = -1$ or at $x = 5$.

$$f(5) - f(-1) = \int_{-1}^5 f'(t) dt < 0$$

Since $f(5) < f(-1)$, the absolute maximum value occurs at $x = -1$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for indicating f decreases then increases

1 point is earned for eliminating $x = 5$ for maximum

1 point is earned for absolute minimum at $x = 4$

1 point is earned for absolute maximum at $x = -1$

The function f decreases from $x = -1$ to $x = 4$, then increases from $x = 4$ to $x = 5$.

Therefore, the absolute minimum value for f is at $x = 4$. The absolute maximum value must occur at $x = -1$ or at $x = 5$.

$$f(5) - f(-1) = \int_{-1}^5 f'(t) dt < 0$$

Since $f(5) < f(-1)$, the absolute maximum value occurs at $x = -1$.

Part C

2 points are earned for $g'(x)$

1 point is earned for the tangent line

$$g'(x) = f(x) + xf'(x)$$

5-5b

$$g'(2) = f(2) + 2f'(2) = 6 + 2(-1) = 4$$

$$g(2) = 2f(2) = 12$$

Tangent line is $y = 4(x - 2) + 12$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $g'(x)$

1 point is earned for the tangent line

$$g'(x) = f(x) + xf'(x)$$

$$g'(2) = f(2) + 2f'(2) = 6 + 2(-1) = 4$$

$$g(2) = 2f(2) = 12$$

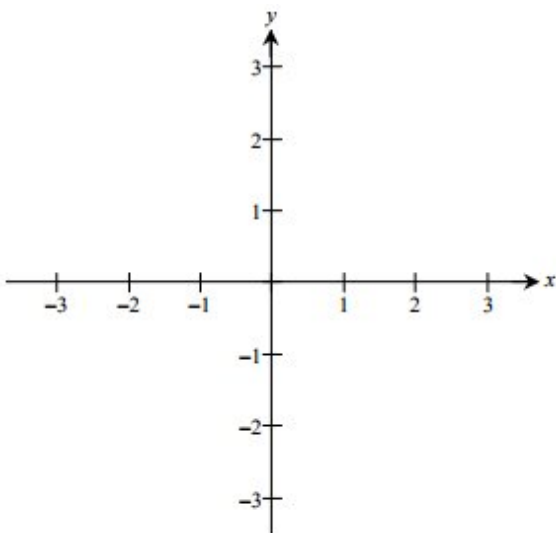
Tangent line is $y = 4(x - 2) + 12$

5-5b

25. Let f be a function that is even and continuous on the closed interval $[-3, 3]$. The function f and its derivatives have the properties indicated in the table below.

x	0	$0 < x < 1$	1	$1 < x < 2$	2	$2 < x < 3$
$f(x)$	1	Positive	0	Negative	-1	Negative
$f'(x)$	Undefined	Negative	0	Negative	Undefined	Positive
$f''(x)$	Undefined	Positive	0	Negative	Undefined	Negative

- (a) Find the x -coordinate of each point at which f attains an absolute maximum value or an absolute minimum value. For each x -coordinate you give, state whether f attains an absolute maximum or an absolute minimum.
- (b) Find the x -coordinate of each point of inflection on the graph of f . Justify your answer.
- (c) In the xy -plane provided below, sketch the graph of a function with all the given characteristics of f .



Part A

1 point is earned for first point

1 point is earned for second point

1 point is earned for third point

1 point is **deducted** for each additional point or incorrect point

Absolute maximum at $x = 0$

Absolute minimum at $x = \pm 2$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

5-5b

1 point is earned for first point

1 point is earned for second point

1 point is earned for third point

1 point is **deducted** for each additional point or incorrect point

Absolute maximum at $x = 0$

Absolute minimum at $x = \pm 2$

Part B

1 point is earned for first point identified

1 point is earned for second point identified

1 point is earned for justification (required on $[0,3)$ only)

1 point is **deducted** for each additional point or incorrect point

The points of inflection are at $x = \pm 1$ because the sign of $f''(x)$ changes at $x = 1$ and f is even.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for first point identified

1 point is earned for second point identified

1 point is earned for justification (required on $[0,3)$ only)

1 point is **deducted** for each additional point or incorrect point

The points of inflection are at $x = \pm 1$ because the sign of $f''(x)$ changes at $x = 1$ and f is even.

Part C

3 points are earned for sketch without error

1 point is **deducted** for each error:

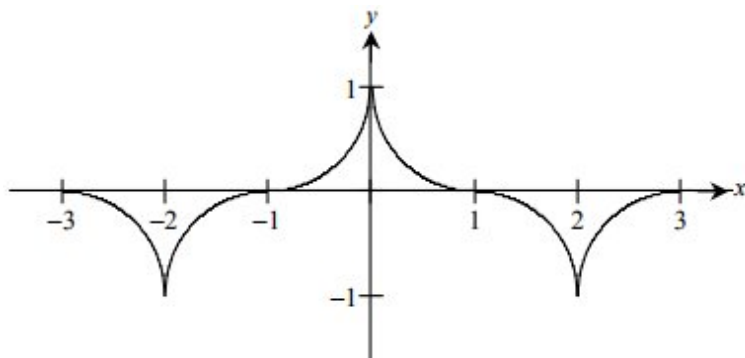
· symmetry

· slopes on $[0,3)$

5-5b

- concavity on $[0,3)$
- discontinuity on $[0,3)$
- $f(0), f(1), f(2)$ incorrect or extra intercept on $[0,3)$

Note: No more than 1 point cant be lost in any one category.



0	1	2	3
---	---	---	---

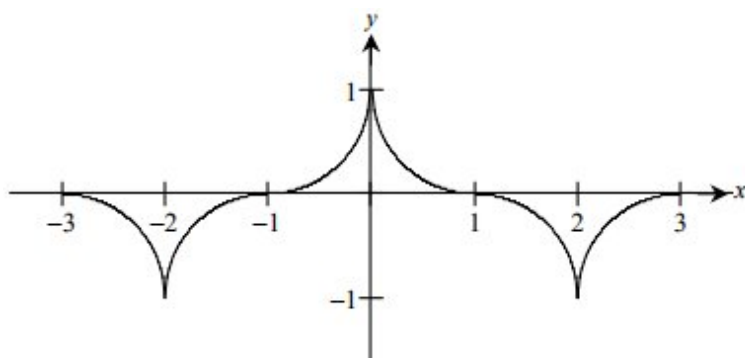
The student response earns all of the following points:

3 points are earned for sketch without error

1 point is **deducted** for each error:

- symmetry
- slopes on $[0,3)$
- concavity on $[0,3)$
- discontinuity on $[0,3)$
- $f(0), f(1), f(2)$ incorrect or extra intercept on $[0,3)$

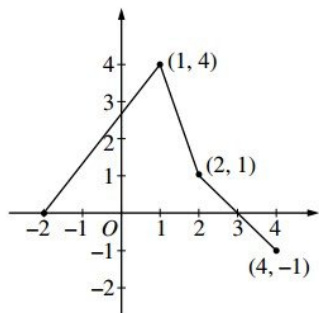
Note: No more than 1 point cant be lost in any one category.



5-5b

5-5b

26.



The graph of the function f , consisting of three line segments, is given above. Let $g(x) = \int_1^x f(t)dt$.

- Compute $g(4)$ and $g(-2)$.
- Find the instantaneous rate of change of g , with respect to x , at $x = 1$.
- Find the absolute minimum value of g on the closed interval $[-2, 4]$. Justify your answer.
- The second derivative of g is not defined at $x = 1$ and $x = 2$. How many of these values are x -coordinates of points of inflection of the graph of g ? Justify your answer.

Part A

1 point is earned for: $g(4)$

1 point is earned for: $g(-2)$

$$g(4) = \int_1^4 f(t)dt = \frac{3}{2} + 1 + \frac{1}{2} - \frac{1}{2} = \frac{5}{2}$$

$$g(-2) = \int_1^{-2} f(t)dt = -\frac{1}{2}(12) = -6$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $g(4)$

1 point is earned for: $g(-2)$

5-5b

$$g(4) = \int_1^4 f(t) dt = \frac{3}{2} + 1 + \frac{1}{2} - \frac{1}{2} = \frac{5}{2}$$

$$g(-2) = \int_1^{-2} f(t) dt = -\frac{1}{2}(12) = -6$$

Part B

1 point is earned for the answer

$$g'(1) = f(1) = 4$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$g'(1) = f(1) = 4$$

Part C

1 point is earned for interior analysis

1 point is earned for endpoint analysis

1 point is earned for the answer

g is increasing on $[-2, 3]$ and decreasing on $[3, 4]$.

Therefore, g has absolute minimum at an endpoint of $[-2, 4]$.

Since $g(-2) = -6$ and $g(4) = \frac{5}{2}$, the absolute minimum value is -6 .



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for interior analysis

5-5b

1 point is earned for endpoint analysis

1 point is earned for the answer

g is increasing on $[-2, 3]$ and decreasing on $[3, 4]$.

Therefore, g has absolute minimum at an endpoint of $[-2, 4]$.

Since $g(-2) = -6$ and $g(4) = \frac{5}{2}$, the absolute minimum value is -6 .

Part D

1 point is earned for the choice of $x = 1$ only

1 point is earned for showing $(1, g(1))$ is a point of inflection

1 point is earned for showing $(2, g(2))$ is not a point of inflection

One; $x = 1$

On $(-2, 1)$, $g''(x) = f'(x) > 0$

On $(1, 2)$, $g''(x) = f'(x) < 0$

On $(2, 4)$, $g''(x) = f'(x) < 0$

Therefore $(1, g(1))$ is a point of inflection and $(2, g(2))$ is not.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the choice of $x = 1$ only

1 point is earned for showing $(1, g(1))$ is a point of inflection

1 point is earned for showing $(2, g(2))$ is not a point of inflection

One; $x = 1$

On $(-2, 1)$, $g''(x) = f'(x) > 0$

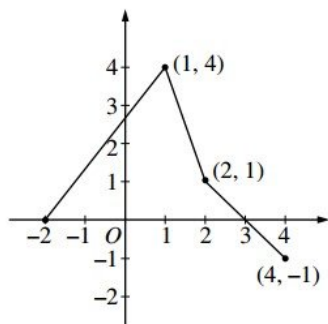
On $(1, 2)$, $g''(x) = f'(x) < 0$

On $(2, 4)$, $g''(x) = f'(x) < 0$

5-5b

Therefore $(1, g(1))$ is a point of inflection and $(2, g(2))$ is not.

27.



The graph of function f , consisting of three line segments, is given above. Let $g(x) = \int_1^x f(t) dt$.

Find the absolute minimum value of g on the closed interval $[-2, 4]$. Justify your answer.

Part C

1 point is earned for interior analysis

1 point is earned for endpoint analysis

1 point is earned for the answer

g is increasing on $[-2, 3]$ and decreasing on $[3, 4]$.

Therefore, g has absolute minimum at an endpoint of $[-2, 4]$.

Since $g(-2) = -6$ and $g(4) = \frac{5}{2}$, the absolute minimum value is -6 .



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for interior analysis

1 point is earned for endpoint analysis

1 point is earned for the answer

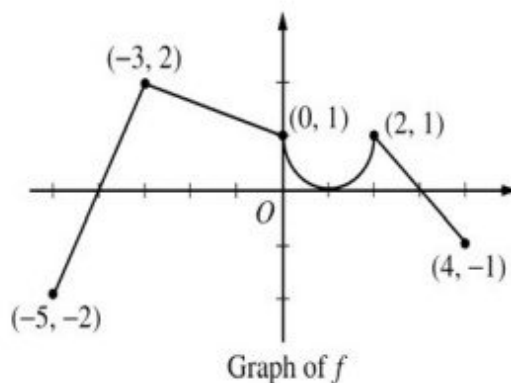
g is increasing on $[-2, 3]$ and decreasing on $[3, 4]$.

Therefore, g has absolute minimum at an endpoint of $[-2, 4]$.

Since $g(-2) = -6$ and $g(4) = \frac{5}{2}$, the absolute minimum value is -6 .

5-5b

28.



The graph of the function f shown above consists of a semicircle and three line segments. Let g be the function given by $g(x) = \int_{-3}^x f(t) dt$.

- Find $g(0)$ and $g'(0)$.
- Find all values of x in the open interval $(-5, 4)$ at which g attains a relative maximum. Justify your answer.
- Find the absolute minimum value of g on the closed interval $[-5, 4]$. Justify your answer.
- Find all values of x in the open interval $(-5, 4)$ at which the graph of g has a point of inflection.

Part A

1 point is earned for $g(0)$

1 point is earned for $g'(0)$

$$g(0) = \int_{-3}^0 f(t) dt = \frac{1}{2}(3)(2 + 1) = \frac{9}{2}$$

$$g'(0) = f(0) = 1$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g(0)$

1 point is earned for $g'(0)$

$$g(0) = \int_{-3}^0 f(t) dt = \frac{1}{2}(3)(2 + 1) = \frac{9}{2}$$

$$g'(0) = f(0) = 1$$

5-5b

Part B

1 point is earned for $x = 3$

1 point is earned for justification

g has a relative maximum at $x = 3$.

This is the only x -value where $g' = f$ changes from positive to negative.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $x = 3$

1 point is earned for justification

g has a relative maximum at $x = 3$.

This is the only x -value where $g' = f$ changes from positive to negative.

Part C

1 point is earned for identifying $x = -4$ as a candidate

1 point is earned for $g(-4) = -1$

1 point is earned for justification and answer

The only x -value where f changes from negative to positive is $x = -4$. The other candidates for the location of the absolute minimum value are the endpoints.

$$g(-5) = 0$$

$$g(-4) = \int_{-3}^{-4} f(t) dt = -1$$

$$g(4) = \frac{9}{2} + \left(2 - \frac{\pi}{2}\right) = \frac{13 - \pi}{2}$$

So the absolute minimum value of g is -1 .

5-5b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for identifying $x = -4$ as a candidate

1 point is earned for $g(-4) = -1$

1 point is earned for justification and answer

The only x -value where f changes from negative to positive is $x = -4$. The other candidates for the location of the absolute minimum value are the endpoints.

$$g(-5) = 0$$

$$g(-4) = \int_{-3}^{-4} f(t) dt = -1$$

$$g(4) = \frac{9}{2} + \left(2 - \frac{\pi}{2}\right) = \frac{13 - \pi}{2}$$

So the absolute minimum value of g is -1 .

Part D

Two points are earned for the correct values

<-1> each missing or extra value

$$x = -3, 1, 2$$



0	1	2
---	---	---

The student response earns all of the following points:

Two points are earned for the correct values

<-1> each missing or extra value

$$x = -3, 1, 2$$

5-5b

29. A particle moves along the x -axis so that its velocity at time t , $0 \leq t \leq 5$, is given by $v(t) = 3(t - 1)(t - 3)$. At time $t = 2$, the position of the particle is $x(2) = 0$.

- (a) Find the minimum acceleration of the particle.
 (b) Find the total distance traveled by the particle.
 (c) Find the average velocity of the particle over the interval $0 \leq t \leq 5$.

Part A

1 point is earned for finding $a(t)$

1 point is earned for recognizing min at $t = 0$

1 point is earned for evaluation

$$v(t) = 3t^2 - 12t + 9$$

$$a(t) = 6t - 12$$

a is increasing, so a is minimum at $t = 0$

$a(0) = -12$ is minimum value of a .



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for finding $a(t)$

1 point is earned for recognizing min at $t = 0$

1 point is earned for evaluation

$$v(t) = 3t^2 - 12t + 9$$

$$a(t) = 6t - 12$$

a is increasing, so a is minimum at $t = 0$

$a(0) = -12$ is minimum value of a .

Part B

2 points are earned for integral expression

5-5b

1 point is earned for anti differentiation

1 point is earned for evaluation

Method 1



$$\begin{aligned}
 d &= \int_0^1 (3t^2 - 12t + 9)dt - \int_1^3 (3t^2 - 12t + 9)dt + \int_3^5 (3t^2 - 12t + 9)dt \\
 &= [t^3 - 6t^2 + 9t]_0^1 - [t^3 - 6t^2 + 9t]_1^3 + [t^3 - 6t^2 + 9t]_3^5 \\
 &= 4 - (-4) + 20 = 28
 \end{aligned}$$

or

Method 2

$$x(t) = t^3 - 6t^2 + 9t - 2$$

$$[(\text{or}) x(t) = t^3 - 6t^2 + 9t + C]$$

$$x(0) = -2$$

$$x(1) = 2$$

$$x(3) = -2$$

$$x(5) = 18$$

$$\text{Total distance} = 4 + 4 + 20 = 28$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for integral expression

1 point is earned for anti differentiation

1 point is earned for evaluation

5-5b

Method 1



$$\begin{aligned}
 d &= \int_0^1 (3t^2 - 12t + 9)dt - \int_1^3 (3t^2 - 12t + 9)dt + \int_3^5 (3t^2 - 12t + 9)dt \\
 &= [t^3 - 6t^2 + 9t]_0^1 - [t^3 - 6t^2 + 9t]_1^3 + [t^3 - 6t^2 + 9t]_3^5 \\
 &= 4 - (-4) + 20 = 28
 \end{aligned}$$

or

Method 2

$$x(t) = t^3 - 6t^2 + 9t - 2$$

$$[(\text{or}) x(t) = t^3 - 6t^2 + 9t + C]$$

$$x(0) = -2$$

$$x(1) = 2$$

$$x(3) = -2$$

$$x(5) = 18$$

$$\text{Total distance} = 4 + 4 + 20 = 28$$

Part C

1 point is earned for applying definition of average velocity

1 point is earned for evaluation

Method 1

$$\begin{aligned}
 &\frac{\int_0^5 (3t^2 - 12t + 9)dt}{5 - 0} \\
 &= \frac{1}{5} [t^3 - 6t^2 + 9t]_0^5 = \frac{1}{5}(20) = 4
 \end{aligned}$$

or

5-5b

Method 2

$$\frac{x(5)-x(0)}{5-0} = \frac{18-(-2)}{5} = 4$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for applying definition of average velocity

1 point is earned for evaluation

Method 1

$$\frac{\int_0^5 (3t^2 - 12t + 9) dt}{5-0}$$

$$= \frac{1}{5} [t^3 - 6t^2 + 9t]_0^5 = \frac{1}{5} (20) = 4$$

or

Method 2

$$\frac{x(5)-x(0)}{5-0} = \frac{18-(-2)}{5} = 4$$

5-5b

30. A particle moves along the x -axis so that its velocity at time t , $0 \leq t \leq 5$, is given by $v(t) = 3(t-1)(t-3)$. At time $t=2$, the position of the particle is $x(2)=0$.

Find the minimum acceleration of the particle

Part A

1 point is earned for finding $a(t)$

1 point is earned for recognizing min at $t=0$

1 point is earned for evaluation

$$v(t) = 3t^2 - 12t + 9$$

$$a(t) = 6t - 12$$

a is increasing, so a is minimum at $t = 0$

$a(0) = -12$ is minimum value of a .



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for finding $a(t)$

1 point is earned for recognizing min at $t=0$

1 point is earned for evaluation

$$v(t) = 3t^2 - 12t + 9$$

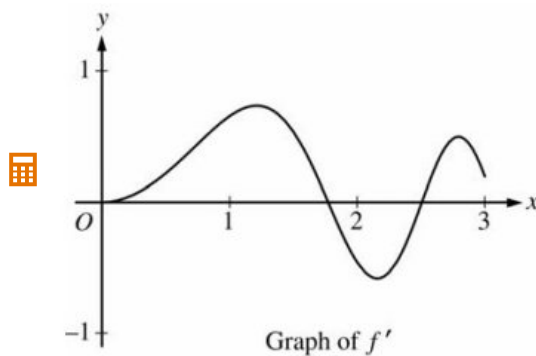
$$a(t) = 6t - 12$$

a is increasing, so a is minimum at $t = 0$

$a(0) = -12$ is minimum value of a .

5-5b

31.



Let f be the function defined for $x \geq 0$ with $f(0) = 5$ and f' , the first derivative of f , given by $f'(x) = e^{(-x/4)} \sin(x^2)$. The graph of $y = f'(x)$ is shown above.

(a) Use the graph of f' to determine whether the graph of f is concave up, concave down, or neither on the interval $1.7 < x < 1.9$. Explain your reasoning.

(b) On the interval $0 \leq x \leq 3$, find the value of x at which f has an absolute maximum. Justify your answer.

(c) Write an equation for the line tangent to the graph of f at $x = 2$.

Part A

1 point is earned for the answer

1 point is earned for the reason

On the interval $1.7 < x < 1.9$, f' is decreasing and thus f is concave down on this interval.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the reason

On the interval $1.7 < x < 1.9$, f' is decreasing and thus f is concave down on this interval.

Part B

1 point is earned if identified $\sqrt{\pi}$ and 3 as candidates **or** indicates that the graph of f increases, decreases, then increases

1 point is earned if justified $f(\sqrt{\pi}) > f(3)$

5-5b

1 point is earned for answer

$$f'(x) = 0 \text{ when } x = 0, \sqrt{\pi}, \sqrt{2\pi}, \sqrt{3\pi}, \dots$$

On $[0, 3]$ f' changes from positive to negative only at $\sqrt{\pi}$. The absolute maximum must occur at $x = \sqrt{\pi}$ or at an endpoint.

$$f(0) = 5$$

$$f(\sqrt{\pi}) = f(0) + \int_0^{\sqrt{\pi}} f'(x) dx = 5.67911$$

$$f(3) = f(0) + \int_0^3 f'(x) dx = 5.57893$$

This shows that f has an absolute maximum at $x = \sqrt{\pi}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if identified $\sqrt{\pi}$ and 3 as candidates **or** indicates that the graph of f increases, decreases, then increases

1 point is earned if justified $f(\sqrt{\pi}) > f(3)$

1 point is earned for answer

$$f'(x) = 0 \text{ when } x = 0, \sqrt{\pi}, \sqrt{2\pi}, \sqrt{3\pi}, \dots$$

On $[0, 3]$ f' changes from positive to negative only at $\sqrt{\pi}$. The absolute maximum must occur at $x = \sqrt{\pi}$ or at an endpoint.

$$f(0) = 5$$

5-5b

$$f(\sqrt{\pi}) = f(0) + \int_0^{\sqrt{\pi}} f'(x) dx = 5.67911$$

$$f(3) = f(0) + \int_0^3 f'(x) dx = 5.57893$$

This shows that f has an absolute maximum at $x = \sqrt{\pi}$.

Part C

1 point is earned for $f(2)$ expression with integral

1 point is earned for $f(2)$ expression including $f(0)$ term

1 point is earned for $f'(2)$

1 point is earned for equation

$$f(2) = f(0) + \int_0^2 f'(x) dx = 5.62342$$

$$f'(2) = e^{-0.5} \sin(4) = -0.45902$$

$$y - 5.623 = (-0.459)(x - 2)$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $f(2)$ expression with integral

1 point is earned for $f(2)$ expression including $f(0)$ term

1 point is earned for $f'(2)$

1 point is earned for equation

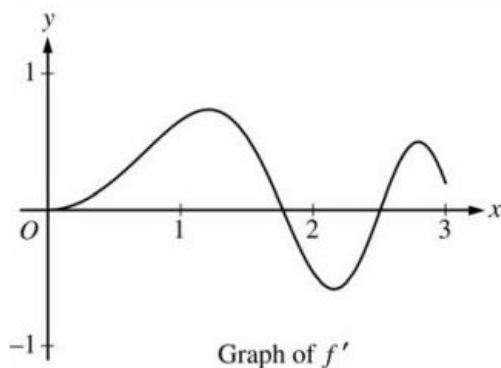
$$f(2) = f(0) + \int_0^2 f'(x) dx = 5.62342$$

$$f'(2) = e^{-0.5} \sin(4) = -0.45902$$

$$y - 5.623 = (-0.459)(x - 2)$$

5-5b

32.



Let f be the function defined for $x \geq 0$ with $f(0)=5$ and f' , the first derivative of f , given by $f'(x) = e^{x/4} \sin(x^2)$. The graph of $y = f'(x)$ is shown above.



On the interval $0 \leq x \leq 3$, find the value of x at which f has an absolute maximum. Justify your answer.

FRQ Score

1 point is earned if identified $\sqrt{\pi}$ and 3 as candidates **or** indicates that the graph of f increases, decreases, then increases

1 point is earned if justified $f(\sqrt{\pi}) > f(3)$

1 point is earned for answer

$$f'(x)=0 \text{ when } x=0, \sqrt{\pi}, \sqrt{2\pi}, \sqrt{3\pi}, \dots$$

On $[0, 3]$ f' changes from positive to negative only at $\sqrt{\pi}$. The absolute maximum must occur at $x = \sqrt{\pi}$ or at an endpoint.

$$f(0)=5$$

$$f(\sqrt{\pi}) = f(0) + \int_0^{\sqrt{\pi}} f'(x) dx = 5.67911$$

$$f(3) = f(0) + \int_0^3 f'(x) dx = 5.57893$$

This shows that f has an absolute maximum at $x = \sqrt{\pi}$

5-5b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if identified $\sqrt{\pi}$ and 3 as candidates **or** indicates that the graph of f increases, decreases, then increases

1 point is earned if justified $f(\sqrt{\pi}) > f(3)$

1 point is earned for answer

$f'(x)=0$ when $x=0$,

On $[0, 3]$ f' changes from positive to negative only at $\sqrt{\pi}$. The absolute maximum must occur at $x = \sqrt{\pi}$ or at an endpoint.

$f(0)=5$

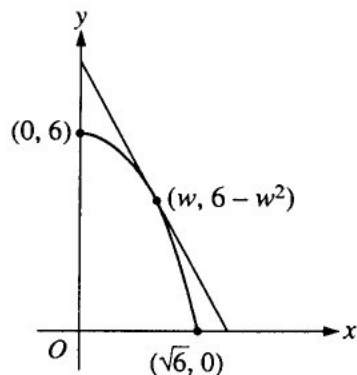
$$f(\sqrt{\pi}) = f(0) + \int_0^{\sqrt{\pi}} f'(x)dx = 5.67911$$

$$f(3) = f(0) + \int_0^3 f'(x)dx = 5.57893$$

This shows that f has an absolute maximum at $x = \sqrt{\pi}$

5-5b

33.



Let $f(x) = 6 - x^2$. For $0 < w < \sqrt{6}$, let $A(w)$ be the area of the triangle formed by the coordinate axes and the line tangent to the graph of f at the point $(w, 6 - w^2)$.

(a) Find $A(1)$.

(b) For what value of w is $A(w)$ a minimum?

Part A

1 point is earned for the correct answer $f'(1) = -2$

$$f(x) = 6 - x^2; f'(x) = -2x$$

$$f'(1) = -2$$

1 point is earned for correctly finding equation of line or x intercept or y intercept

$$y - 5 = -2(x - 1) \text{ or } y = -2x + 7$$

x -intercept: $7/2$ y -intercept: 7

1 point is earned for the correct answer

$$A(1) = \frac{1}{2} \left(\frac{7}{2} \right) (7) = \frac{49}{4}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer $f'(1) = -2$

$$f(x) = 6 - x^2; f'(x) = -2x$$

$$f'(1) = -2$$

5-5b

1 point is earned for correctly finding equation of line or x intercept or y intercept

$$y - 5 = -2(x - 1) \text{ or } y = -2x + 7$$

x -intercept: $7/2$ y -intercept: 7

1 point is earned for the correct answer

$$A(1) = \frac{1}{2} \left(\frac{7}{2} \right) (7) = \frac{49}{4}$$

Part B

1 point is earned for the correct equation of line

$$f'(w) = -2w; \quad y - (6 - w^2) = -2w(x - w)$$

1 point is earned for correctly expressing x intercept or y intercept in terms of w

$$x\text{-intercept} : \frac{6+w^2}{2w}; \quad y\text{-intercept} : 6 + w^2$$

1 point is earned for the correct answer for $A(w)$

$$A(w) = \frac{(6+w^2)^2}{4w}$$

1 point is earned for the correct answer for $A'(w)$

$$A'(w) = \frac{4w(2(6+w^2)(2w)) - 4(6+w^2)^2}{16w^2}$$

1 point is earned for correctly solving $A'(w) = 0$

$$A'(w) = 0 \text{ When } (6 + w^2)(3w^2 - 6) = 0$$

$$w = \sqrt{2}$$

1 point is earned for correctly showing solution yields a minimum



Note: $A(w)$ must be of the form $\frac{P(w)}{Q(w)}$, neither constant, to be eligible for derivative point

5-5b

OR

If student is using Theorem that $A(w)$ is minimized at w_1 , where $f(w_1)$ is the average value of the function:

1 point is earned for the correct average value set-up

1 point is earned for the correct antidifferentiation and evaluation

1 point is earned for correctly setting $6 - w^2 = \text{average value}$

1 point is earned for correctly solving for w_1 in domain $(0, \sqrt{6})$

2 points are earned for the clear statement of Theorem

<-1> error in statement

<-2> No statement



0	1	2	3	4	5	6
---	---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct equation of line

$$f'(w) = -2w; \quad y - (6 - w^2) = -2w(x - w)$$

1 point is earned for correctly expressing x intercept or y intercept in terms of w

$$x\text{-intercept} : \frac{6+w^2}{2w}; \quad y\text{-intercept} : 6 + w^2$$

1 point is earned for the correct answer for $A(w)$

$$A(w) = \frac{(6+w^2)^2}{4w}$$

1 point is earned for the correct answer for $A'(w)$

$$A'(w) = \frac{4w(2(6+w^2)(2w)) - 4(6+w^2)^2}{16w^2}$$

1 point is earned for correctly solving $A'(w) = 0$

5-5b

$$A'(w) = 0 \text{ When } (6 + w^2)(3w^2 - 6) = 0$$

$$w = \sqrt{2}$$

1 point is earned for correctly showing solution yields a minimum



Note: $A(w)$ must be of the form $\frac{P(w)}{Q(w)}$, neither constant, to be eligible for derivative point

OR

If student is using Theorem that $A(w)$ is minimized at w_1 , where $f(w_1)$ is the average value of the function:

1 point is earned for the correct average value set-up

1 point is earned for the correct antidifferentiation and evaluation

1 point is earned for correctly setting $6 - w^2 = \text{average value}$

1 point is earned for correctly solving for w_1 in domain $(0, \sqrt{6})$

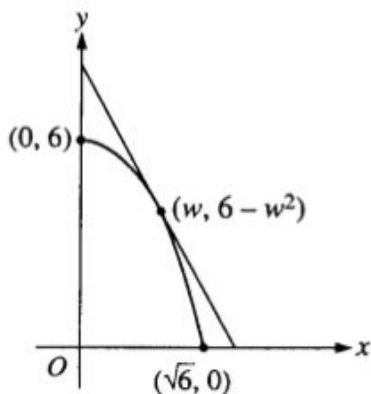
2 points are earned for the clear statement of Theorem

<-1> error in statement

<-2> No statement

5-5b

34.



Let $f(x) = 6 - x^2$. For $0 < w < \sqrt{6}$, let $A(w)$ be the area of the triangle formed by the coordinate axes and the line tangent to the graph of f at the point $(w, 6 - w^2)$.

For what value of w is $A(w)$ a minimum?

Part B

1 point is earned for the correct equation of line

$$f'(w) = -2w; y - (6 - w^2) = -2w(x - w)$$

1 point is earned for correctly expressing x intercept or y intercept in terms of w

$$x - \text{intercept} : \frac{6 + w^2}{2w}; y - \text{intercept} : 6 + w^2$$

1 point is earned for the correct answer for $A(w)$

$$A(w) = \frac{(6 + w^2)^2}{4w}$$

1 point is earned for the correct answer for $A'(w)$

$$A'(w) = \frac{4w(2(6 + w^2)(2w)) - 4(6 + w^2)^2}{16w^2}$$

1 point is earned for correctly solving $A'(w) = 0$

$$A'(w) = 0 \quad (6 + w^2) \quad \text{when} \quad (3w^2 - 6) = 0$$

$$w = \sqrt{2}$$

1 point is earned for correctly showing solution yields a minimum



5-5b

Note: $A(w)$ must be of the form $\frac{P(w)}{Q(w)}$, neither constant, to be eligible for derivative point

OR

If student is using Theorem that $A(w)$ is minimized at w_1 , where $f(w_1)$ is the average value of the function:

1 point is earned for the correct average value set-up

1 point is earned for the correct antidifferentiation and evaluation

1 point is earned for correctly setting $6 - w^2 = \text{average value}$

1 point is earned for correctly solving for w_1 in domain $(0, \sqrt{6})$

2 points are earned for the clear statement of Theorem

< -1 > error in statement

< -2 > no statement



0	1	2	3	4	5	6
---	---	---	---	---	---	---

The student response earns six of the following points:

1 point is earned for the correct equation of line

$$f'(w) = -2w; y - (6 - w^2) = -2w(x - w)$$

1 point is earned for correctly expressing x intercept or y intercept in terms of w

$$x - \text{intercept} : \frac{6+w^2}{2w}; y - \text{intercept} : 6 + w^2$$

1 point is earned for the correct answer for $A(w)$

$$A(w) = \frac{(6+w^2)^2}{4w}$$

1 point is earned for the correct answer for $A'(w)$

$$A'(w) = \frac{4w(2(6+w^2)(2w)) - 4(6+w^2)^2}{16w^2}$$

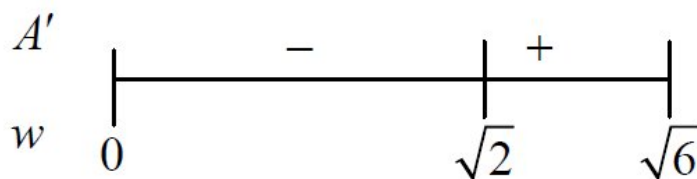
1 point is earned for correctly solving $A'(w) = 0$

$$A'(w) = 0 \quad (6 + w^2) \text{ when } (3w^2 - 6) = 0$$

$$w = \sqrt{2}$$

5-5b

1 point is earned for correctly showing solution yields a minimum



Note: $A(w)$ must be of the form $\frac{P(w)}{Q(w)}$, neither constant, to be eligible for derivative point

OR

If student is using Theorem that $A(w)$ is minimized at w_1 , where $f(w_1)$ is the average value of the function:

1 point is earned for the correct average value set-up

1 point is earned for the correct antidifferentiation and evaluation

1 point is earned for correctly setting $6 - w^2 = \text{average value}$

1 point is earned for correctly solving for w_1 in domain $(0, \sqrt{6})$

2 points are earned for the clear statement of Theorem

< -1 > error in statement

< -2 > no statement

5-5b

35.  Let f and g be the functions given by $f(x) = e^x$ and $g(x) = \ln x$.
- (a) Find the area of the region enclosed by the graphs of f and g between $x = \frac{1}{2}$ and $x = 1$.
- (b) Find the volume of the solid generated when the region enclosed by the graphs of f and g between $x = \frac{1}{2}$ and $x = 1$ is revolved about the line $y = 4$.
- (c) Let h be the function given by $h(x) = f(x) - g(x)$. Find the absolute minimum value of $h(x)$ on the closed interval $\frac{1}{2} \leq x \leq 1$, and find the absolute maximum value of $h(x)$ on the closed interval $\frac{1}{2} \leq x \leq 1$. Show the analysis that leads to your answers.

Part A

1 point is earned for integral

1 point is earned for the answer

$$\text{Area} = \int_{1/2}^1 (e^x - \ln x) dx = 1.222 \text{ or } 1.223$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$\text{Area} = \int_{1/2}^1 (e^x - \ln x) dx = 1.222 \text{ or } 1.223$$

Part B

1 point is earned for limits and constant

2 points are earned for integrand

<-1> each error

1 point is earned for the answer

5-5b

$$\begin{aligned}\text{Volume} &= \pi \int_{1/2}^1 \left((4 - \ln x)^2 - (4 - e^x)^2 \right) dx \\ &= 7.515\pi \text{ or } 23.609\end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for limits and constant

2 points are earned for integrand

<-1> each error

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_{1/2}^1 \left((4 - \ln x)^2 - (4 - e^x)^2 \right) dx \\ &= 7.515\pi \text{ or } 23.609\end{aligned}$$

Part C

1 point is earned for considering $h'(x) = 0$

1 point is earned for identifying critical point and endpoints as candidates

1 point is earned for the answers

$$\begin{aligned}h'(x) &= f'(x) - g'(x) = e^x - \frac{1}{x} = 0 \\ x &= 0.567143\end{aligned}$$

Absolute minimum value and absolute maximum value occur at the critical point or at the endpoints.

$$h(0.567143) = 2.330$$

$$h(0.5) = 2.3418$$

$$h(1) = 2.718$$

The absolute minimum is 2.330.

The absolute maximum is 2.718.

5-5b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $h'(x) = 0$

1 point is earned for identifying critical point and endpoints as candidates

1 point is earned for the answers

$$h'(x) = f'(x) - g'(x) = e^x - \frac{1}{x} = 0$$
$$x = 0.567143$$

Absolute minimum value and absolute maximum value occur at the critical point or at the endpoints.

$$h(0.567143) = 2.330$$

$$h(0.5) = 2.3418$$

$$h(1) = 2.718$$

The absolute minimum is 2.330.

The absolute maximum is 2.718.

5-5b

36. Let f and g be the functions given by $f(x) = e^x$ and $g(x) = \ln x$.



Let h be the function given by $h(x) = f(x) - g(x)$. Find the absolute minimum value of $h(x)$ on the closed interval $\frac{1}{2} \leq x \leq 1$, and find the absolute maximum of $h(x)$ on the closed interval $\frac{1}{2} \leq x \leq 1$. Show the analysis that leads to your answers.

Part C

1 point is earned for considers $h'(x) = 0$

1 point is earned for identifies critical point and endpoints as candidates

1 point is earned for the answers

$$h'(x) = f'(x) - g'(x) = e^x - \frac{1}{x} = 0$$

$$x = 0.567143$$

Absolute minimum value and absolute maximum value occur at the critical point or at the endpoints.

$$h(0.567143) = 2.330$$

$$h(0.5) = 2.34418$$

$$h(1) = 2.718$$

The absolute minimum is 2.330.

The absolute maximum is 2.718.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considers $h'(x) = 0$

1 point is earned for identifies critical point and endpoints as candidates

1 point is earned for the answers

$$h'(x) = f'(x) - g'(x) = e^x - \frac{1}{x} = 0$$

$$x = 0.567143$$

Absolute minimum value and absolute maximum value occur at the critical point or at the endpoints.

5-5b

$$h(0.567143) = 2.330$$

$$h(0.5) = 2.34418$$

$$h(1) = 2.718$$

The absolute minimum is 2.330.

The absolute maximum is 2.718.

5-5b

37.  A water tank at Camp Newton holds 1200 gallons of water at time $t = 0$. During the time interval $0 \leq t \leq 18$ hours, water is pumped into the tank at the rate

$$W(t) = 95\sqrt{t} \sin^2\left(\frac{t}{6}\right) \text{ gallons per hour.}$$

During the same time interval, water is removed from the tank at the rate

$$R(t) = 275 \sin^2\left(\frac{t}{3}\right) \text{ gallons per hour.}$$

- (a) Is the amount of water in the tank increasing at time $t = 15$? Why or why not?
 (b) To the nearest whole number, how many gallons of water are in the tank at time $t = 18$?
 (c) At what time t , $0 \leq t \leq 18$, is the amount of water in the tank at an absolute minimum? Show the work that leads to your conclusion.
 (d) For $t > 18$, no water is pumped into the tank, but water continues to be removed at the rate $R(t)$ until the tank becomes empty. Let k be the time at which the tank becomes empty. Write, but do not solve, an equation involving an integral expression that can be used to find the value of k .

Part A

1 point is earned for the answer with reason

No; the amount of water is not increasing at $t = 15$ since $W(15) - R(15) = -121.09 < 0$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with reason

No; the amount of water is not increasing at $t = 15$ since $W(15) - R(15) = -121.09 < 0$.

Part B

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$1200 + \int_0^{18} (W(t) - R(t))dt = 1309.788$$

1310 gallons

5-5b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$1200 + \int_0^{18} (W(t) - R(t))dt = 1309.788$$

1310 gallons

Part C

1 point is earned for the interior critical points

1 point is earned for the amount of water is least at $t = 6.494$ or 6.495

1 point is earned for the analysis for absolute minimum

$$W(t) - R(t) = 0$$

$$t = 0, 6.4948, 12.9748$$

t (hours)	gallons of water
0	1200
6.495	525
12.975	1697
18	1310

The values at the endpoints and the critical points show that the absolute minimum occurs when $t = 6.494$ or 6.495 .



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the interior critical points

1 point is earned for the amount of water is least at $t = 6.494$ or 6.495

5-5b

1 point is earned for the analysis for absolute minimum

$$W(t) - R(t) = 0$$

$$t = 0, 6.4948, 12.9748$$

t (hours)	gallons of water
0	1200
6.495	525
12.975	1697
18	1310

The values at the endpoints and the critical points show that the absolute minimum occurs when $t = 6.494$ or 6.495 .

Part D

1 point is earned for the limits

1 point is earned for the equation

$$\int_{18}^k R(t)dt = 1310$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the equation

$$\int_{18}^k R(t)dt = 1310$$

5-5b

38. A water tank at Camp Newton holds 1200 gallons of water at time $t = 0$. During the time interval $0 \leq t \leq 18$ hours, water is pumped into the tank at the rate

$$W(t) = 95\sqrt{t}\sin^2\left(\frac{t}{6}\right) \text{ gallons per hour.}$$

During the same time interval, water is removed from the tank at the rate

$$R(t) = 275\sin^2\left(\frac{t}{3}\right) \text{ gallons per hour.}$$



At what time t , $0 \leq t \leq 18$, is the amount of water in the tank at an absolute minimum? Show the work that leads to your conclusion.

Part C

1 point is earned for the interior critical points

1 point is earned for the amount of water is least at $t=6.494$ or 6.495

1 point is earned for the analysis for absolute minimum

$$W(t)-R(t)=0 \quad t=0,$$

6.4948, 12.9748

t (hours)	gallons of water
0	1200
6.495	525
12.975	1697
18	1310

The values at the endpoints and the critical points show that the absolute minimum occurs when $t=6.494$ or 6.495 .



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the interior critical points

1 point is earned for the amount of water is least at $t=6.494$ or 6.495

1 point is earned for the analysis for absolute minimum

$$W(t)-R(t)=0 \quad t=0,$$

6.4948, 12.9748

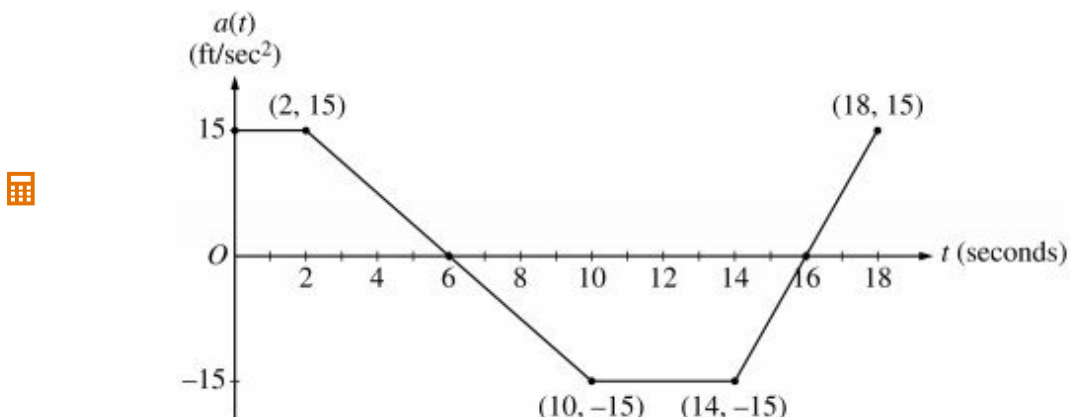
5-5b

t (hours)	gallons of water
0	1200
6.495	525
12.975	1697
18	1310

The values at the endpoints and the critical points show that the absolute minimum occurs when $t=6.494$ or 6.495 .

5-5b

39.



A car is traveling on a straight road with velocity 55 ft/sec at time $t = 0$. For $0 \leq t \leq 18$ seconds, the car's acceleration $a(t)$, in ft/sec^2 , is the piecewise linear function defined by the graph above.

- Is the velocity of the car increasing at $t = 2$ seconds? Why or why not?
- At what time in the interval $0 \leq t \leq 18$, other than $t = 0$, is the velocity of the car 55 ft/sec? Why?
- On the time interval $0 \leq t \leq 18$, what is the car's absolute maximum velocity, in ft/sec, and at what time does it occur? Justify your answer.
- At what times in the interval $0 \leq t \leq 18$, if any, is the car's velocity equal to zero? Justify your answer.

Part A

1 point is earned for the answer and reason

Since $v'(2) = a(2)$ and $a(2) = 15 > 0$, the velocity is increasing at $t = 2$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer and reason

Since $v'(2) = a(2)$ and $a(2) = 15 > 0$, the velocity is increasing at $t = 2$.

Part B

1 point is earned for: $t = 12$

1 point is earned for the reason

At time $t = 12$ because $v(12) - v(0) = \int_0^{12} a(t) dt = 0$.

5-5b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $t = 12$

1 point is earned for the reason

At time $t = 12$ because $v(12) - v(0) = \int_0^{12} a(t)dt = 0$.

Part C

1 point is earned for: $t = 6$

1 point is earned for the absolute maximum velocity

1 point is earned for identifying $t = 6$ and $t = 18$ as candidates or indicating that v increases, decreases, then increases.

1 point is earned for eliminating $t = 18$

The absolute maximum velocity is 115 ft/sec at $t = 6$.

The absolute maximum must occur at $t = 6$ or at an endpoint.

$$\begin{aligned}
 v(6) &= 55 + \int_0^6 a(t)dt \\
 &= 55 + 2(15) + \frac{1}{2}(4)(15) = 115 > v(0) \\
 \int_6^{18} a(t)dt &< 0 \text{ so } v(18) < v(6)
 \end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for: $t = 6$

1 point is earned for the absolute maximum velocity

5-5b

1 point is earned for identifying $t = 6$ and $t = 18$ as candidates or indicating that v increases, decreases, then increases.

1 point is earned for eliminating $t = 18$

The absolute maximum velocity is 115 ft/sec at $t = 6$.

The absolute maximum must occur at $t = 6$ or at an endpoint.

$$\begin{aligned} v(6) &= 55 + \int_0^6 a(t) dt \\ &= 55 + 2(15) + \frac{1}{2}(4)(15) = 115 > v(0) \\ \int_6^{18} a(t) dt &< 0 \text{ so } v(18) < v(6) \end{aligned}$$

Part D

1 point is earned for the answer

1 point is earned for the reason

The car's velocity is never equal to 0. The absolute minimum occurs at $t = 16$ where

$$v(6) = 115 + \int_6^{16} a(t) dt = 115 - 105 = 10 > 0.$$



0	1	2
---	---	---

The student response earns all of the following points:

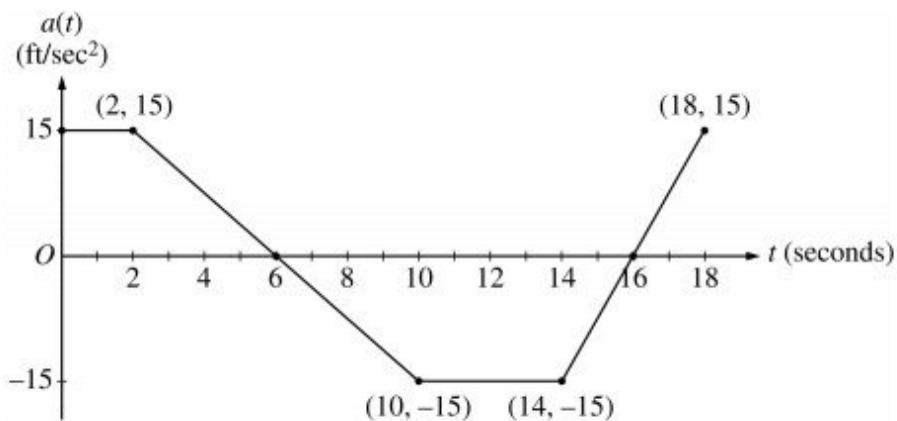
1 point is earned for the answer

1 point is earned for the reason

The car's velocity is never equal to 0. The absolute minimum occurs at $t = 16$ where

$$v(6) = 115 + \int_6^{16} a(t) dt = 115 - 105 = 10 > 0.$$

5-5b



A car is traveling on a straight road with velocity 55 ft/sec at time $t = 0$. For $0 \leq t \leq 18$ seconds, the car's acceleration $a(t)$, in ft/sec², is the piecewise linear function defined by the graph above

40. On the time interval $0 \leq t \leq 18$, what is the car's absolute maximum velocity, in ft/sec, and at what time does it occur? Justify your answer.

Part C

1 point is earned for: $t = 6$

1 point is earned for the absolute maximum velocity

1 point is earned for identifies $t = 6$ and $t = 18$ as candidates or indicates that v increases, decreases, then increases.

1 point is earned for eliminates $t = 18$

The absolute maximum velocity is 115 ft/sec at $t = 6$.

The absolute maximum must occur at $t = 6$ or at an endpoint.

$$\begin{aligned} v(6) &= 55 + \int_0^6 a(t) dt \\ &= 55 + 2(15) + \frac{1}{2}(4)(15) = 115 > v(0) \\ \int_6^{18} a(t) dt &< 0, \text{ so, } v(18) < v(16) \end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for: $t = 6$

5-5b

1 point is earned for the absolute maximum velocity

1 point is earned for identifies $t = 6$ and $t = 18$ as candidates or indicates that v increases, decreases, then increases.

1 point is earned for eliminates $t = 18$

The absolute maximum velocity is 115 ft/sec at $t = 6$.

The absolute maximum must occur at $t = 6$ or at an endpoint.

$$\begin{aligned} v(6) &= 55 + \int_0^6 a(t) dt \\ &= 55 + 2(15) + \frac{1}{2}(4)(15) = 115 > v(0) \\ \int_6^{18} a(t) dt &< 0, \text{ so, } v(18) < v(16) \end{aligned}$$

41. At what time in the interval $0 \leq t \leq 18$, if any, is the car's velocity equal to zero? Justify your answer.

Part D

1 point is earned for the answer

1 point is earned for the reason

The car's velocity is never equal to 0. The absolute minimum occurs at $t = 16$ where

$$v(16) = 115 + \int_6^{16} a(t) dt = 115 - 105 = 10 > 0.$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the reason

The car's velocity is never equal to 0. The absolute minimum occurs at $t = 16$ where

$$v(16) = 115 + \int_6^{16} a(t) dt = 115 - 105 = 10 > 0.$$

5-5b

42. A particle moves along the x -axis with position at time t given by $x(t) = e^{-t} \sin t$ for $0 \leq t \leq 2\pi$. Find the time t at which the particle is farthest to the left. Justify your answer.

Part A

2 points are earned for: $x'(t)$

1 point is earned for sets $x'(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

$$x'(t) = -e^{-t} \sin t + e^{-t} \cos t = -e^{-t} (\cos t - \sin t)$$

$x'(t) = 0$, when, $\cos t = \sin t$. Therefore, $x'(t) = 0$, on

$$0 \leq t \leq 2\pi, \text{ for, } t = \frac{\pi}{4} \text{ and, } t = \frac{5\pi}{4}$$

The candidates for the absolute minimum are at

$$t = 0, \frac{\pi}{4}, \frac{5\pi}{4} \text{ and, } 2\pi.$$

t	$x(t)$
0	$e^0 \sin(0) = 0$
$\frac{\pi}{4}$	$e^{-\frac{\pi}{4}} \sin\left(\frac{\pi}{4}\right) > 0$
$\frac{5\pi}{4}$	$e^{-\frac{5\pi}{4}} \sin\left(\frac{5\pi}{4}\right) < 0$
2π	$e^{-2\pi} \sin(2\pi) = 0$

The particle is farthest to the left when $t = \frac{5\pi}{4}$.



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

2 points are earned for: $x'(t)$

1 point is earned for sets $x'(t) = 0$

5-5b

1 point is earned for the answer

1 point is earned for the justification

$$x'(t) = -e^{-t} \sin t + e^{-t} \cos t = -e^{-t} (\cos t - \sin t)$$

$$x'(t) = 0, \text{ when, } \cos t = \sin t. \text{ Therefore, } x'(t) = 0, \text{ on}$$

$$0 \leq t \leq 2\pi, \text{ for, } t = \frac{\pi}{4} \text{ and, } t = \frac{5\pi}{4}$$

The candidates for the absolute minimum are at

$$t = 0, \frac{\pi}{4}, \frac{5\pi}{4} \text{ and, } 2\pi.$$

t	$x(t)$
0	$e^0 \sin(0) = 0$
$\frac{\pi}{4}$	$e^{-\frac{\pi}{4}} \sin\left(\frac{\pi}{4}\right) > 0$
$\frac{5\pi}{4}$	$e^{-\frac{5\pi}{4}} \sin\left(\frac{5\pi}{4}\right) < 0$
2π	$e^{-2\pi} \sin(2\pi) = 0$

The particle is farthest to the left when $t = \frac{5\pi}{4}$.

5-5b

43.

t (hours)	0	2	5	7	8
$E(t)$ (hundreds of entries)	0	4	13	21	23

A zoo sponsored a one-day contest to name a new baby elephant. Zoo visitors deposited entries in a special box between noon ($t=0$) and 8 P.M. ($t=8$). The number of entries in the box t hours after noon is modeled by a differentiable function E for $0 \leq t \leq 8$. Values of $E(t)$, in hundreds of entries, at various times t are shown in the table above.

 According to the model from part (c), at what time were the entries being processed most quickly? Justify your answer.

Part D

1 point is earned for considering $P'(t)=0$

1 point is earned for identifying candidates

1 point is earned for answering with justification

$P'(t)=0$ when $t=9.183503$ and $t=10.816497$

t	$P(t)$
8	0
9.183503	5.088662
10.816497	2.911338
12	8

Entries are being processed most quickly at time $t=12$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $P'(t)=0$

1 point is earned for identifying candidates

1 point is earned for answering with justification

$P'(t)=0$ when $t=9.183503$ and $t=10.816497$

5-5b

t	$P(t)$
8	0
9.183503	5.088662
10.816497	2.911338
12	8

Entries are being processed most quickly at time $t=12$

5-5b

44. The rate at which people enter an auditorium for a rock concert is modeled by the function R given by $R(t) = 1380t^2 - 675t^3$ for $0 \leq t \leq 2$ hours; $R(t)$ is measured in people per hour. No one is in the auditorium at time $t = 0$, when the doors open. The doors close and the concert begins at time $t = 2$.



Find the time when the rate at which people enter the auditorium is a maximum. Justify your answer.

Part B

1 point is earned if considers $R'(t)=0$

1 point is earned for interior critical point

1 point is earned for the answer plus justification

$R'(t) = 0$ when $t = 0$ and $t = 1.36296$ The maximum rate may occur at 0, $a = 1.36296$, or 2.

$R(0)=0$

$R(a) = 854.527$

$R(2)=120$

The maximum rate occurs when $t=1.362$ or $t=1.363$.



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned if considers $R'(t)=0$

1 point is earned for interior critical point

1 point is earned for the answer plus justification

$R'(t) = 0$ when $t = 0$ and $t = 1.36296$ The maximum rate may occur at 0, $a = 1.36296$, or 2.

$R(0)=0$

$R(a) = 854.527$

$R(2)=120$

The maximum rate occurs when $t=1.362$ or $t=1.363$.

5-5b

45. A particle moves along the x -axis so that its velocity at time t is given by

$$v(t) = -(t + 1) \sin\left(\frac{t^2}{2}\right).$$

At time $t = 0$, the particle is at position $x = 1$.



During the time interval $0 \leq t \leq 3$, what is the greatest distance between the particle and the origin? Show the work that leads to your answer.

Part D

1 point is earned for the $\pm$ distance that the particle travels while velocity is negative

$$\int_0^{\sqrt{2\pi}} v(t) dt = -3.265$$

$$x(\sqrt{2\pi}) = x(0) + \int_0^{\sqrt{2\pi}} v(t) dt = -2.265$$

1 point is earned for the correct answer

Since the total distance from $t = 0$ to $t = 3$ is 4.334, the particle is still to the left of the origin at $t = 3$. Hence the greatest distance from the origin is 2.265.



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the $\pm$ distance that the particle travels while velocity is negative

$$\int_0^{\sqrt{2\pi}} v(t) dt = -3.265$$

$$x(\sqrt{2\pi}) = x(0) + \int_0^{\sqrt{2\pi}} v(t) dt = -2.265$$

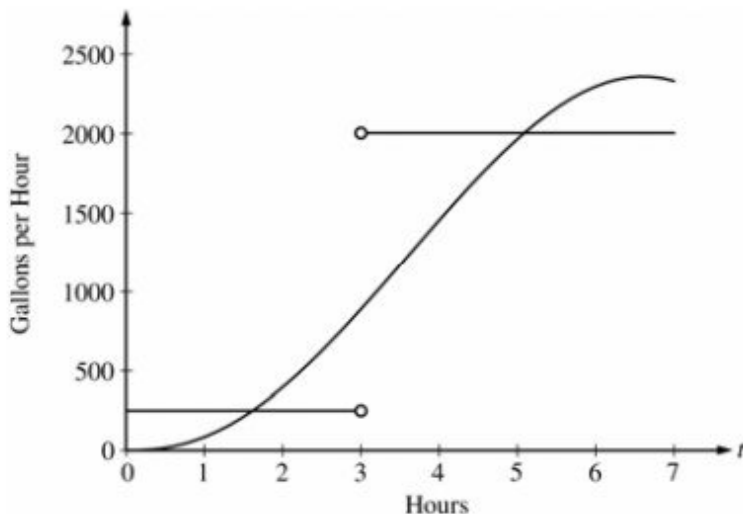
1 point is earned for the correct answer

Since the total distance from $t = 0$ to $t = 3$ is 4.334, the particle is still to the left of the origin at $t = 3$. Hence the greatest distance from the origin is 2.265.

5-5b

46. The amount of water in a storage tank, in gallons, is modeled by a continuous function on the time interval $0 \leq t \leq 7$, where t is measured in hours. In this model, rates are given as follows:

- i. The rate at which water enters the tank is $f(t) = 100t^2 \sin(\sqrt{t})$ gallons per hour for $0 \leq t \leq 7$.
- ii. The rate at which water enters the tank is $g(t) = \begin{cases} 250 & \text{for } 0 \leq t < 3 \\ 2000 & \text{for } 3 < t \leq 7 \end{cases}$ gallons per hour.



The graphs of f and g , which intersect at $t = 1.617$ and $t = 5.076$, are shown in the figure above. At time $t = 0$, the amount of water in the tank is 5000 gallons.

 For $0 \leq t \leq 7$, at what time t is the amount of water in the tank greatest? To the nearest gallon, compute the amount of water at this time. Justify your answer.

Part C

1 point is earned for identifies $t = 3$ as a candidate

1 point is earned for the integrand

1 point is earned for the amount of water at $t = 3$

1 point is earned for the amount of water at $t = 7$

1 point is earned for the conclusion

Since $f(t) - g(t)$ changes sign from positive to negative only at $t = 3$, the candidates for the absolute maximum are $t = 0$, 3 , and 7 .

5-5b

t (hours)	gallons of water
0	5000
3	$5000 + \int_0^3 f(t) dt - 250(3) = 5126.591$
7	$5126.591 + \int_3^7 f(t) dt - 2000(4) = 4513.807$

The amount of water in the tank is greatest at 3 hours. At that time, the amount of water in the tank, rounded to the nearest gallon, is 5127 gallons.



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for identifies $t = 3$ as a candidate

1 point is earned for the integrand

1 point is earned for the amount of water at $t = 3$

1 point is earned for the amount of water at $t = 7$

1 point is earned for the conclusion

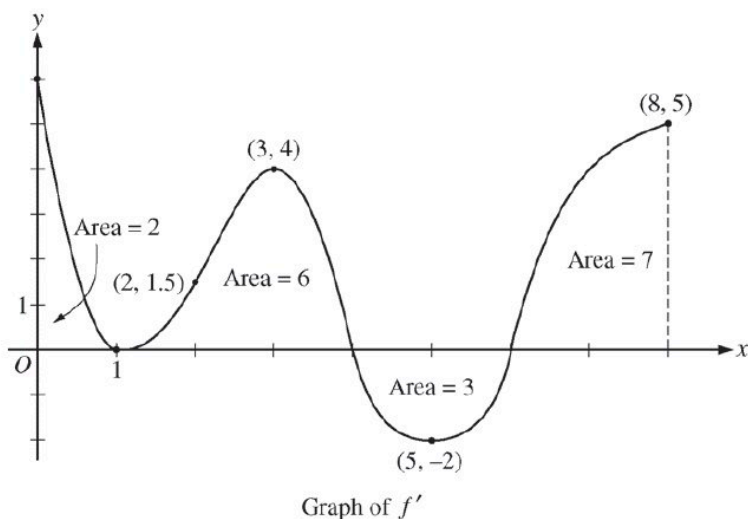
Since $f(t) - g(t)$ changes sign from positive to negative only at $t = 3$, the candidates for the absolute maximum are $t = 0$, 3 , and 7 .

t (hours)	gallons of water
0	5000
3	$5000 + \int_0^3 f(t) dt - 250(3) = 5126.591$
7	$5126.591 + \int_3^7 f(t) dt - 2000(4) = 4513.807$

The amount of water in the tank is greatest at 3 hours. At that time, the amount of water in the tank, rounded to the nearest gallon, is 5127 gallons.

5-5b

47.



The figure above shows the graph of f' the derivative of a twice-differentiable function f , on the closed interval $0 \leq x \leq 8$. The graph of f' has horizontal tangent lines at $x=1$, $x=3$, and $x=5$. The areas of the regions between the graph of f' and the x -axis are labeled in the figure. The function f is defined for all real numbers and satisfied $f(8)=4$.

Determine the absolute minimum value of f on the closed interval $0 \leq x \leq 8$. Justify your answer.

Part B

1 point is earned for considering $x=0$ and $x=6$

1 point is earned for the answer

1 point is earned for justification

From part (a), the absolute minimum occurs either at end point

$$f(0) = f(8) + \int_0^8 f'(x) dx$$

$$= f(8) - \int_0^8 f'(x) dx = 4 - 12 = -8$$

$$f(6) = f(8) + \int_8^6 f'(x) dx$$

$$= f(8) - \int_6^8 f'(x) dx = 4 - 7 = -3$$

$$f(8) = 4$$

5-5b

The absolute minimum value of f on the closed interval $[0, 8]$ is -8.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $x=0$ and $x=6$

1 point is earned for the answer

1 point is earned for justification

From part (a), the absolute minimum occurs either at $x=6$ or at an endpoint.

$$f(0) = f(8) + \int_0^8 f'(x) dx$$

$$= f(8) - \int_0^8 f'(x) dx = 4 - 12 = -8$$

$$f(6) = f(8) + \int_8^6 f'(x) dx$$

$$= f(8) - \int_6^8 f'(x) dx = 4 - 7 = -3$$

$$f(8) = 4$$

The absolute minimum value of f on the closed interval $[0, 8]$ is -8.

5-5b

48. On a certain workday, the rate, in tons per hour, at which unprocessed gravel arrives at a gravel processing plant is modeled by $G(t) = 90 + 45\cos(t^2/18)$, where t is measured in hours and $0 \leq t \leq 8$. At the beginning of the workday ($t = 0$), the plant has 500 tons of unprocessed gravel. During the hours of operation, $0 \leq t \leq 8$, the plant processes gravel at a constant rate of 100 tons per hour.



What is the maximum amount of unprocessed gravel at the plant during the hours of operation on this workday? Justify your answer.

Part D

1 point is earned for considering $A'(t)=0$

1 point is earned for the answer

1 point is earned for the justification

The amount of unprocessed gravel at time t is given by

$$A(t) = 500 + \int_0^t (G(s) - 100) ds.$$

$$A'(t) = G(t) - 100 = 0 \Rightarrow t = 4.923480$$

t	$A(t)$
0	500
4.92348	635.376123
8	525.551089

The maximum amount of unprocessed gravel at the plant during this workday is 635.376 tons.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $A'(t)=0$

1 point is earned for the answer

5-5b

1 point is earned for the justification

The amount of unprocessed gravel at time t is given by

$$A(t) = 500 + \int_0^t (G(s) - 100) ds.$$

$$A'(t) = G(t) - 100 = 0 \Rightarrow t = 4.923480$$

t	$A(t)$
0	500
4.92348	635.376123
8	525.551089

The maximum amount of unprocessed gravel at the plant during this workday is 635.376 tons.

5-5b

49. The functions f and g are given by $f(x) = \int_0^{3x} \sqrt{4+t^2} dt$ and $g(x) = f(\sin x)$.

Write, but do not evaluate, an integral expression that represents the maximum value of g on the interval $0 \leq x \leq \pi$. Justify your answer.

Part C

1 point is earned if sets $g'(x)=0$

1 point is earned if justifies maximum at $\frac{\pi}{2}$

1 point is earned for integral expression at $g(\frac{\pi}{2})$

For $0 < x < \pi$, $g'(x)=0$ only at $x = \frac{\pi}{2}$.

$$g(0)=g(\pi)=0$$

$$g(\frac{\pi}{2}) = \int_0^3 \sqrt{4+t^2} dt > 0$$

The maximum value of g on $[0, \pi]$ is $\int_0^3 \sqrt{4+t^2} dt$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if sets $g'(x)=0$

1 point is earned if justifies maximum at $\frac{\pi}{2}$

1 point is earned for integral expression at $g(\frac{\pi}{2})$

For $0 < x < \pi$, $g'(x)=0$ only at $x = \frac{\pi}{2}$.

$$g(0)=g(\pi)=0$$

$$g(\frac{\pi}{2}) = \int_0^3 \sqrt{4+t^2} dt > 0$$

The maximum value of g on $[0, \pi]$ is $\int_0^3 \sqrt{4+t^2} dt$

5-5b

5-5b

50. The functions f and g are given by $f(x) = \int_0^{3x} \sqrt{4+t^2} dt$ and $g(x) = f(\sin x)$.
- (a) Find $f'(x)$ and $g'(x)$.
- (b) Write an equation for the line tangent to the graph of $y = g(x)$ at $x = \pi$.
- (c) Write, but do not evaluate, an integral expression that represents the maximum value of g on the interval $0 \leq x \leq \pi$. Justify your answer.

Part A

2 points are earned for $f'(x)$

2 points are earned for $g'(x)$

$$\begin{aligned} f'(x) &= 3\sqrt{4 + (3x)^2} \\ g'(x) &= f'(\sin x) \cdot \cos x \\ &= 3\sqrt{4 + (3\sin x)^2} \cdot \cos x \end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for $f'(x)$

2 points are earned for $g'(x)$

$$\begin{aligned} f'(x) &= 3\sqrt{4 + (3x)^2} \\ g'(x) &= f'(\sin x) \cdot \cos x \\ &= 3\sqrt{4 + (3\sin x)^2} \cdot \cos x \end{aligned}$$

Part B

1 point is earned for $g(\pi)$ or $g'(\pi)$

1 point is earned for the tangent line equation

$$g(\pi) = 0, g'(\pi) = -6$$

$$\text{Tangent line: } y = -6(x - \pi)$$

5-5b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g(\pi)$ or $g'(\pi)$

1 point is earned for the tangent line equation

$$g(\pi) = 0, g'(\pi) = -6$$

$$\text{Tangent line: } y = -6(x - \pi)$$

Part C

1 point is earned if sets $g'(x) = 0$

1 point is earned if justifies maximum at $\frac{\pi}{2}$

1 point is earned for integral expression at $g\left(\frac{\pi}{2}\right)$

For $0 < x < \pi$, $g'(x) = 0$ only at $x = \frac{\pi}{2}$.

$$g(0) = g(\pi) = 0$$

$$g\left(\frac{\pi}{2}\right) = \int_0^3 \sqrt{4 + t^2} dt > 0$$

The maximum value of g on $[0, \pi]$ is $\int_0^3 \sqrt{4 + t^2} dt$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if sets $g'(x) = 0$

1 point is earned if justifies maximum at $\frac{\pi}{2}$

1 point is earned for integral expression at $g\left(\frac{\pi}{2}\right)$

5-5b

For $0 < x < \pi$, $g'(x) = 0$ only at $x = \frac{\pi}{2}$.

$$g(0) = g(\pi) = 0$$

$$g\left(\frac{\pi}{2}\right) = \int_0^3 \sqrt{4+t^2} dt > 0$$

The maximum value of g on $[0, \pi]$ is $\int_0^3 \sqrt{4+t^2} dt$.

5-5b

51.  The rate at which rainwater flows into a drainpipe is modeled by the function R , where $R(t) = 20 \sin\left(\frac{t^2}{35}\right)$ cubic feet per hour, t is measured in hours, and $0 \leq t \leq 8$. The pipe is partially blocked, allowing water to drain out the other end of the pipe at a rate modeled by $D(t) = -0.04t^3 + 0.4t^2 + 0.96t$ cubic feet per hour, for $0 \leq t \leq 8$. There are 30 cubic feet of water in the pipe at time $t = 0$.
- How many cubic feet of rainwater flow into the pipe during the 8-hour time interval $0 \leq t \leq 8$?
 - Is the amount of water in the pipe increasing or decreasing at time $t = 3$ hours? Give a reason for your answer.
 - At what time t , $0 \leq t \leq 8$, is the amount of water in the pipe at a minimum? Justify your answer.
 - The pipe can hold 50 cubic feet of water before overflowing. For $t > 8$, water continues to flow into and out of the pipe at the given rates until the pipe begins to overflow. Write, but do not solve, an equation involving one or more integrals that gives the time w when the pipe will begin to overflow.

Part A

1 point is earned for the Integrand

1 point is earned for the answer

$$\int_0^8 R(t) dt = 76.570$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the Integrand

1 point is earned for the answer

$$\int_0^8 R(t) dt = 76.570$$

Part B

1 point is earned for considering $R(3)$ and $D(3)$

1 point is earned for the answer and reason

$$R(3) - D(3) = -0.313632 < 0$$

5-5b

Since $R(3) < D(3)$, the amount of water in the pipe is decreasing at time $t = 3$ hours.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considering $R(3)$ and $D(3)$

1 point is earned for the answer and reason

$$R(3) - D(3) = -0.313632 < 0$$

Since $R(3) < D(3)$, the amount of water in the pipe is decreasing at time $t = 3$ hours.

Part C

1 point is earned for considering $R(t) - D(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

The amount of water in the pipe at time t , $0 \leq t \leq 8$, is $30 + \int_0^t [R(x) - D(x)]dx$.

$$R(t) - D(t) = 0 \Rightarrow t = 0, 3.271658$$

t	Amount of water in the pipe
0	30
3.271658	27.964561
8	48.543686

The amount of water in the pipe is a minimum at time $t = 3.272$ (or 3.271) hours.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $R(t) - D(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

5-5b

The amount of water in the pipe at time t , $0 \leq t \leq 8$, is $30 + \int_0^t [R(x) - D(x)]dx$.

$$R(t) - D(t) = 0 \Rightarrow t = 0, 3.271658$$

t	Amount of water in the pipe
0	30
3.271658	27.964561
8	48.543686

The amount of water in the pipe is a minimum at time $t = 3.272$ (or 3.271) hours.

Part D

1 point is earned for the integral

1 point is earned for the equation

$$30 + \int_0^w [R(t) - D(t)]dt = 50$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the equation

$$30 + \int_0^w [R(t) - D(t)]dt = 50$$

5-5b

52. The rate at which rainwater flows into a drainpipe is modeled by the function R , where $R(t) = 20 \sin\left(\frac{t^2}{35}\right)$ cubic feet per hour, t is measured in hours, and $0 \leq t \leq 8$. The pipe is partially blocked, allowing water to drain out the other end of the pipe at a rate modeled by $D(t) = -0.04t^3 + 0.4t^2 + 0.96t$ or $0 \leq t \leq 8$. There are 30 cubic feet of water in the pipe at time $t=0$.



At what time t , $0 \leq t \leq 8$, is the amount of water in the pipe at a minimum? Justify your answer.

Part C

1 point is earned for considering $R(t) - D(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

The amount of water in the pipe at time t , $0 \leq t \leq 8$, is $30 + \int_0^t [R(x) - D(x)] dx$.

$$R(t) - D(t) = 0 \Rightarrow t = 0, 3.271658$$

t	Amount of water in the pipe
0	30
3.271658	27.964561
8	48.543686

The amount of water in the pipe is a minimum at time $t = 3.272$ (or 3.271) hours.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $R(t) - D(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

The amount of water in the pipe at time t , $0 \leq t \leq 8$, is $30 + \int_0^t [R(x) - D(x)] dx$

$$R(t) - D(t) = 0 \Rightarrow t = 0, 3.271658$$

t	Amount of water in the pipe
0	30
3.271658	27.964561
8	48.543686

5-5b

The amount of water in the pipe is a minimum at time $t = 3.272$ (or 3.271) hours.

5-5b

53.

t (hours)	0	2	5	7	8
$E(t)$ (hundreds of entries)	0	4	13	21	23

A zoo sponsored a one-day contest to name a new baby elephant. Zoo visitors deposited entries in a special box between noon ($t = 0$) and 8 P.M. ($t = 8$). The number of entries in the box t hours after noon is modeled by a differentiable function E for $0 \leq t \leq 8$. Values of $E(t)$, in hundreds of entries, at various times t are shown in the table above.

(a) Use the data in the table to approximate the rate, in hundreds of entries per hour, at which entries were being deposited at time $t = 6$. Show the computations that lead to your answer.

(b) Use a trapezoidal sum with the four subintervals given by the table to approximate the value of

$18 \int_0^8 E(t) dt$. Using correct units, explain the meaning of $18 \int_0^8 E(t) dt$ in terms of the number of entries.

(c) At 8 P.M., volunteers began to process the entries. They processed the entries at a rate modeled by the function P , where $P(t) = t^3 - 30t^2 + 298t - 976$ hundreds of entries per hour for $8 \leq t \leq 12$. According to the model, how many entries had not yet been processed by midnight ($t = 12$)?

(d) According to the model from part (c), at what time were the entries being processed most quickly? Justify your answer.

Part A

1 point is earned for answer

$$E'(6) \approx \frac{E(7) - E(5)}{7 - 5} = 4 \text{ hundred entries per hour}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer

$$E'(6) \approx \frac{E(7) - E(5)}{7 - 5} = 4 \text{ hundred entries per hour}$$

Part B

1 point is earned for trapezoidal sum

1 point is earned for approximation

5-5b

1 point is earned for meaning

$$\begin{aligned}\frac{1}{8} \int_0^8 E(t) dt &\approx \\ \frac{1}{8} \left(2 \cdot \frac{E(0)+E(2)}{2} + 3 \cdot \frac{E(2)+E(5)}{2} + 2 \cdot \frac{E(5)+E(7)}{2} + 1 \cdot \frac{E(7)+E(8)}{2} \right) \\ &= 10.687 \text{ or } 10.688\end{aligned}$$

$$\frac{1}{8} \int_0^8 E(t) dt \text{ is the average number of hundreds of entries in the box between noon and 8 P.M.}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for trapezoidal sum

1 point is earned for approximation

1 point is earned for meaning

$$\begin{aligned}\frac{1}{8} \int_0^8 E(t) dt &\approx \\ \frac{1}{8} \left(2 \cdot \frac{E(0)+E(2)}{2} + 3 \cdot \frac{E(2)+E(5)}{2} + 2 \cdot \frac{E(5)+E(7)}{2} + 1 \cdot \frac{E(7)+E(8)}{2} \right) \\ &= 10.687 \text{ or } 10.688\end{aligned}$$

$$\frac{1}{8} \int_0^8 E(t) dt \text{ is the average number of hundreds of entries in the box between noon and 8 P.M.}$$

Part C

1 point is earned for integral

1 point is earned for the answer

$$23 - \int_8^{12} P(t) dt = 23 - 16 = 7 \text{ hundred entries}$$

5-5b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$23 - \int_8^{12} P(t) dt = 23 - 16 = 7 \text{ hundred entries}$$

Part D

1 point is earned for considering $P'(t) = 0$

1 point is earned for identifying candidates

1 point is earned for answering with justification

$P'(t) = 0$ when $t = 9.183503$ and $t = 10.816497$.

t	$P(t)$
8	0
9.183503	5.088662
10.816497	2.911338
12	8

Entries are being processed most quickly at time $t = 12$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $P'(t) = 0$

1 point is earned for identifying candidates

1 point is earned for answering with justification

$P'(t) = 0$ when $t = 9.183503$ and $t = 10.816497$.

5-5b

t	$P(t)$
8	0
9.183503	5.088662
10.816497	2.911338
12	8

Entries are being processed most quickly at time $t = 12$.

5-5b

54.  The rate at which people enter an amusement park on a given day is modeled by the function E defined by

$$E(t) = \frac{15600}{(t^2 - 24t + 160)}.$$

The rate at which people leave the same amusement park on the same day is modeled by the function L defined by

$$L(t) = \frac{9890}{(t^2 - 38t + 370)}.$$

Both $E(t)$ and $L(t)$ are measured in people per hour and time t is measured in hours after midnight. These functions are valid for $9 \leq t \leq 23$, the hours during which the park is open. At time $t = 9$, there are no people in the park.

(a) How many people have entered the park by 5:00 P.M. ($t = 17$)? Round your answer to the nearest whole number.

(b) The price of admission to the park is \$15 until 5:00 P.M. ($t = 17$). After 5:00 P.M., the price of admission to the park is \$11. How many dollars are collected from admissions to the park on the given day? Round your answer to the nearest whole number.

(c) Let $H(t) = \int_9^t (E(x) - L(x))dx$ for $9 \leq t \leq 23$. The value of $H(17)$ to the nearest whole number is 3725. Find the value of $H'(17)$, and explain the meaning of $H(17)$ and $H'(17)$ in the context of the amusement park.

(d) At what time t , for $9 \leq t \leq 23$, does the model predict that the number of people in the park is a maximum?

Part A

1 point is earned for limits

1 point is earned for integrand

1 point is earned for the answer

$$\int_9^{17} E(t)dt = 6004.270$$

6004 people entered the park by 5 pm.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for limits

5-5b

1 point is earned for integrand

1 point is earned for the answer

$$\int_9^{17} E(t) dt = 6004.270$$

6004 people entered the park by 5 pm.

Part B

1 point is earned for the setup

$$15 \int_9^{17} E(t) dt + 11 \int_{17}^{23} E(t) dt = 104048.165$$

The amount collected was \$104,048.

or

$$\int_{17}^{23} E(t) dt = 1271.283$$

1271 people entered the park between 5 pm and 11 pm, so the amount collected was

$$\$15 \cdot (6004) + \$11 \cdot (1271) = \$104,041.$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the setup

$$15 \int_9^{17} E(t) dt + 11 \int_{17}^{23} E(t) dt = 104048.165$$

The amount collected was \$104,048.

or

5-5b

$$\int_{17}^{23} E(t) dt = 1271.283$$

1271 people entered the park between 5 pm and 11 pm, so the amount collected was

$$\$15 \cdot (6004) + \$11 \cdot (1271) = \$104,041.$$

Part C

1 point is earned for value $H'(17)$

2 points are earned for meanings

1: meaning of $H(17)$

1: meaning of $H'(17)$

<-1> if no reference to $t = 17$

$$H'(17) = E(17) - L(17) = -380.281$$

There were 3725 people in the park at $t = 17$. The number of people in the park was decreasing at the rate of approximately 380 people/hr at time $t = 17$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for value $H'(17)$

2 points are earned for meanings

1: meaning of $H(17)$

1: meaning of $H'(17)$

<-1> if no reference to $t = 17$

$$H'(17) = E(17) - L(17) = -380.281$$

There were 3725 people in the park at $t = 17$. The number of people in the park was decreasing at the rate of approximately 380 people/hr at time $t = 17$.

Part D

5-5b

1 point is earned for $E(t) - L(t) = 0$

1 point is earned for the answer

$$H'(t) = E(t) - L(t) = 0$$

$$t = 15.794 \text{ or } 15.795$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $E(t) - L(t) = 0$

1 point is earned for the answer

$$H'(t) = E(t) - L(t) = 0$$

$$t = 15.794 \text{ or } 15.795$$

5-5b

55. The rate at which people enter an amusement park on a given day is modeled by the function E defined by

$$E(t) = \frac{15600}{(t^2 - 24t + 160)}$$

The rate at which people leave the same amusement park on the same day is modeled by the function L defined by

$$L(t) = \frac{9890}{(t^2 - 38t + 370)}$$

Both $E(t)$ and $L(t)$ are measured in people per hour and time t is measured in hours after midnight. These functions are valid for $9 \leq t \leq 23$, the hours during which the park is open. At time $t = 9$, there are no people in the park.



At what time t , for $9 \leq t \leq 23$, does the model predict that the number of people in the park is a maximum?

Part D

1 point is earned for $E(t) - L(t) = 0$

1 point is earned for the answer

$$H'(t) = E(t) - L(t) = 0$$

$$t = 15.794 \text{ or } 15.795$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $E(t) - L(t) = 0$

1 point is earned for the answer

$$H'(t) = E(t) - L(t) = 0$$

$$t = 15.794 \text{ or } 15.795$$

5-5b

56.  The rate at which people enter an auditorium for a rock concert is modeled by the function R given by $R(t) = 1380t^2 - 675t^3$ for $0 \leq t \leq 2$ hours; $R(t)$ is measured in people per hour. No one is in the auditorium at time $t = 0$, when the doors open. The doors close and the concert begins at time $t = 2$.
- (a) How many people are in the auditorium when the concert begins?
- (b) Find the time when the rate at which people enter the auditorium is a maximum. Justify your answer.
- (c) *The total wait time for all the people in the auditorium is found by adding the time each person waits, starting at the time the person enters the auditorium and ending when the concert begins. The function w models the total wait time for all the people who enter the auditorium before time t . The derivative of w is given by $w'(t) = (2 - t)R(t)$. Find $w(2) - w(1)$, the total wait time for those who enter the auditorium after time $t = 1$.*
- (d) On average, how long does a person wait in the auditorium for the concert to begin? Consider all people who enter the auditorium after the doors open, and use the model for total wait time from part (c).

Part A

1 point is earned for integral

1 point is earned for answer

$$\int_0^2 R(t) dt = 980 \text{ people}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for answer

$$\int_0^2 R(t) dt = 980 \text{ people}$$

Part B

1 point is earned if considers $R'(t) = 0$

1 point is earned for interior critical point

1 point is earned for the answer plus justification

$$R'(t) = 0 \text{ when } t = 0 \text{ and } t = 1.36296$$

5-5b

The maximum rate may occur at 0, $a = 1.36296$, or 2.

$$R(0) = 0$$

$$R(a) = 854.527$$

$$R(2) = 120$$

The maximum rate occurs when $t = 1.362$ or 1.363 .



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if considers $R'(t) = 0$

1 point is earned for interior critical point

1 point is earned for the answer plus justification

$$R'(t) = 0 \text{ when } t = 0 \text{ and } t = 1.36296$$

The maximum rate may occur at 0, $a = 1.36296$, or 2.

$$R(0) = 0$$

$$R(a) = 854.527$$

$$R(2) = 120$$

The maximum rate occurs when $t = 1.362$ or 1.363 .

Part C

1 point is earned for integral

1 point is earned for answer

$$w(2) - w(1) = \int_1^2 w'(t) dt = \int_1^2 (2 - t)R(t) dt = 387.5$$

The total wait time for those who enter the auditorium after time $t = 1$ is 387.5 hours.



0	1	2
---	---	---

5-5b

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for answer

$$w(2) - w(1) = \int_1^2 w'(t) dt = \int_1^2 (2 - t)R(t) dt = 387.5$$

The total wait time for those who enter the auditorium after time $t = 1$ is 387.5 hours.

Part D

1 point is earned for integral

1 point is earned for answer

$$\frac{1}{980}w(2) = \frac{1}{980} \int_0^2 (2 - t)R(t) dt = 0.77551$$

On average, a person waits 0.775 or 0.776 hour.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for answer

$$\frac{1}{980}w(2) = \frac{1}{980} \int_0^2 (2 - t)R(t) dt = 0.77551$$

On average, a person waits 0.775 or 0.776 hour.

5-5b

57. Water is pumped into an underground tank at a constant rate of 8 gallons per minute. Water leaks out of the tank at the rate of $\sqrt{t+1}$ gallons per minute, for $0 \leq t \leq 120$ minutes. At time $t = 0$, the tank contains 30 gallons of water.

At what time t , for $0 \leq t \leq 120$, is the amount of water in the tank a maximum? Justify your answer.

Part D

1 point is earned for set $A'(t) = 0$

1 point is earned for solves for t

1 point is earned for the justification

$$A'(t) = 8 - \sqrt{t+1} = 0 \text{ when } t = 63$$

$A'(t)$ is positive for $0 < t < 63$ and negative for $63 < t < 120$. Therefore there is a maximum at $t = 63$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for set $A'(t) = 0$

1 point is earned for solves for t

1 point is earned for the justification

$$A'(t) = 8 - \sqrt{t+1} = 0 \text{ when } t = 63$$

$A'(t)$ is positive for $0 < t < 63$ and negative for $63 < t < 120$. Therefore there is a maximum at $t = 63$.

5-5b

58. The tide removes sand from Sandy Point Beach at a rate modeled by the function R , given by

$$R(t) = 2 + 5 \sin\left(\frac{4\pi t}{25}\right).$$

A pumping station adds sand to the beach at a rate modeled by the function S , given by $S(t) = \frac{15t}{1+3t}$.

Both $R(t)$ and $S(t)$ have units of cubic yards per hour and t is measured in hours for $0 \leq t \leq 6$. At time $t = 0$, the beach contains 2500 cubic yards of sand.

 For $0 \leq t \leq 6$, at what time t is the amount of sand on the beach a minimum? What is the minimum value? Justify your answers.

Part D

1 point is earned for sets $Y'(t) = 0$

1 point is earned for the critical t -value

1 point is earned for the answer with justification

$Y'(t) = 0$ when $S(t) - R(t) = 0$.

The only value in $[0, 6]$ to satisfy $S(t) = R(t)$ is $a = 5.117865$.

t	$Y(t)$
0	2500
a	2492.3694
6	2493.2766

The amount of sand is a minimum when $t = 5.117$ or 5.118 hours. The minimum value is 2492.369 cubic yards.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for sets $Y'(t) = 0$

1 point is earned for the critical t -value

1 point is earned for the answer with justification

$Y'(t) = 0$ when $S(t) - R(t) = 0$.

The only value in $[0, 6]$ to satisfy $S(t) = R(t)$ is $a = 5.117865$.

5-5b

t	$Y(t)$
0	2500
a	2492.3694
6	2493.2766

The amount of sand is a minimum when $t = 5.117$ or 5.118 hours. The minimum value is 2492.369 cubic yards.

5-5b

59. For $0 \leq t \leq 31$, the rate of change of the number of mosquitoes on Tropical Island at time t days is modeled by $R(t) = 5\sqrt{t} \cos\left(\frac{t}{5}\right)$ mosquitoes per day. There are 1000 mosquitoes on Tropical Island at time $t=0$.



To the nearest whole number, what is the maximum number of mosquitoes for $0 \leq t \leq 31$? Show the analysis that leads to your conclusion.

Part D

1 point is earned for the integral

1 point is earned for the answer

1 point is earned for computing interior critical points

1 point is earned for completing analysis

$R(t)=0$ when $t=0$, $t=2.5\pi$, or $t=7.5\pi$

$R(t)>0$ on $0<t<2.5\pi$

$R(t)<0$ on $2.5\pi<t<7.5\pi$

$R(t)>0$ on $7.5\pi<t<31$

The absolute maximum number of mosquitoes occurs at $t=2.5\pi$ or at $t=31$.

$$1000 + \int_0^{2.5\pi} R(t)dt = 1039.357$$

There are 964 mosquitoes at $t=31$, so the maximum number of mosquitoes is 1039, to the nearest whole number.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

1 point is earned for computing interior critical points

1 point is earned for completing analysis

$R(t)=0$ when $t=0$, $t=2.5\pi$, or $t=7.5\pi$

$R(t)>0$ on $0<t<2.5\pi$

$R(t)<0$ on $2.5\pi<t<7.5\pi$

5-5b

$$R(t) > 0 \text{ on } 7.5\pi < t < 31$$

The absolute maximum number of mosquitoes occurs at $t = 2.5\pi$ or at $t = 31$.

$$1000 + \int_0^{2.5\pi} R(t) dt = 1039.357,$$

There are 964 mosquitoes at $t = 31$, so the maximum number of mosquitoes is 1039, to the nearest whole number.

5-5b

60. A storm washed away sand from a beach, causing the edge of the water to get closer to a nearby road. The rate at which the distance between the road and the edge of the water was changing during the storm is modeled by $f(t) = \sqrt{t} + \cos t - 3$ meters per hour, t hours after the storm began. The edge of the water was 35 meters from the road when the storm began, and the storm lasted 5 hours. The derivative of $f(t)$ is $f'(t) = \frac{1}{2\sqrt{t}} - \sin t$.



At what time during the 5 hours of the storm was the distance between the road and the edge of the water decreasing most rapidly? Justify your answer.

Part C

1 point is earned if considers $f'(t)=0$

1 point is earned for the answer

1 point is earned for justification

$f'(t) = 0$ when $t = 0.66187$ and $t = 2.84038$

The minimum of f for $0 \leq t \leq 5$ may occur at 0, 0.66187, 2.84038, or 5.

$f(0) = -2$

$f(0.66187) = -1.39760$

$f(2.84038) = -2.26963$

$f(5) = -0.48027$

The distance between the road and the edge of the water was decreasing most rapidly at time $t = 2.840$ hours after the storm began.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if considers $f'(t)=0$

1 point is earned for the answer

1 point is earned for justification

$f'(t) = 0$ when $t = 0.66187$ and $t = 2.84038$

The minimum of f for $0 \leq t \leq 5$ may occur at 0, 0.66187, 2.84038, or 5.

5-5b

$$f(0) = -2$$

$$f(0.66187) = -1.39760$$

$$f(2.84038) = -2.26963$$

$$f(5) = -0.48027$$

The distance between the road and the edge of the water was decreasing most rapidly at time $t = 2.840$ hours after the storm began.

5-5b

61. Let $f(x)=12-x^2$ for $x \geq 0$ and $f(x) \geq 0$.

An isosceles triangle whose base is the interval from $(0,0)$ to $(c,0)$ has its vertex on the graph of f . For what value of c does the triangle have maximum area? Justify your answer.

Part B

1 point is earned for $A = \frac{1}{2}C(12 - \frac{C^2}{4})$

1 point is earned for computing $\frac{dA}{dC}$ & setting it to 0

1 point is earned for answer

1 point is earned for justification with end points

$$A = \frac{1}{2} c \cdot f(c/2) = \frac{1}{2} c \left(12 - \frac{c^2}{4}\right)$$

$$= 6c - \frac{c^3}{4} \text{ on } [0, 4\sqrt{3}]$$

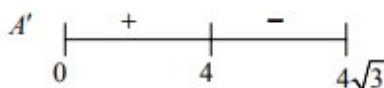
$$\frac{da}{dc} = 6 - 3\frac{c^2}{8}; 6 - 3\frac{c^2}{8} = 0 \text{ when } c = 4.$$

.

Candidate test

c	A
0	0
4	16 ← Max
$4\sqrt{3}$	0

First derivative



second derivative

$\frac{d^2A}{dc^2}\bigg|_{c=4} = -3 < 0$ so $c=4$ gives a relative max. $c=4$ is the only critical value in the domain interval, therefore maximum



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for $A = \frac{1}{2}C(12 - \frac{C^2}{4})$

1 point is earned for computing $\frac{dA}{dC}$ & setting it to 0

5-5b

1 point is earned for answer

1 point is earned for justification with end points

$$A = \frac{1}{2} c \cdot f(c/2) = \frac{1}{2} c \left(12 - \frac{c^2}{4}\right)$$

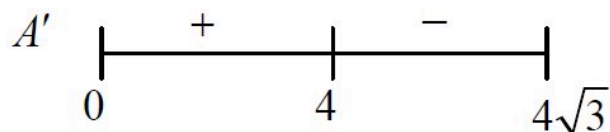
$$= 6c - \frac{c^3}{4} \text{ on } [0, 4\sqrt{3}]$$

$$\frac{da}{dc} = 6 - 3\frac{c^2}{8}; 6 - 3\frac{c^2}{8} = 0 \text{ when } c = 4.$$

Candidate test

c	A
0	0
4	16 ← Max
$4\sqrt{3}$	0

First derivative



second derivative

$$\left. \frac{d^2 A}{dc^2} \right|_{c=4} = -3 < 0 \text{ so } c=4 \text{ gives a relative max.}$$

$c=4$ is the only critical value in the domain interval, therefore maximum

5-5b

62. Mighty Cable Company manufactures cable that sells for \$120 per meter. For a cable of fixed length, the cost of producing a portion of the cable varies with its distance from the beginning of the cable. Mighty reports that the cost to produce a portion of a cable that is x meters from the beginning of the cable is $6\sqrt{x}$ dollars per meter. (Note: Profit is defined to be the difference between the amount of money received by the company for selling the cable and the company's cost of producing the cable.)

Find the maximum profit that Mighty could earn on the sale of one cable. Justify your answer.

Part D

1 point is earned for $P'(k)=0$

1 point is earned for $k=400$

1 point is earned for answer

1 point is earned for justification

Let $P(k)$ be the profit for a cable of length k . $P'(k) = 120 - 6\sqrt{k} = 0$ when $k=400$.

This is the only critical point for P , and P' changes from positive to negative at $k=400$. Therefore, the maximum profit is $P(400)=16,000$ dollars.



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for $P'(k)=0$

1 point is earned for $k=400$

1 point is earned for answer

1 point is earned for justification

Let $P(k)$ be the profit for a cable of length k .

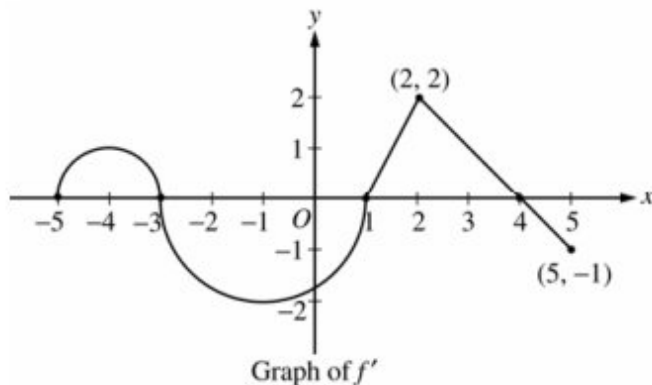
$P'(k) = 120 - 6\sqrt{k} = 0$ when $k=400$.

This is the only critical point for P , and P' changes from positive to negative at $k=400$.

Therefore, the maximum profit is $P(400)=16,000$ dollars.

5-5b

63.



Let f be a function defined on the closed interval $-5 \leq x \leq 5$ with $f(1)=3$. The graph of f' the derivative of f , consists of two semicircles and two line segments, as shown above.

Find the absolute minimum value of $f(x)$ over the closed interval $-5 \leq x \leq 5$. Explain your reasoning.

Part D

1 point is earned if identifies $x=1$ as a candidate

1 point is earned if considers endpoint

1 point is earned for value and explanation

Candidates for the absolute minimum are where f' changes from negative to positive (at $x=1$) and at the endpoints ($x=-5, 5$).

$$f(-5) = 3 + \int_1^{-5} f'(x) dx = 3 - \frac{\pi}{2} + 2\pi > 3$$

$$f(1) = 3$$

$$f(5) = 3 + \int_1^5 f'(x) dx = 3 + \frac{3 \cdot 2}{2} - \frac{1}{2} > 3$$

The absolute minimum value of f on $[-5, 5]$ is $f(1)=3$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if identifies $x=1$ as a candidate

1 point is earned if considers endpoint

1 point is earned for value and explanation

5-5b

Candidates for the absolute minimum are where f' changes from negative to positive (at $x=1$) and at the endpoints ($x=-5,5$).

$$f(-5) = 3 + \int_1^{-5} f'(x)dx = 3 - \frac{\pi}{2} + 2\pi > 3$$

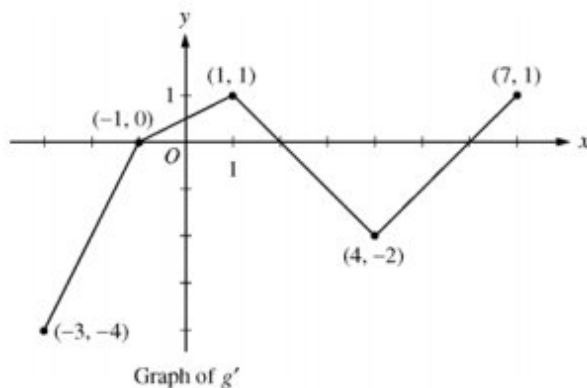
$$f(1) = 3$$

$$f(5) = 3 + \int_1^5 f'(x)dx = 3 + \frac{3 \cdot 2}{2} - \frac{1}{2} > 3$$

The absolute minimum value of f on $[-5, 5]$ is $f(1)=3$.

5-5b

64.



Let g be a continuous function with $g(2) = 5$. The graph of the piecewise-linear function g' , the derivative of g , is shown above for $-3 \leq x \leq 7$.

Find the absolute maximum value of g on the interval $-3 \leq x \leq 7$. Justify your answer.

Part B

1 point is earned if identifies $x=2$ as a candidate

1 point is earned if considers endpoints

1 point is earned for maximum value and justification

The only sign change of g' from positive to negative in the interval is at $x=2$.

$$g(-3) = 5 + \int_2^{-3} g'(x) dx = 5 + \left(-\frac{3}{2}\right) + 4 = \frac{15}{2}$$

$$g(2) = 5$$

$$g(7) = 5 + \int_2^7 g'(x) dx = 5 + (-4) + \frac{1}{2} = \frac{3}{2}$$

The maximum value of g for $-3 \leq x \leq 7$ is $\frac{15}{2}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if identifies $x=2$ as a candidate

1 point is earned if considers endpoints

5-5b

1 point is earned for maximum value and justification

The only sign change of g' from positive to negative in the interval is at $x=2$.

$$g(-3) = 5 + \int_2^{-3} g'(x) dx = 5 + \left(-\frac{3}{2}\right) + 4 = \frac{15}{2}$$

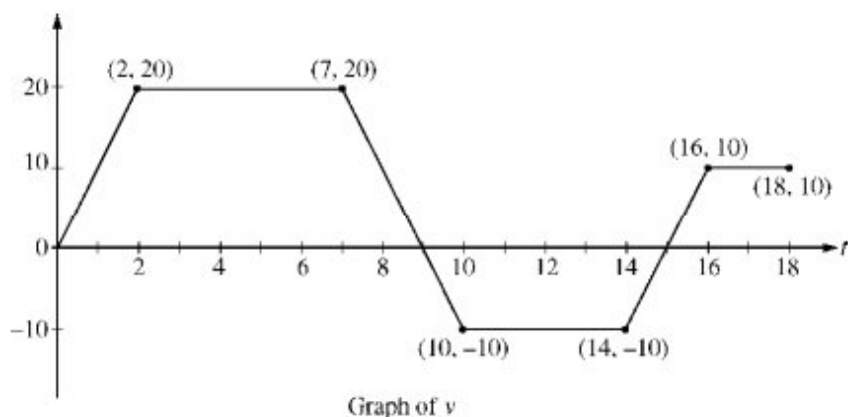
$$g(2) = 5$$

$$g(7) = 5 + \int_2^7 g'(x) dx = 5 + (-4) + \frac{1}{2} = \frac{3}{2}$$

The maximum value of g for $-3 \leq x \leq 7$ is $\frac{15}{2}$.

5-5b

65.



A squirrel starts at building A at time $t=0$ and travels along a straight, horizontal wire connected to building B . For $0 \leq t \leq 18$, the squirrel's velocity is modeled by the piecewise-linear function defined by the graph above.

At what time in the interval $0 \leq t \leq 18$ is the squirrel farthest from building A ? How far from building A is the squirrel at this time?

Part B

1 point is earned for identifying candidates

1 point is earned for answers

Velocity is 0 at $t=0$, $t=9$, and $t=15$.

t	position at time t
0	0
9	$\frac{9+5}{2} \cdot 20 = 140$
15	$140 - \frac{6+4}{2} \cdot 10 = 90$
18	$90 + \frac{3+2}{2} \cdot 10 = 115$

The squirrel is farthest from building A at time $t=9$;

its greatest distance from the building is 140.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for identifying candidates

5-5b

1 point is earned for answers

Velocity is 0 at $t=0$, $t=9$, and $t=15$.

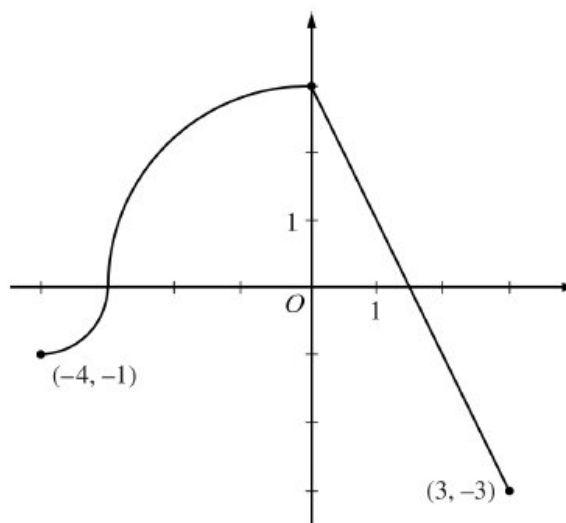
t	position at time t
0	0
9	$\frac{9+5}{2} \cdot 20 = 140$
15	$140 - \frac{6+4}{2} \cdot 10 = 90$
18	$90 + \frac{3+2}{2} \cdot 10 = 115$

The squirrel is farthest from building A at time $t=9$;

its greatest distance from the building is 140.

5-5b

66.

Graph of f

The continuous function f is defined on the interval $-4 \leq x \leq 3$. The graph of f consists of two quarter circles and one line segment, as shown in the figure above. Let $g(x) = 2x + \int_0^x f(t) dt$.

Determine the x -coordinate of the point at which g has an absolute maximum on the interval $-4 \leq x \leq 3$. Justify your answer.

Part B

1 point is earned for considering $g'(x)=0$

1 point is earned for identifying interior candidate

1 point is earned for answer with justification

$g'(x)=0$ when $f(x)=-2$. This occurs at $x = \frac{5}{2}$

$g'(x) > 0$ for $-4 < x < \frac{5}{2}$ and $g'(x) < 0$ for $\frac{5}{2} < x < 3$

Therefore g has an absolute maximum at $x = \frac{5}{2}$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for considering $g'(x)=0$

1 point is earned for identifying interior candidate

1 point is earned for answer with justification

5-5b

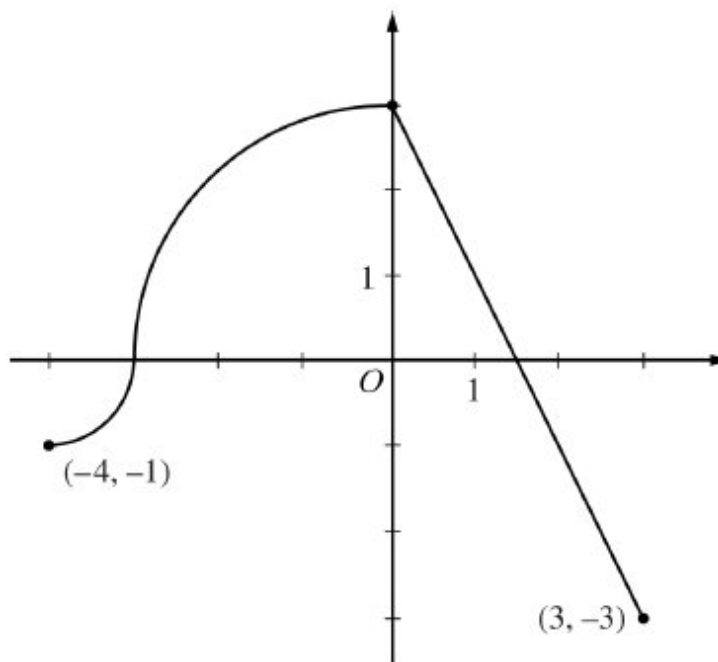
$g'(x)=0$ when $f(x)=-2$. This occurs at $x = \frac{5}{2}$

$g'(x) > 0$ for $-4 < x < \frac{5}{2}$ and $g'(x) < 0$ for $\frac{5}{2} < x < 3$

Therefore g has an absolute maximum at $x = \frac{5}{2}$

5-5b

67.

Graph of f

The continuous function f is defined on the interval $-4 \leq x \leq 3$. The graph of f consists of two quarter circles and one line segment, as shown in the figure above. Let $g(x) = 2x + \int_0^x f(t) dt$.

- Find $g(-3)$. Find $g'(x)$ and evaluate $g'(-3)$.
- Determine the x -coordinate of the point at which g has an absolute maximum on the interval $-4 \leq x \leq 3$. Justify your answer.
- Find all values of x on the interval $-4 < x < 3$ for which the graph of g has a point of inflection. Give a reason for your answer.
- Find the average rate of change of f on the interval $-4 \leq x \leq 3$. There is no point c , $-4 < c < 3$, for which $f'(c)$ is equal to that average rate of change. Explain why this statement does not contradict the Mean Value Theorem.

Part A1 point is earned for $g(-3)$ 1 point is earned for $g'(x)$ 1 point is earned for $g'(-3)$

$$g(-3) = 2(-3) + \int_0^{-3} f(t) dt = -6 - \frac{9\pi}{4}$$

$$g'(x) = 2 + f(x)$$

5-5b

$$g'(-3) = 2 + f(-3) = 2$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g(-3)$

1 point is earned for $g'(x)$

1 point is earned for $g'(-3)$

$$g(-3) = 2(-3) + \int_0^{-3} f(t)dt = -6 - \frac{9\pi}{4}$$

$$g'(x) = 2 + f(x)$$

$$g'(-3) = 2 + f(-3) = 2$$

Part B

1 point is earned for considering $g'(x) = 0$

1 point is earned for identifying interior candidate

1 point is earned for answer with justification

$g'(x) = 0$ when $f(x) = -2$. This occurs at $x = \frac{5}{2}$.

$g'(x) > 0$ for $-4 < x < \frac{5}{2}$ and $g'(x) < 0$ for $\frac{5}{2} < x < 3$.

Therefore g has an absolute maximum at $x = \frac{5}{2}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $g'(x) = 0$

1 point is earned for identifying interior candidate

1 point is earned for answer with justification

5-5b

$g'(x) = 0$ when $f(x) = -2$. This occurs at $x = \frac{5}{2}$.

$g'(x) > 0$ for $-4 < x < \frac{5}{2}$ and $g'(x) < 0$ for $\frac{5}{2} < x < 3$.

Therefore g has an absolute maximum at $x = \frac{5}{2}$.

Part C

1 point is earned for answer with reason

$g''(x) = f'(x)$ changes sign only at $x = 0$. Thus the graph of g has a point of inflection at $x = 0$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer with reason

$g''(x) = f'(x)$ changes sign only at $x = 0$. Thus the graph of g has a point of inflection at $x = 0$.

Part D

1 point is earned for average rate of change

1 point is earned for explanation

The average rate of change of f on the interval $-4 \leq x \leq 3$ is $\frac{f(3) - f(-4)}{3 - (-4)} = -\frac{2}{7}$.

To apply the Mean Value Theorem, f must be differentiable at each point in the interval $-4 < x < 3$. However, f is not differentiable at $x = -3$ and $x = 0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for average rate of change

1 point is earned for explanation

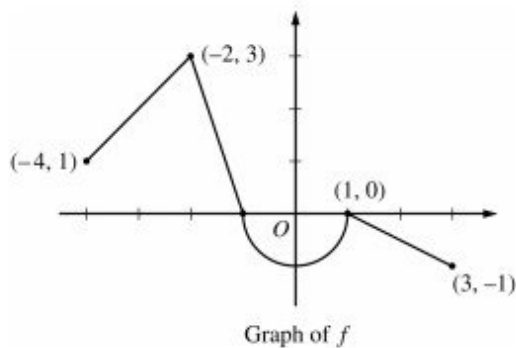
5-5b

The average rate of change of f on the interval $-4 \leq x \leq 3$ is $\frac{f(3) - f(-4)}{3 - (-4)} = -\frac{2}{7}$.

To apply the Mean Value Theorem, f must be differentiable at each point in the interval $-4 < x < 3$. However, f is not differentiable at $x = -3$ and $x = 0$.

5-5b

68.



Let f be the continuous function defined on $[-4, 3]$ whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let g be the function given by $g(x) = \int_1^x f(t) dt$.

- Find the values of $g(2)$ and $g(-2)$.
- For each of $g'(-3)$ and $g''(-3)$, find the value or state that it does not exist.
- Find the x -coordinate of each point at which the graph of g has a horizontal tangent line. For each of these points, determine whether g has a relative minimum, relative maximum, or neither a minimum nor a maximum at the point. Justify your answers.
- For $-4 < x < 3$, find all values of x for which the graph of g has a point of inflection. Explain your reasoning.

Part A

1 point is earned for $g(2)$

1 point is earned for $g(-2)$

$$g(2) = \int_1^2 f(t) dt = -\frac{1}{2}(1) \left(\frac{1}{2}\right) = -\frac{1}{4}$$

$$\begin{aligned} g(-2) &= \int_1^{-2} f(t) dt = -\int_{-2}^1 f(t) dt \\ &= -\left(\frac{3}{2} - \frac{\pi}{2}\right) = \left(\frac{\pi}{2} - \frac{3}{2}\right) \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g(2)$

1 point is earned for $g(-2)$

5-5b

$$g(2) = \int_1^2 f(t) dt = -\frac{1}{2}(1) \left(\frac{1}{2}\right) = -\frac{1}{4}$$

$$g(-2) = \int_1^{-2} f(t) dt = -\int_{-2}^1 f(t) dt$$

$$= -\left(\frac{3}{2} - \frac{\pi}{2}\right) = \left(\frac{\pi}{2} - \frac{3}{2}\right)$$

Part B

1 point is earned for $g'(-3)$

1 point is earned for $g''(-3)$

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$

$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g'(-3)$

1 point is earned for $g''(-3)$

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$

$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$

Part C

1 point is earned if considers $g'(x) = 0$

1 point is earned for $x = -1$ and $x = 1$

1 point is earned for answers with justification

The graph of g has a horizontal tangent line where $g'(x) = f(x) = 0$. This occurs at $x = -1$ and $x = 1$.

$g'(x)$ changes sign from positive to negative at $x = -1$. Therefore, g has a relative maximum at $x = -1$.

$g'(x)$ does not change sign at $x = 1$. Therefore, g has neither a relative maximum nor a relative minimum at $x = 1$.

5-5b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if considers $g'(x) = 0$

1 point is earned for $x = -1$ and $x = 1$

1 point is earned for answers with justification

The graph of g has a horizontal tangent line where $g'(x) = f(x) = 0$. This occurs at $x = -1$ and $x = 1$.

$g'(x)$ changes sign from positive to negative at $x = -1$. Therefore, g has a relative maximum at $x = -1$.

$g'(x)$ does not change sign at $x = 1$. Therefore, g has neither a relative maximum nor a relative minimum at $x = 1$.

Part D

1 point is earned for the answer

1 point is earned for the explanation

The graph of g has a point of inflection at each of $x = -2$, $x = 0$, and $x = 1$ because $g''(x) = f'(x)$ changes sign at each of these values.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the explanation

The graph of g has a point of inflection at each of $x = -2$, $x = 0$, and $x = 1$ because $g''(x) = f'(x)$ changes sign at each of these values.

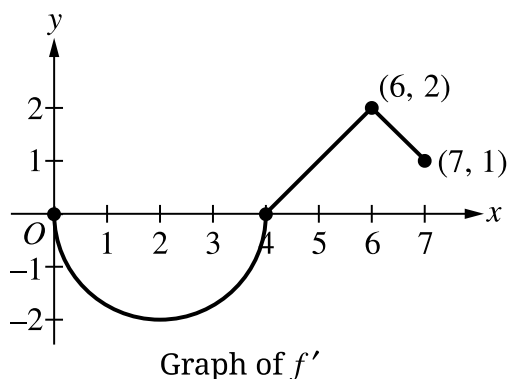
5-5b

69. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let f be a differentiable function with $f(4) = 3$. On the interval $0 \leq x \leq 7$, the graph of f' , the derivative of f , consists of a semicircle and two line segments, as shown in the figure.

- (a) Find $f(0)$ and $f(5)$.
- (b) Find the x -coordinates of all points of inflection of the graph of f for $0 < x < 7$. Justify your answer.
- (c) Let g be the function defined by $g(x) = f(x) - x$. On what intervals, if any, is g decreasing for $0 \leq x \leq 7$? Show the analysis that leads to your answer.
- (d) For the function g defined in part (c), find the absolute minimum value on the interval $0 \leq x \leq 7$. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with answers of only $f(0) = \pm 2\pi$, or only $f(5) = \frac{1}{2}$, or both earns 1 of the 3 points.

A response displaying $f(5) = \frac{7}{2}$ and a missing or incorrect value for $f(0)$ earns 2 of the 3 points.

The second and third points can be earned in either order.

Read unlabeled values from left to right and from top to bottom as $f(0)$ and $f(5)$. A single value must be labeled as either $f(0)$ or $f(5)$ in order to earn any points.

5-5b



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Area of either region – OR – $\int_0^4 f'(x) \, dx$ – OR – $\int_4^5 f'(x) \, dx$
- ☐ $f(0)$
- ☐ $f(5)$

Solution:

$$f(0) = f(4) + \int_4^0 f'(x) \, dx = 3 - \int_0^4 f'(x) \, dx = 3 + 2\pi$$

$$f(5) = f(4) + \int_4^5 f'(x) \, dx = 3 + \frac{1}{2} = \frac{7}{2}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that gives only one of $x = 2$ or $x = 6$, along with a correct justification, earns 1 of the 2 points.

A response that claims that there is a point of inflection at any value other than $x = 2$ or $x = 6$ earns neither point.

To earn the second point a response must use correct reasoning based on the graph of f' . Examples of correct reasoning include:

- Correctly discussing the signs of the slopes of the graph of f'
- Citing $x = 2$ and $x = 6$ as the locations of local extrema on the graph of f'

Examples of reasoning not (sufficiently) connected to the graph of f' include:

- Reasoning based on sign changes in f'' unless the connection is made between the sign of f'' and the slopes of the graph of f'
- Reasoning based only on the concavity of the graph of f

The second point cannot be earned by use of vague or undefined terms such as “it” or “the function” or “the derivative.”

Responses that report inflection points as ordered pairs must report the points $(2, 3 + \pi)$ and $(6, 5)$ in order to earn the first point. If the y -coordinates are reported incorrectly, the response remains eligible for the second point.

5-5b



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Justification

Solution:

The graph of f has a point of inflection at each of $x = 2$ and $x = 6$, because $f'(x)$ changes from decreasing to increasing at $x = 2$ and from increasing to decreasing at $x = 6$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point can be earned for $f'(x) \leq 1$ or the equivalent, in words or symbols.

Endpoints do not need to be included in the interval to be eligible for the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $g'(x) = f'(x) - 1$
- ☐ Interval with reason

Solution:

$$g'(x) = f'(x) - 1$$

$$f'(x) - 1 \leq 0 \Rightarrow f'(x) \leq 1$$

The graph of g is decreasing on the interval $0 \leq x \leq 5$ because $g'(x) \leq 0$ on this interval.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A justification that uses a local argument, such as “ g' changes from negative to positive (or g changes from decreasing to increasing) at $x = 5$ ” must also state that $x = 5$ is the only critical point.

5-5b

If $g'(x) = 0$ (or equivalent) is not declared explicitly, a response that isolates $x = 5$ as the only critical number belonging to $(0, 7)$ earns the first point.

A response that imports $g'(x) = f'(x)$ from part (c) is eligible for the first point but not the second.

In this case, consideration of $x = 4$ as the only critical number on $(0, 7)$ earns the first point.

Solution using Candidates Test:

$$g'(x) = f'(x) - 1 = 0 \Rightarrow x = 5, x = 7$$

x	$g(x)$
0	$3 + 2\pi$
5	$-\frac{3}{2}$
7	$-\frac{1}{2}$

The absolute minimum value of g on the interval $0 \leq x \leq 7$ is $-\frac{3}{2}$.

When using a Candidates Test, a response may import an incorrect value of $f(0) = g(0) > -\frac{3}{2}$ from part (a). The second point can only be earned for an answer of $-\frac{3}{2}$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $g'(x) = 0$
- ☐ Answer with justification

Solution:

g is continuous, $g'(x) < 0$ for $0 < x < 5$, and $g'(x) > 0$ for $5 < x < 7$.

Therefore, the absolute minimum occurs at $x = 5$, and $g(5) = f(5) - 5 = \frac{7}{2} - 5 = -\frac{3}{2}$ is the absolute minimum value of g .

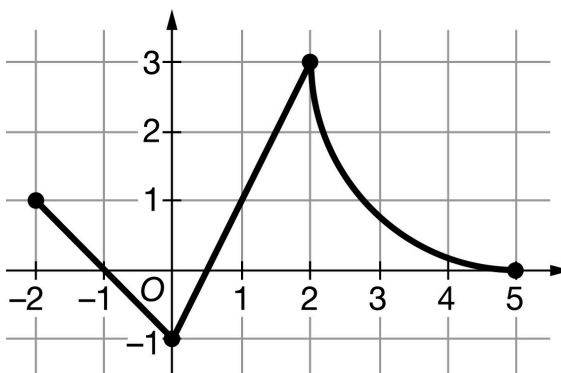
5-5b

70. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The continuous function f is defined on the closed interval $-6 \leq x \leq 5$. The figure above shows a portion of the graph of f , consisting of two line segments and a quarter of a circle centered at the point $(5, 3)$. It is known that the point $(3, 3 - \sqrt{5})$ is on the graph of f .

- If $\int_{-6}^5 f(x) \, dx = 7$, find the value of $\int_{-6}^{-2} f(x) \, dx$. Show the work that leads to your answer.
- Evaluate $\int_3^5 (2f'(x) + 4) \, dx$.
- The function g is given by $g(x) = \int_{-2}^x f(t) \, dt$. Find the absolute maximum value of g on the interval $-2 \leq x \leq 5$. Justify your answer.
- Find $\lim_{x \rightarrow 1} \frac{10^x - 3f'(x)}{f(x) - \arctan x}$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for showing evidence of using the property of adding integrals. This point can be earned symbolically or numerically with correct values of the integrals.

Simplification is not required to earn the second point or the third point

5-5b



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $\int_{-6}^5 f(x) dx = \int_{-6}^{-2} f(x) dx + \int_{-2}^5 f(x) dx$ integral
- ☐ $\int_{-2}^5 f(x) dx$
- ☐ answer

Solution:

$$\begin{aligned}\int_{-6}^5 f(x) dx &= \int_{-6}^{-2} f(x) dx + \int_{-2}^5 f(x) dx \\ \Rightarrow 7 &= \int_{-6}^{-2} f(x) dx + 2 + \left(9 - \frac{9\pi}{4}\right) \\ \Rightarrow \int_{-6}^{-2} f(x) dx &= 7 - \left(11 - \frac{9\pi}{4}\right) = \frac{9\pi}{4} - 4\end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with an antiderivative of $f'(x)$ and use of the limits of integration. Responses with errors in the integrand, such as a missing coefficient, earn the first point; such errors are considered for the second point.

Simplification is not required to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ answer

Solution:

5-5b

$$\begin{aligned}
 \int_3^5 (2f'(x) + 4) \, dx &= 2 \int_3^5 f'(x) \, dx + \int_3^5 4 \, dx \\
 &= 2(f(5) - f(3)) + 4(5 - 3) \\
 &= 2\left(0 - (3 - \sqrt{5})\right) + 8 \\
 &= 2(-3 + \sqrt{5}) + 8 = 2 + 2\sqrt{5}
 \end{aligned}$$

OR

$$\begin{aligned}
 \int_3^5 (2f'(x) + 4) \, dx &= [2f(x) + 4x]_{x=3}^{x=5} \\
 &= (2f(5) + 20) - (2f(3) + 12) \\
 &= (2 \cdot 0 + 20) - (2(3 - \sqrt{5}) + 12) \\
 &= 2 + 2\sqrt{5}
 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for connecting $g'(x) = f(x)$ analytically or verbally.

The second point is earned with identification of $x = -1$ regardless of what other candidates are considered.

The third point is earned for calculating $g(-1)$ and $g(5)$, declaring the absolute maximum to be $11 - \frac{9\pi}{4}$, and correctly reasoning for excluding $g(-2)$. Note that the ordered pair $(5, 11 - \frac{9\pi}{4})$ by itself is not a declaration of the absolute maximum.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $g'(x) = f(x)$
☐ identifies $x = -1$ as a candidate
☐ answer with justification

Solution:

$$g'(x) = f(x) = 0 \Rightarrow x = -1, x = \frac{1}{2}, x = 5$$

5-5b

x	$g(x)$
-2	0
-1	$\frac{1}{2}$
$\frac{1}{2}$	$-\frac{1}{4}$
5	$11 - \frac{9\pi}{4}$

On the interval $-2 \leq x \leq 5$, the absolute maximum value of g is $g(5) = 11 - \frac{9\pi}{4}$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer. Simplification, including $\arctan 1 = \frac{\pi}{4}$, is not required.



0

1

The student response accurately includes the criteria below.

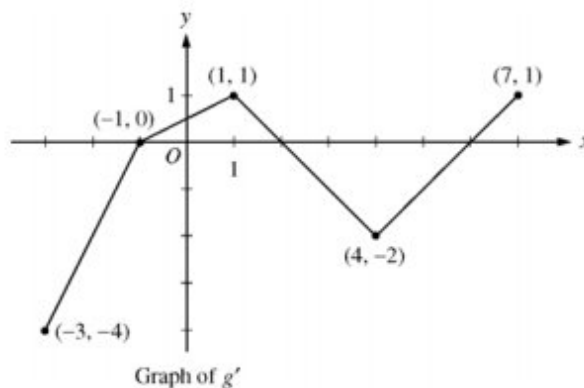
☐ · answer

Solution:

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{10^x - 3f'(x)}{f(x) - \arctan x} &= \frac{10^1 - 3f'(1)}{f(1) - \arctan 1} \\ &= \frac{10 - 3 \cdot 2}{1 - \arctan 1} = \frac{4}{1 - \frac{\pi}{4}} \end{aligned}$$

5-5b

71.



Let g be a continuous function with $g(2) = 5$. The graph of the piecewise-linear function g' , the derivative of g , is shown above for $-3 \leq x \leq 7$.

- Find the x -coordinate of all points of inflection of the graph of $y = g(x)$ for $-3 < x < 7$. Justify your answer.
- Find the absolute maximum value of g on the interval $-3 \leq x \leq 7$. Justify your answer.
- Find the average rate of change of $g(x)$ on the interval $-3 \leq x \leq 7$.
- Find the average rate of change of $g'(x)$ on the interval $-3 \leq x \leq 7$. Does the Mean Value Theorem applied on the interval $-3 \leq x \leq 7$ guarantee a value of c , for $-3 < c < 7$, such that $g''(c)$ is equal to this average rate of change? Why or why not?

Part A

1 point is earned for x -values

1 point is earned for the justification

g' changes from increasing to decreasing at $x = 1$;

g' changes from decreasing to increasing at $x = 4$.

Points of inflection for the graph of $y = g(x)$ occur at $x = 1$ and $x = 4$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for x -values

1 point is earned for the justification

g' changes from increasing to decreasing at $x = 1$;

g' changes from decreasing to increasing at $x = 4$.

5-5b

Points of inflection for the graph of $y = g(x)$ occur at $x = 1$ and $x = 4$.

Part B

1 point is earned if identifies $x = 2$ as a candidate

1 point is earned if considers endpoints

1 point is earned for maximum value and justification

The only sign change of g' from positive to negative in the interval is at $x = 2$.

$$g(-3) = 5 + \int_2^{-3} g'(x) dx = 5 + \left(-\frac{3}{2}\right) + 4 = \frac{15}{2}$$

$$g(2) = 5$$

$$g(7) = 5 + \int_2^7 g'(x) dx = 5 + (-4) + \frac{1}{2} = \frac{3}{2}$$

The maximum value of g for $-3 \leq x \leq 7$ is $\frac{15}{2}$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if identifies $x = 2$ as a candidate

1 point is earned if considers endpoints

1 point is earned for maximum value and justification

The only sign change of g' from positive to negative in the interval is at $x = 2$.

$$g(-3) = 5 + \int_2^{-3} g'(x) dx = 5 + \left(-\frac{3}{2}\right) + 4 = \frac{15}{2}$$

$$g(2) = 5$$

$$g(7) = 5 + \int_2^7 g'(x) dx = 5 + (-4) + \frac{1}{2} = \frac{3}{2}$$

The maximum value of g for $-3 \leq x \leq 7$ is $\frac{15}{2}$.

Part C

5-5b

1 point is earned for the difference quotient

1 point is earned for the answer

$$\frac{g(7) - g(-3)}{7 - (-3)} = \frac{\frac{3}{2} - \frac{15}{2}}{10} = -\frac{3}{5}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the difference quotient

1 point is earned for the answer

$$\frac{g(7) - g(-3)}{7 - (-3)} = \frac{\frac{3}{2} - \frac{15}{2}}{10} = -\frac{3}{5}$$

Part D

1 point is earned for average value of $g'(x)$

1 point is earned for answer "No" with reason

$$\frac{g'(7) - g'(-3)}{7 - (-3)} = \frac{1 - (-4)}{10} = \frac{1}{2}$$

No, the MVT does *not* guarantee the existence of a value c with the stated properties because g' is not differentiable for at least one point in $-3 < x < 7$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for average value of $g'(x)$

1 point is earned for answer "No" with reason

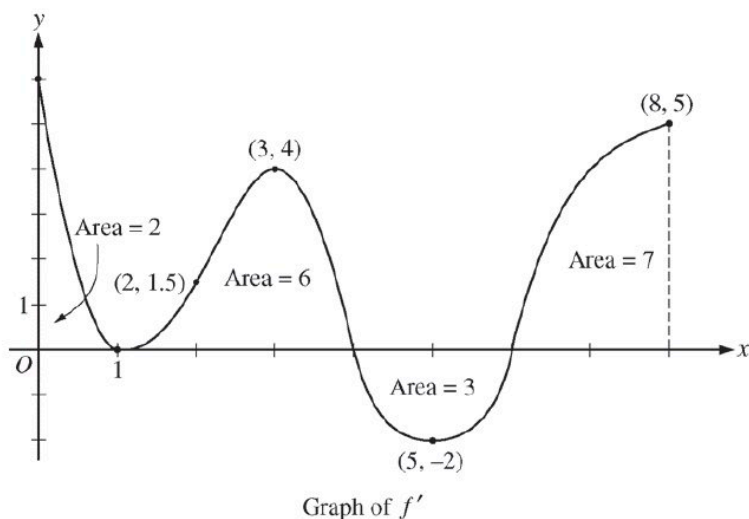
5-5b

$$\frac{g'(7) - g'(-3)}{7 - (-3)} = \frac{1 - (-4)}{10} = \frac{1}{2}$$

No, the MVT does *not* guarantee the existence of a value c with the stated properties because g' is not differentiable for at least one point in $-3 < x < 7$.

5-5b

72.



The figure above shows the graph of f' , the derivative of a twice-differentiable function f , on the closed interval $0 \leq x \leq 8$. The graph of f' has horizontal tangent lines at $x = 1$, $x = 3$, and $x = 5$. The areas of the regions between the graph of f' and the x -axis are labeled in the figure. The function f is defined for all real numbers and satisfies $f(8) = 4$.

- Find all values of x on the open interval $0 < x < 8$ for which the function f has a local minimum. Justify your answer.
- Determine the absolute minimum value of f on the closed interval $0 \leq x \leq 8$. Justify your answer.
- On what open intervals contained in $0 < x < 8$ is the graph of f both concave down and increasing? Explain your reasoning.
- The function g is defined by $g(x) = (f(x))^3$. If $f(3) = -5/2$, find the slope of the line tangent to the graph of g at $x = 3$.

Part A

1 point is earned for the answer with justification

$x = 6$ is the only critical point at which f' changes sign from negative to positive. Therefore, f has a local minimum at $x = 6$.



0

1

The student response earns all of the following points:

1 point is earned for the answer with justification

$x = 6$ is the only critical point at which f' changes sign from negative to positive. Therefore, f has a local minimum at $x = 6$.

Part B

5-5b

1 point is earned for considering $x = 0$ and $x = 6$

1 point is earned for the answer

1 point is earned for justification

From part (a), the absolute minimum occurs either at $x = 6$ or at an endpoint.

$$\begin{aligned} f(0) &= f(8) + \int_8^0 f'(x) dx \\ &= f(8) - \int_0^8 f'(x) dx = 4 - 12 = -8 \end{aligned}$$

$$\begin{aligned} f(6) &= f(8) + \int_8^6 f'(x) dx \\ &= f(8) - \int_6^8 f'(x) dx = 4 - 7 = -3 \end{aligned}$$

$$f(8) = 4$$

The absolute minimum value of f on the closed interval $[0, 8]$ is -8 .



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $x = 0$ and $x = 6$

1 point is earned for the answer

1 point is earned for justification

From part (a), the absolute minimum occurs either at $x = 6$ or at an endpoint.

$$\begin{aligned} f(0) &= f(8) + \int_8^0 f'(x) dx \\ &= f(8) - \int_0^8 f'(x) dx = 4 - 12 = -8 \end{aligned}$$

$$\begin{aligned} f(6) &= f(8) + \int_8^6 f'(x) dx \\ &= f(8) - \int_6^8 f'(x) dx = 4 - 7 = -3 \end{aligned}$$

5-5b

$$f(8) = 4$$

The absolute minimum value of f on the closed interval $[0, 8]$ is -8.

Part C

1 point is earned for the answer

1 point is earned for the explanation

The graph of f is concave down and increasing on $0 < x < 1$ and $3 < x < 4$, because f' is decreasing and positive on these intervals.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the explanation

The graph of f is concave down and increasing on $0 < x < 1$ and $3 < x < 4$, because f' is decreasing and positive on these intervals.

Part D

2 points are earned for $g'(x)$

1 point is earned for the answer

$$g'(x) = 3[f(x)]^2 \cdot f'(x)$$

$$g'(3) = 3[f(3)]^2 \cdot f'(3) = 3\left(-\frac{5}{2}\right)^2 \cdot 4 = 75$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $g'(x)$

1 point is earned for the answer

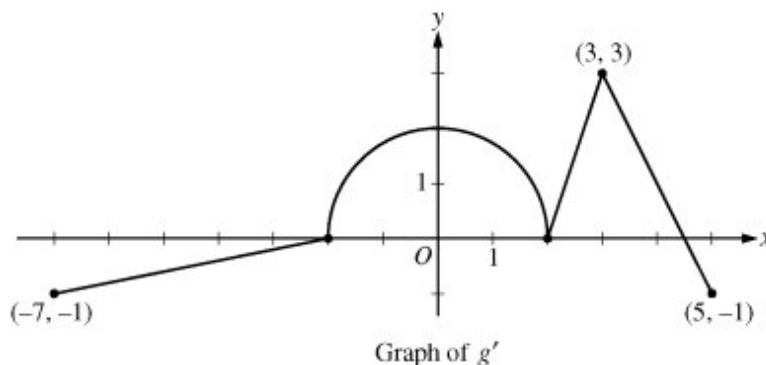
5-5b

$$g'(x) = 3[f(x)]^2 \cdot f'(x)$$

$$g'(3) = 3[f(3)]^2 \cdot f'(3) = 3\left(-\frac{5}{2}\right)^2 \cdot 4 = 75$$

5-5b

73.



The function g is defined and differentiable on the closed interval $[-7, 5]$ and satisfies $g(0) = 5$. The graph of $y = g'(x)$, the derivative of g , consists of a semicircle and three line segments, as shown in the figure above.

- (a) Find $g(3)$ and $g(-2)$.
- (b) Find the x -coordinate of each point of inflection of the graph of $y = g(x)$ on the interval $-7 < x < 5$. Explain your reasoning.
- (c) The function h is defined by $h(x) = g(x) - \frac{1}{2}x^2$. Find the x -coordinate of each critical point of h , where $-7 < x < 5$, and classify each critical point as the location of a relative minimum, relative maximum, or neither a minimum nor a maximum. Explain your reasoning.

Part A

1 point is earned for using $g(0) = 5$

1 point is earned for $g(3)$

1 point is earned for $g(-2)$

$$g(3) = 5 + \int_0^3 g'(x) dx = 5 + \frac{\pi \cdot 2^2}{2} + \frac{3}{2} = \frac{13}{2} + \pi$$

$$g(-2) = 5 + \int_0^{-2} g'(x) dx = 5 - \pi$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for using $g(0) = 5$

1 point is earned for $g(3)$

1 point is earned for $g(-2)$

5-5b

$$g(3) = 5 + \int_0^3 g'(x)dx = 5 + \frac{\pi \cdot 2^2}{2} + \frac{3}{2} = \frac{13}{2} + \pi$$

$$g(-2) = 5 + \int_0^{-2} g'(x)dx = 5 - \pi$$

Part B

1 point is earned for identifying $x = 0, 2, 3$

1 point is earned for explanation

The graph of $y = g(x)$ has points of inflection at $x = 0, x = 2$, and $x = 3$ because g' changes from increasing to decreasing at $x = 0$ and $x = 3$, and g' changes from decreasing to increasing at $x = 2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for identifying $x = 0, 2, 3$

1 point is earned for explanation

The graph of $y = g(x)$ has points of inflection at $x = 0, x = 2$, and $x = 3$ because g' changes from increasing to decreasing at $x = 0$ and $x = 3$, and g' changes from decreasing to increasing at $x = 2$.

Part C

1 point is earned for $h'(x)$

1 point is earned for identifying $x = \sqrt{2}, 3$

1 point is earned for answer for $\sqrt{2}$ with analysis

1 point is earned for answer for 3 with analysis

$$h'(x) = g'(x) - x = 0 \Rightarrow g'(x) = x$$

On the interval $-2 \leq x \leq 2$, $g'(x) = \sqrt{4 - x^2}$.

5-5b

On this interval, $g'(x) = x$ when $x = \sqrt{2}$.

The only other solution to $g'(x) = x$ is $x = 3$.

$$h'(x) = g'(x) - x > 0 \text{ for } 0 \leq x < \sqrt{2}$$

$$h'(x) = g'(x) - x \leq 0 \text{ for } \sqrt{2} < x \leq 5$$

Therefore h has a relative maximum at $x = \sqrt{2}$, and h has neither a minimum nor a maximum at $x = 3$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $h'(x)$

1 point is earned for identifying $x = \sqrt{2}$, 3

1 point is earned for answer for $\sqrt{2}$ with analysis

1 point is earned for answer for 3 with analysis

$$h'(x) = g'(x) - x = 0 \Rightarrow g'(x) = x$$

On the interval $-2 \leq x \leq 2$, $g'(x) = \sqrt{4 - x^2}$.

On this interval, $g'(x) = x$ when $x = \sqrt{2}$.

The only other solution to $g'(x) = x$ is $x = 3$.

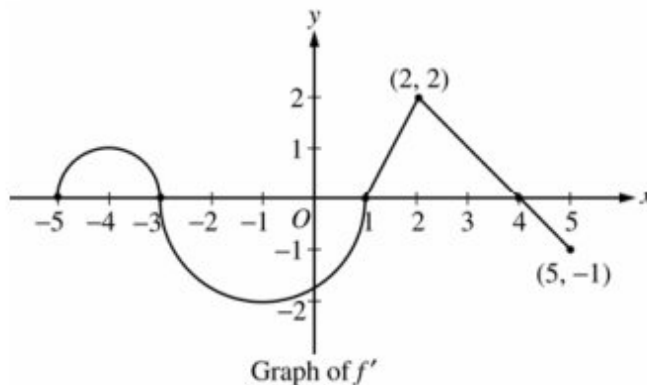
$$h'(x) = g'(x) - x > 0 \text{ for } 0 \leq x < \sqrt{2}$$

$$h'(x) = g'(x) - x \leq 0 \text{ for } \sqrt{2} < x \leq 5$$

Therefore h has a relative maximum at $x = \sqrt{2}$, and h has neither a minimum nor a maximum at $x = 3$.

5-5b

74.



Let f be a function defined on the closed interval $-5 \leq x \leq 5$ with $f(1) = 3$. The graph of f' the derivative of f , consists of two semicircles and two line segments, as shown above.

- For $-5 < x < 5$, find all values x at which f has a relative maximum. Justify your answer.
- For $-5 < x < 5$, find all values x at which the graph of f has a point of inflection. Justify your answer.
- Find all intervals on which the graph of f is concave up and also has positive slope. Explain your reasoning.
- Find the absolute minimum value of $f(x)$ over the closed interval $-5 \leq x \leq 5$. Explain your reasoning.

Part A

1 point is earned for the x -values

1 point is earned for the justification

$$f'(x) = 0 \text{ at } x = -3, 1, 4$$

f' changes from positive to negative at -3 and 4.

Thus, f has a relative maximum at $x = -3$ and at $x = 4$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the x -values

1 point is earned for the justification

$$f'(x) = 0 \text{ at } x = -3, 1, 4$$

f' changes from positive to negative at -3 and 4.

5-5b

Thus, f has a relative maximum at $x = -3$ and at $x = 4$.

Part B

1 point is earned for the x -values

1 point is earned for the justification

f' changes from increasing to decreasing, or vice versa, at $x = -4$, -1 , and 2 . Thus, the graph of f has points of inflection when $x = -4$, -1 , and 2 .



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the x -values

1 point is earned for the justification

f' changes from increasing to decreasing, or vice versa, at $x = -4$, -1 , and 2 . Thus, the graph of f has points of inflection when $x = -4$, -1 , and 2 .

Part C

1 point is earned if considers intervals

1 point is earned for value and explanation

The graph of f is concave up with positive slope where f' is increasing and positive: $-5 < x < -4$ and $1 < x < 2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned if considers intervals

1 point is earned for value and explanation

The graph of f is concave up with positive slope where f' is increasing and positive: $-5 < x < -4$ and $1 < x < 2$.

Part D

1 point is earned if identifies $x = 1$ as a candidate

5-5b

1 point is earned for considering endpoint

1 point is earned for value and explanation

Candidates for the absolute minimum are where f' changes from negative to positive (at $x = 1$) and at the endpoints ($x = -5, 5$).

$$f(-5) = 3 + \int_1^{-5} f(x) \, dx = 3 - \frac{\pi}{2} + 2\pi > 3$$

$$f(1) = 3$$

$$f(5) = 3 + \int_1^{-5} f'(x) \, dx = 3 + \frac{3 \cdot 2}{2} - \frac{1}{2} > 3$$

The absolute minimum value of f on $[-5, 5]$ is $f(1) = 3$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if identifies $x = 1$ as a candidate

1 point is earned for considering endpoint

1 point is earned for value and explanation

Candidates for the absolute minimum are where f' changes from negative to positive (at $x = 1$) and at the endpoints ($x = -5, 5$).

$$f(-5) = 3 + \int_1^{-5} f(x) \, dx = 3 - \frac{\pi}{2} + 2\pi > 3$$

$$f(1) = 3$$

$$f(5) = 3 + \int_1^{-5} f'(x) \, dx = 3 + \frac{3 \cdot 2}{2} - \frac{1}{2} > 3$$

The absolute minimum value of f on $[-5, 5]$ is $f(1) = 3$.

5-5b

75. Consider the function f defined by $f(x) = e^x \cos x$ with domain $[0, 2\pi]$.
- (a) Find the absolute maximum and minimum values of $f(x)$.
 - (b) Find the intervals on which f is increasing.
 - (c) Find the x -coordinate of each point of inflection of the graph of f .

Part A

1 point is earned for derivative

1 point is earned for finding critical values

1 point is earned for evaluating at critical points and ends

1 point is earned for choosing absolute max

1 point is earned for absolute min

$$\begin{aligned} f'(x) &= -e^x \sin x + e^x \cos x \\ &= e^x [\cos x - \sin x] \end{aligned}$$

$$f'(x) = 0 \text{ when } \sin x = \cos x, x = \frac{\pi}{4}, \frac{5\pi}{4}$$

x	$f(x)$
0	1
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2} e^{\pi/4}$
$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2} e^{5\pi/4}$
2π	$e^{2\pi}$

$$\text{Max: } e^{2\pi}; \text{ Min: } -\frac{\sqrt{2}}{2} e^{5\pi/4}$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for derivative

5-5b

1 point is earned for finding critical values

1 point is earned for evaluating at critical points and ends

1 point is earned for choosing absolute max

1 point is earned for absolute min

$$\begin{aligned} f'(x) &= -e^x \sin x + e^x \cos x \\ &= e^x [\cos x - \sin x] \end{aligned}$$

$$f'(x) = 0 \text{ when } \sin x = \cos x, x = \frac{\pi}{4}, \frac{5\pi}{4}$$

x	$f(x)$
0	1
$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2} e^{\pi/4}$
$\frac{5\pi}{4}$	$-\frac{\sqrt{2}}{2} e^{5\pi/4}$
2π	$e^{2\pi}$

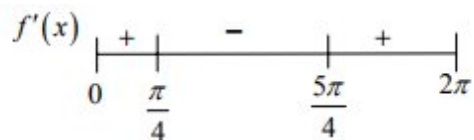
$$\text{Max: } e^{2\pi}; \text{ Min: } -\frac{\sqrt{2}}{2} e^{5\pi/4}$$

Part B

1 point is earned for positive intervals

1 point is earned for negative intervals

1 point is **deducted** for intervals outside of domain



Increasing on $[0, \frac{\pi}{4}]$, $[\frac{5\pi}{4}, 2\pi]$

5-5b

✓

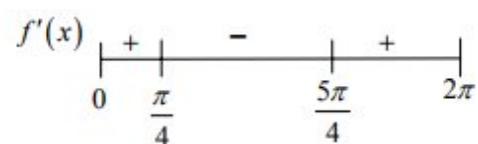
0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for positive intervals

1 point is earned for negative intervals

1 point is **deducted** for intervals outside of domain



Increasing on $[0, \frac{\pi}{4}]$, $[\frac{5\pi}{4}, 2\pi]$

Part C

1 point is earned for $f''(x)$

1 point is earned for answer

$$\begin{aligned} f''(x) &= e^x [-\sin x - \cos x] + e^x [\cos x - \sin x] \\ &= -2e^x \sin x \end{aligned}$$

$$f''(x) = 0 \text{ when } x = 0, \pi, 2\pi$$

Point of inflection at $x = \pi$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f''(x)$

1 point is earned for answer

$$\begin{aligned} f''(x) &= e^x [-\sin x - \cos x] + e^x [\cos x - \sin x] \\ &= -2e^x \sin x \end{aligned}$$

5-5b $f''(x) = 0$ when $x = 0, \pi, 2\pi$ Point of inflection at $x = \pi$

5-5b

76. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

From 5 A.M. to 10 A.M., the rate at which vehicles arrive at a certain toll plaza is given by $A(t) = 450\sqrt{\sin(0.62t)}$, where t is the number of hours after 5 A.M. and $A(t)$ is measured in vehicles per hour. Traffic is flowing smoothly at 5 A.M. with no vehicles waiting in line.

(a) Write, but do not evaluate, an integral expression that gives the total number of vehicles that arrive at the toll plaza from 6 A.M. ($t = 1$) to 10 A.M. ($t = 5$).

(b) Find the average value of the rate, in vehicles per hour, at which vehicles arrive at the toll plaza from 6 A.M. ($t = 1$) to 10 A.M. ($t = 5$).

(c) Is the rate at which vehicles arrive at the toll plaza at 6 A.M. ($t = 1$) increasing or decreasing? Give a reason for your answer.

(d) A line forms whenever $A(t) \geq 400$. The number of vehicles in line at time t , for $a \leq t \leq 4$, is given by $N(t) = \int_a^t (A(x) - 400) dx$, where a is the time when a line first begins to form. To the nearest whole number, find the greatest number of vehicles in line at the toll plaza in the time interval $a \leq t \leq 4$. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must be a definite integral with correct lower and upper limits to earn this point.

Because $|A(t)| = A(t)$ for $1 \leq t \leq 5$, a response of $\int_1^5 450\sqrt{\sin(0.62t)} dt$ or $\int_1^5 |A(t)| dt$ earns the point.

A response missing dt or using dx is eligible to earn the point.

A response with a copy error in the expression for $A(t)$ will earn the point only in the presence of $\int_1^5 A(t) dt$.

5-5b



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

The total number of vehicles that arrive at the toll plaza from 6 a.m. to 10 a.m. is given by $\int_1^5 A(t) dt$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The use of the average value formula, indicating that $a = 1$ and $b = 5$, can be presented in single or multiple steps to earn the first point. For example, the following response earns both points: $\int_1^5 A(t) dt = 1502.147865$, so the average value is 375.536966 .

A response that presents a correct integral along with the correct average value, but provides incorrect or incomplete communication, earns 1 out of 2 points. For example, the following response earns 1 out of 2 points:

$\int_1^5 A(t) dt = 1502.147865 = 375.536966$. so the average value is 375.536966 .

The answer must be correct to three decimal places. For example, $\frac{1}{5-1} \int_1^5 A(t) dt = 375.536966 \approx 376$ earns only the first point.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is generally eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode,

$\frac{1}{4} \int_1^5 A(t) dt = 79.416068$.

Special case: $\frac{1}{5} \int_1^5 A(t) dt = 300.429573$ earns 1 out of 2 points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Uses average value formula: $\frac{1}{b-a} \int_a^b A(t) dt$

☐ Answer

5-5b

Solution:

$$\text{Average} = \frac{1}{5-1} \int_1^5 A(t) dt = 375.536966$$

The average rate at which vehicles arrive at the toll plaza from 6 a.m. to 10 a.m. is 375.537 (or 375.536) vehicles per hour.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response need not present the value of $A'(1)$. The second line of the model solution earns both points.

An incorrect value assigned to $A'(1)$ earns the first point (but will not earn the second point).

Without a reference to $t = 1$, the first point is earned by any of the following:

- 148.947 accurate to the number of decimals presented, with zero up to three decimal places (i.e., 149, 148, 148.9, 148.95, or 148.94)
- $A'(t) = 148.947$ by itself

To be eligible for the second point, the first point must be earned.

To earn the second point, there must be a reference to $t = 1$.

Degree mode: $A'(1) = 23.404311$



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $A'(1)$
- ☐ Answer with reason

Solution:

$$A'(1) = 148.947272$$

Because $A'(1) > 0$, the rate at which the vehicles arrive at the toll plaza is increasing.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-5b

It is not necessary to indicate that $A(t) = 400$ to earn the first point, although this statement alone would earn the first point.

A response of “ $A(t) \geq 400$ when $1.469372 \leq t \leq 3.597713$ ” will earn the first 2 points. A response of “ $A(t) \geq 400$ ” along with the presence of exactly one of the two numbers above will earn the first point, but not the second. A response of “ $A(t) \geq 400$ ” by itself will not earn either of the first 2 points.

To earn the second point the values for a and b must be accurate to the number of decimals presented, with at least one and up to three decimal places. These may appear only in a candidates table, as limits of integration, or on a number line.

A response with incorrect notation involving t or x is eligible to earn all 4 points.

A response that does not earn the first point is still eligible for the remaining 3 points.

To earn the third point, a response must present the greatest number of vehicles. This point is earned for answers of either 71 or 71.254*** only.

A correct justification earns the fourth point, even if the third point is not earned because of a decimal presentation error.

When using a Candidates Test, the response must include the values for $N(a)$, $N(b)$, and $N(4)$ to earn the fourth point. These values must be correct to the number of decimals presented, with up to three decimal places. (Correctly rounded integer values are acceptable.)

Alternate solution for the third and fourth points:

For $a \leq t \leq b$, $A(t) \geq 400$. For $b \leq t \leq 4$, $A(t) \leq 400$.

Thus, $N(t) = \int_a^t (A(x) - 400)dx$ is greatest at $t = b$.

$N(b) = 71.254129$, and the greatest number of vehicles in line is 71.

Degree mode: The response is only eligible to earn the first point because in degree mode $A(t) < 400$.



0	1	2	3	4
---	---	---	---	---

The student response accurately includes **all four** of the criteria below.

- ☐ Considers $N'(t) = 0$
- ☐ $t = a$ and $t = b$
- ☐ Answer
- ☐ Justification

5-5b

Solution:

$$N'(t) = A(t) - 400 = 0$$

$$\Rightarrow A(t) = 400 \Rightarrow t = 1.469372, t = 3.597713$$

$$a = 1.469372$$

$$b = 3.597713$$

t	$N(t) = \int_a^t (A(x) - 400) dx$
a	0
b	71.254129
4	62.338346

The greatest number of vehicles in line is 71.

5-5b

77.  For $0 \leq t \leq 31$, the rate of change of the number of mosquitoes on Tropical Island at time t days is modeled by $R(t) = 5\sqrt{t} \cos\left(\frac{t}{5}\right)$ mosquitoes per day. There are 1000 mosquitoes on Tropical Island at time $t = 0$.
- Show that the number of mosquitoes is increasing at time $t = 6$.
 - At time $t = 6$, is the number of mosquitoes increasing at an increasing rate, or is the number of mosquitoes increasing at a decreasing rate? Give a reason for your answer.
 - According to the model, how many mosquitoes will be on the island at time $t = 31$? Round your answer to the nearest whole number.
 - To the nearest whole number, what is the maximum number of mosquitoes for $0 \leq t \leq 31$? Show the analysis that leads to your conclusion.

Part A

1 point is earned for showing that $R(6) > 0$

Since $R(6) = 4.438 > 0$, the number of mosquitoes is increasing at $t = 6$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for showing that $R(6) > 0$

Since $R(6) = 4.438 > 0$, the number of mosquitoes is increasing at $t = 6$.

Part B

1 point is earned for considering $R'(6)$

1 point is earned for the answer with reason

$$R'(6) = -1.913$$

Since $R'(6) < 0$, the number of mosquitoes is increasing at a decreasing rate at $t = 6$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considering $R'(6)$

1 point is earned for the answer with reason

5-5b

$$R'(6) = -1.913$$

Since $R'(6) < 0$, the number of mosquitoes is increasing at a decreasing rate at $t = 6$.

Part C

1 point is earned for the integral

1 point is earned for the answer

$$1000 + \int_0^{31} R(t) dt = 964.335$$

To the nearest whole number, there are 964 mosquitoes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$1000 + \int_0^{31} R(t) dt = 964.335$$

To the nearest whole number, there are 964 mosquitoes.

Part D

2 points are earned for absolute maximum value where 1 point is earned for the integral and 1 point is earned for the answer

2 points are earned for analysis where 1 point is earned for computing interior critical points and 1 point is earned for completing analysis

$$R(t) = 0 \text{ when } t = 0, t = 2.5\pi, \text{ or } t = 7.5\pi$$

$$R(t) > 0 \text{ on } 0 < t < 2.5\pi$$

$$R(t) < 0 \text{ on } 2.5\pi < t < 7.5\pi$$

$$R(t) > 0 \text{ on } 7.5\pi < t < 31$$

The absolute maximum number of mosquitoes occurs at $t = 2.5\pi$ or at $t = 31$.

5-5b

$$1000 + \int_0^{2.5\pi} R(t) dt = 1039.357,$$

There are 964 mosquitoes at $t = 31$, so the maximum number of mosquitoes is 1039, to the nearest whole number.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for absolute maximum value where 1 point is earned for the integral and 1 point is earned for the answer

2 points are earned for analysis where 1 point is earned for computing interior critical points and 1 point is earned for completing analysis

$$R(t) = 0 \text{ when } t = 0, t = 2.5\pi, \text{ or } t = 7.5\pi$$

$$R(t) > 0 \text{ on } 0 < t < 2.5\pi$$

$$R(t) < 0 \text{ on } 2.5\pi < t < 7.5\pi$$

$$R(t) > 0 \text{ on } 7.5\pi < t < 31$$

The absolute maximum number of mosquitoes occurs at $t = 2.5\pi$ or at $t = 31$.

$$1000 + \int_0^{2.5\pi} R(t) dt = 1039.357,$$

There are 964 mosquitoes at $t = 31$, so the maximum number of mosquitoes is 1039, to the nearest whole number.

5-5b

78. Let f be the function defined by $f(x) = \ln(2 + \sin x)$ for $\pi \leq x \leq 2\pi$.

(a) Find the absolute maximum value and the absolute minimum value of f . Show the analysis that leads to your conclusion.

(b) Find the x -coordinate of each inflection point on the graph of f . Justify your answer.

Part A

1 point is earned for $f'(x)$

1 point is earned for solutions for $f'(x) = 0$ which includes at least one solution

1 point is earned for evaluation of f at π and 2π

1 point is earned for evaluation of f at student's critical point(s)

1 point is earned for answer

$$f'(x) = \frac{1}{2 + \sin x} \cos x; \text{ In } [\pi, 2\pi], \cos x = 0 \text{ when } x = 3\pi/2;$$

x	$f(x)$
π	$\ln(2) = 0.693$
2π	$\ln(2)$
$\frac{3\pi}{2}$	$\ln(1) = 0$

absolute maximum value is $\ln 2$

absolute minimum value is 0



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $f'(x)$

1 point is earned for solutions for $f'(x) = 0$ which includes at least one solution

1 point is earned for evaluation of f at π and 2π

1 point is earned for evaluation of f at student's critical point(s)

5-5b

1 point is earned for answer

$$f'(x) = \frac{1}{2 + \sin x} \cos x; \text{ In } [\pi, 2\pi], \cos x = 0 \text{ when } x = 3\pi/2;$$

x	$f'(x)$
π	$\ln(2) = 0.693$
2π	$\ln(2)$
$\frac{3\pi}{2}$	$\ln(1) = 0$

absolute maximum value is $\ln 2$

absolute minimum value is 0

Part B

1 point is earned for $f''(x)$

2 points are earned for solution for $f''(x) = 0$

1 point is earned for specific justification

$$\begin{aligned} f''(x) &= \frac{(-\sin x)(2 + \sin x) - \cos x \cos x}{(2 + \sin x)^2} \\ &= \frac{-2 \sin x - 1}{(2 + \sin x)^2}; \end{aligned}$$

$$f''(x) = 0 \text{ when } \sin x = -1/2$$

$$x = 7\pi/6, 11\pi/6$$

sign of f''	-	+	-
concavity	down	up	down
	π	$\frac{7\pi}{6}$	$\frac{11\pi}{6}$
			2π

$x = 7\pi/6$ and $11\pi/6$ since concavity changes as indicated at these points

5-5b



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $f''(x)$

2 points are earned for solution for $f''(x) = 0$

1 point is earned for specific justification

$$f''(x) = \frac{(-\sin x)(2 + \sin x) - \cos x \cos x}{(2 + \sin x)^2}$$

$$= \frac{-2 \sin x - 1}{(2 + \sin x)^2};$$

$$f''(x) = 0 \text{ when } \sin x = -1/2$$

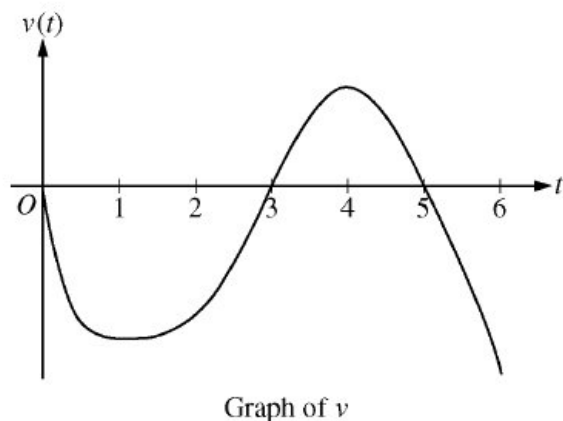
$$x = 7\pi/6, 11\pi/6$$

sign of f''	-	+	-
concavity	down	up	down
	π	$\frac{7\pi}{6}$	$\frac{11\pi}{6}$
			2π

$x = 7\pi/6$ and $11\pi/6$ since concavity changes as indicated at these points

5-5b

79.



A particle moves along the x -axis so that its velocity at time t , for $0 \leq t \leq 6$, is given by a differentiable function v whose graph is shown above. The velocity is 0 at $t = 0$, $t = 3$, and $t = 5$, and the graph has horizontal tangents at $t = 1$ and $t = 4$. The areas of the regions bounded by the t -axis and the graph of v on the intervals $[0,3]$, $[3,5]$, and $[5,6]$ are 8, 3, and 2 respectively. At time $t = 0$, the particle is $x = -2$.

- (a) For $0 \leq t \leq 6$, find *both the time and the position of the particle when the particle is farthest to the left. Justify your answer.*
- (b) For how many values of t , where $0 \leq t \leq 6$, is the particle at $x = -8$? Explain your reasoning.
- (c) On the interval $2 < t < 3$, is the speed of the particle increasing or decreasing? Give a reason for your answer.
- (d) During what time intervals, if any, is the acceleration of the particle negative? Justify your answer.

Part A

1 point is earned for identifying $t = 3$ as a candidate

1 point is earned for considering $\int_0^6 v(t) dt$

1 point is earned for conclusion

Since $v(t) < 0$ for $0 < t < 3$ and $5 < t < 6$, and $v(t) > 0$ for $3 < t < 5$, we consider $t = 3$ and $t = 6$.

$$x(3) = -2 + \int_0^3 v(t) dt = -2 - 8 = -10$$

$$x(6) = -2 + \int_0^6 v(t) dt = -2 - 8 + 3 - 2 = -9$$

Therefore, the particle is farthest left at time $t = 3$ when its position is $x(3) = -10$.

5-5b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for identifying $t = 3$ as a candidate

1 point is earned for considering $\int_0^6 v(t)dt$

1 point is earned for conclusion

Since $v(t) < 0$ for $0 < t < 3$ and $5 < t < 6$, and $v(t) > 0$ for $3 < t < 5$, we consider $t = 3$ and $t = 6$.

$$x(3) = -2 + \int_0^3 v(t)dt = -2 - 8 = -10$$

$$x(6) = -2 + \int_0^6 v(t)dt = -2 - 8 + 3 - 2 = -9$$

Therefore, the particle is farthest left at time $t = 3$ when its position is $x(3) = -10$.

Part B

1 point for positioning at $t = 3$, $t = 5$, and $t = 6$

1 point for description of motion

1 point for conclusion

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

By the Intermediate Value Theorem, there are three values of t for which the particle is at $x(t) = -8$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point for positioning at $t = 3$, $t = 5$, and $t = 6$

1 point for description of motion

1 point for conclusion

5-5b

The particle moves continuously and monotonically from $x(0) = -2$ to $x(3) = -10$. Similarly, the particle moves continuously and monotonically from $x(3) = -10$ to $x(5) = -7$ and also from $x(5) = -7$ to $x(6) = -9$.

By the Intermediate Value Theorem, there are three values of t for which the particle is at $x(t) = -8$.

Part C

1 point is earned for answer with reason

The speed is decreasing on the interval $2 < t < 3$ since on this interval $v < 0$ and v is increasing.



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer with reason

The speed is decreasing on the interval $2 < t < 3$ since on this interval $v < 0$ and v is increasing.

Part D

1 point for answer

1 point for justification

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.



0	1	2
---	---	---

The student response earns all of the following points:

1 point for answer

1 point for justification

The acceleration is negative on the intervals $0 < t < 1$ and $4 < t < 6$ since velocity is decreasing on these intervals.

5-5b

80.  A particle moves along the x -axis so that its velocity at time t is given by

$$v(t) = -(t + 1) \sin\left(\frac{t^2}{2}\right).$$

At time $t = 0$, the particle is at position $x = 1$.

- (a) Find the acceleration of the particle at time $t = 2$. Is the speed of the particle increasing at $t = 2$? Why or why not?
- (b) Find all times t in the open interval $0 < t < 3$ when the particle changes direction. Justify your answer.
- (c) Find the total distance traveled by the particle from time $t = 0$ until time $t = 3$.
- (d) During the time interval $0 \leq t \leq 3$, what is the greatest distance between the particle and the origin? Show the work that leads to your answer.

Part A

1 point is earned for correctly solving for $a(2)$

$$a(2) = v'(2) = 1.587 \text{ or } 1.588$$

$$v(2) = -3 \sin(2) < 0$$

1 point is earned for the reason for the speed decreasing

Speed is decreasing since $a(2) > 0$ and $v(2) < 0$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly solving for $a(2)$

$$a(2) = v'(2) = 1.587 \text{ or } 1.588$$

$$v(2) = -3 \sin(2) < 0$$

1 point is earned for the reason for the speed decreasing

Speed is decreasing since $a(2) > 0$ and $v(2) < 0$.

Part B

1 point is earned for the correct answer $t = \sqrt{2\pi}$ only

1 point is earned for the justification

5-5b

$$v(t) = 0 \text{ when } \frac{t^2}{2} = \pi$$

$$t = \sqrt{2\pi} \text{ or } 2.506 \text{ or } 2.507$$

Since $v(t) < 0$ for $0 < t < \sqrt{2\pi}$ and $v(t) > 0$ for $\sqrt{2\pi} < t < 3$, the particle changes directions at $t = \sqrt{2\pi}$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer $t = \sqrt{2\pi}$ only

1 point is earned for the justification

$$v(t) = 0 \text{ when } \frac{t^2}{2} = \pi$$

$$t = \sqrt{2\pi} \text{ or } 2.506 \text{ or } 2.507$$

Since $v(t) < 0$ for $0 < t < \sqrt{2\pi}$ and $v(t) > 0$ for $\sqrt{2\pi} < t < 3$, the particle changes directions at $t = \sqrt{2\pi}$.

Part C

1 point is earned for the correct limits

1 point is earned for the correct integrand

1 point is earned for the correct answer

$$\text{Distance} = \int_0^3 |v(t)| dt = 4.333 \text{ or } 4.334$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct limits

1 point is earned for the correct integrand

1 point is earned for the correct answer

5-5b

$$\text{Distance} = \int_0^3 |v(t)| dt = 4.333 \text{ or } 4.334$$

Part D

1 point is earned for the $\pm$ distance that the particle travels while velocity is negative

$$\int_0^{\sqrt{2\pi}} v(t) dt = -3.265$$

$$x(\sqrt{2\pi}) = x(0) + \int_0^{\sqrt{2\pi}} v(t) dt = -2.265$$

1 point is earned for the correct answer

Since the total distance from $t = 0$ to $t = 3$ is 4.334, the particle is still to the left of the origin at $t = 3$. Hence the greatest distance from the origin is 2.265.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the $\pm$ distance that the particle travels while velocity is negative

$$\int_0^{\sqrt{2\pi}} v(t) dt = -3.265$$

$$x(\sqrt{2\pi}) = x(0) + \int_0^{\sqrt{2\pi}} v(t) dt = -2.265$$

1 point is earned for the correct answer

Since the total distance from $t = 0$ to $t = 3$ is 4.334, the particle is still to the left of the origin at $t = 3$. Hence the greatest distance from the origin is 2.265.

5-5b

81. A particle moves along the x -axis with position at time t given by $x(t) = e^{-t} \sin t$ for $0 \leq t \leq 2\pi$.

(a) Find the time t at which the particle is farthest to the left. Justify your answer.

(b) Find the value of the constant A for which $x(t)$ satisfies the equation $Ax''(t) + x'(t) + x(t) = 0$ for $0 < t < 2\pi$.

Part A

2 points are earned for $x'(t)$

1 point is earned for setting $x'(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

$$x'(t) = -e^{-t} \sin t + e^{-t} \cos t = e^{-t} (\cos t - \sin t)$$

$x'(t) = 0$ when $\cos t = \sin t$. Therefore, $x'(t) = 0$ on

$$0 \leq t \leq 2\pi \text{ for } t = \frac{\pi}{4} \text{ and } t = \frac{5\pi}{4}.$$

The candidates for the absolute minimum are at $t = 0, \frac{\pi}{4}, \frac{5\pi}{4}$, and 2π .

t	$x(t)$
0	$e^0 \sin(0) = 0$
$\frac{\pi}{4}$	$e^{-\frac{\pi}{4}} \sin\left(\frac{\pi}{4}\right) > 0$
$\frac{5\pi}{4}$	$e^{-\frac{5\pi}{4}} \sin\left(\frac{5\pi}{4}\right) < 0$
2π	$e^{-2\pi} \sin(2\pi) = 0$

The particle is farthest to the left when $t = \frac{5\pi}{4}$.



0	1	2	3	4	5
---	---	---	---	---	---

5-5b

The student response earns all of the following points:

2 points are earned for $x'(t)$

1 point is earned for setting $x'(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

$$x'(t) = -e^{-t} \sin t + e^{-t} \cos t = e^{-t} (\cos t - \sin t)$$

$x'(t) = 0$ when $\cos t = \sin t$. Therefore, $x'(t) = 0$ on

$$0 \leq t \leq 2\pi \text{ for } t = \frac{\pi}{4} \text{ and } t = \frac{5\pi}{4}.$$

The candidates for the absolute minimum are at $t = 0, \frac{\pi}{4}, \frac{5\pi}{4}$, and 2π .

t	$x(t)$
0	$e^0 \sin(0) = 0$
$\frac{\pi}{4}$	$e^{-\frac{\pi}{4}} \sin\left(\frac{\pi}{4}\right) > 0$
$\frac{5\pi}{4}$	$e^{-\frac{5\pi}{4}} \sin\left(\frac{5\pi}{4}\right) < 0$
2π	$e^{-2\pi} \sin(2\pi) = 0$

The particle is farthest to the left when $t = \frac{5\pi}{4}$.

Part B

2 points are earned for $x''(t)$

1 point is earned for substituting $x''(t)$, $x'(t)$, and $x(t)$ into $Ax''(t) + x'(t) + x(t)$

1 point is earned for the answer

5-5b

$$\begin{aligned}x''(t) &= -e^{-t}(\cos t - \sin t) + e^{-t}(\sin t - \cos t) \\ &= -2e^{-t} \cos t\end{aligned}$$

$$\begin{aligned}Ax''(t) + x'(t) + x(t) &= A(-2e^{-t} \cos t) + e^{-t}(\cos t - \sin t) + e^{-t} \sin t \\ &= (-2A + 1)e^{-t} \cos t \\ &= 0\end{aligned}$$

Therefore, $A = \frac{1}{2}$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for $x''(t)$

1 point is earned for substituting $x''(t)$, $x'(t)$, and $x(t)$ into $Ax''(t) + x'(t) + x(t)$

1 point is earned for the answer

$$\begin{aligned}x''(t) &= -e^{-t}(\cos t - \sin t) + e^{-t}(\sin t - \cos t) \\ &= -2e^{-t} \cos t\end{aligned}$$

$$\begin{aligned}Ax''(t) + x'(t) + x(t) &= A(-2e^{-t} \cos t) + e^{-t}(\cos t - \sin t) + e^{-t} \sin t \\ &= (-2A + 1)e^{-t} \cos t \\ &= 0\end{aligned}$$

Therefore, $A = \frac{1}{2}$.

5-5b

82.  On a certain workday, the rate, in tons per hour, at which unprocessed gravel arrives at a gravel processing plant is modeled by $G(t) = 90 + 45\cos\left(\frac{t^2}{18}\right)$, where t is measured in hours and $0 \leq t \leq 8$. At the beginning of the workday ($t = 0$), the plant has 500 tons of unprocessed gravel. During the hours of operation, $0 \leq t \leq 8$, the plant processes gravel at a constant rate of 100 tons per hour.
- (a) Find $G'(5)$. Using correct units, interpret your answer in the context of the problem.
- (b) Find the total amount of unprocessed gravel that arrives at the plant during the hours of operation on this workday.
- (c) Is the amount of unprocessed gravel at the plant increasing or decreasing at time $t = 5$ hours? Show the work that leads to your answer.
- (d) What is the maximum amount of unprocessed gravel at the plant during the hours of operation on this workday? Justify your answer.

Part A

1 point is earned for $G'(5)$

1 point is earned for the interpretation with units

$$G'(5) = -24.588 \text{ (or } -24.587)$$

The rate at which gravel is arriving is decreasing by 24.588 (or 24.587) tons per hour per hour at time $t = 5$ hours.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $G'(5)$

1 point is earned for the interpretation with units

$$G'(5) = -24.588 \text{ (or } -24.587)$$

The rate at which gravel is arriving is decreasing by 24.588 (or 24.587) tons per hour per hour at time $t = 5$ hours.

Part B

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^8 G(t) dt = 825.551 \text{ tons}$$

5-5b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^8 G(t) dt = 825.551 \text{ tons}$$

Part C

1 point is earned for comparing $G(5)$ to 100

1 point is earned for the conclusion

$$G(5) = 98.140764 < 100$$

At time $t = 5$, the rate at which unprocessed gravel is arriving is less than the rate at which it is being processed.

Therefore, the amount of unprocessed gravel at the plant is decreasing at time $t = 5$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for comparing $G(5)$ to 100

1 point is earned for the conclusion

$$G(5) = 98.140764 < 100$$

At time $t = 5$, the rate at which unprocessed gravel is arriving is less than the rate at which it is being processed.

Therefore, the amount of unprocessed gravel at the plant is decreasing at time $t = 5$.

Part D

1 point is earned for considering $A'(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

5-5b

The amount of unprocessed gravel at time t is given by $A(t) = 500 + \int_0^t (G(s) - 100)ds$.

$$A'(t) = G(t) - 100 = 0 \Rightarrow t = 4.923480$$

t	$A(t)$
0	500
4.92348	635.376123
8	525.551089

The maximum amount of unprocessed gravel at the plant during this workday is 635.376 tons.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $A'(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

The amount of unprocessed gravel at time t is given by $A(t) = 500 + \int_0^t (G(s) - 100)ds$.

$$A'(t) = G(t) - 100 = 0 \Rightarrow t = 4.923480$$

t	$A(t)$
0	500
4.92348	635.376123
8	525.551089

The maximum amount of unprocessed gravel at the plant during this workday is 635.376 tons.

5-5b

83.  A storm washed away sand from a beach, causing the edge of the water to get closer to a nearby road. The rate at which the distance between the road and the edge of the water was changing during the storm is modeled by $f(t) = \sqrt{t} + \cos t - 3$ meters per hour, t hours after the storm began. The edge of the water was 35 meters from the road when the storm began, and the storm lasted 5 hours. The derivative of $f(t)$ is $f'(t) = \frac{1}{2\sqrt{t}} - \sin t$.

- (a) What was the distance between the road and the edge of the water at the end of the storm?
- (b) Using correct units, interpret the value $f'(4) = 1.007$ in terms of the distance between the road and the edge of the water.
- (c) At what time during the 5 hours of the storm was the distance between the road and the edge of the water decreasing most rapidly? Justify your answer.
- (d) After the storm, a machine pumped sand back onto the beach so that the distance between the road and the edge of the water was growing at a rate of $g(p)$ meters per day, where p is the number of days since pumping began. Write an equation involving an integral expression whose solution would give the number of days that sand must be pumped to restore the original distance between the road and the edge of the water.

Part A

1 point is earned for the integral

1 point is earned for the answer

$$35 + \int_0^5 f(t) dt = 26.494 \text{ or } 26.495 \text{ meters}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$35 + \int_0^5 f(t) dt = 26.494 \text{ or } 26.495 \text{ meters}$$

Part B

1 point is earned for interpretation of $f'(4)$

1 point is earned for units

5-5b

Four hours after the storm began, the rate of change of the distance between the road and the edge of the water is increasing at a rate of 1.007 meters/hours².



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for interpretation of $f'(4)$

1 point is earned for units

Four hours after the storm began, the rate of change of the distance between the road and the edge of the water is increasing at a rate of 1.007 meters/hours².

Part C

1 point is earned if considers $f'(t) = 0$

1 point is earned for the answer

1 point is earned for justification

$f'(t) = 0$ when $t = 0.66187$ and $t = 2.84038$

The minimum of f for $0 \leq t \leq 5$ may occur at 0, 0.66187, 2.84038, or 5.

$f(0) = -2$

$f(0.66187) = -1.39760$

$f(2.84038) = -2.26963$

$f(5) = -0.48027$

The distance between the road and the edge of the water was decreasing most rapidly at time $t = 2.840$ hours after the storm began.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if considers $f'(t) = 0$

5-5b

1 point is earned for the answer

1 point is earned for justification

$$f(t) = 0 \text{ when } t = 0.66187 \text{ and } t = 2.84038$$

The minimum of f for $0 \leq t \leq 5$ may occur at 0, 0.66187, 2.84038, or 5.

$$f(0) = -2$$

$$f(0.66187) = -1.39760$$

$$f(2.84038) = -2.26963$$

$$f(5) = -0.48027$$

The distance between the road and the edge of the water was decreasing most rapidly at time $t = 2.840$ hours after the storm began.

Part D

1 point is earned for the integral of g

1 point is earned for the answer

$$-\int_0^5 f(t)dt = \int_0^x g(p)dp$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral of g

1 point is earned for the answer

$$-\int_0^5 f(t)dt = \int_0^x g(p)dp$$

5-5b

84.  The tide removes sand from Sandy Point Beach at a rate modeled by the function R , given by

$$R(t) = 2 + 5 \sin\left(\frac{4\pi t}{25}\right).$$

A pumping station adds sand to the beach at a rate modeled by the function S , given by

$$S(t) = \frac{15t}{1+3t}.$$

Both $R(t)$ and $S(t)$ have units of cubic yards per hour and t is measured in hours for $0 \leq t \leq 6$. At time $t = 0$, the beach contains 2500 cubic yards of sand.

- How much sand will the tide remove from the beach during this 6-hour period? Indicate units of measure.
- Write an expression for $Y(t)$, the total number of cubic yards of sand on the beach at time t .
- Find the rate at which the total amount of sand on the beach is changing at time $t = 4$.
- For $0 \leq t \leq 6$, at what time t is the amount of sand on the beach a minimum? What is the minimum value? Justify your answers.

Part A

1 point is earned for integral

1 point is earned for the answer with units

$$\int_0^6 R(t) dt = 31.815 \text{ or } 31.816 \text{ yd}^3$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer with units

$$\int_0^6 R(t) dt = 31.815 \text{ or } 31.816 \text{ yd}^3$$

Part B

1 point is earned for the integrand

1 point is earned for the limits

5-5b

1 point is earned for the answer

$$Y(t) = 2500 + \int_0^t (S(x) - R(x))dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$Y(t) = 2500 + \int_0^t (S(x) - R(x))dx$$

Part C

1 point is earned for the answer

$$Y'(t) = S(t) - R(t)$$

$$Y'(4) = S(4) - R(4) = -1.908 \text{ or } -1.909 \text{ yd}^3/\text{hr}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$Y'(t) = S(t) - R(t)$$

$$Y'(4) = S(4) - R(4) = -1.908 \text{ or } -1.909 \text{ yd}^3/\text{hr}$$

Part D

1 point is earned for setting $Y'(t) = 0$

1 point is earned for the critical t -value

1 point is earned for the answer with justification

5-5b

$Y'(t) = 0$ when $S(t) - R(t) = 0$.

The only value in $[0, 6]$ to satisfy $S(t) = R(t)$ is $a = 5.117865$.

t	$Y(t)$
0	2500
a	2492.3694
6	2493.2766

The amount of sand is a minimum when $t = 5.117$ or 5.118 hours. The minimum value is 2492.369 cubic yards.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for setting $Y'(t) = 0$

1 point is earned for the critical t -value

1 point is earned for the answer with justification

$Y'(t) = 0$ when $S(t) - R(t) = 0$.

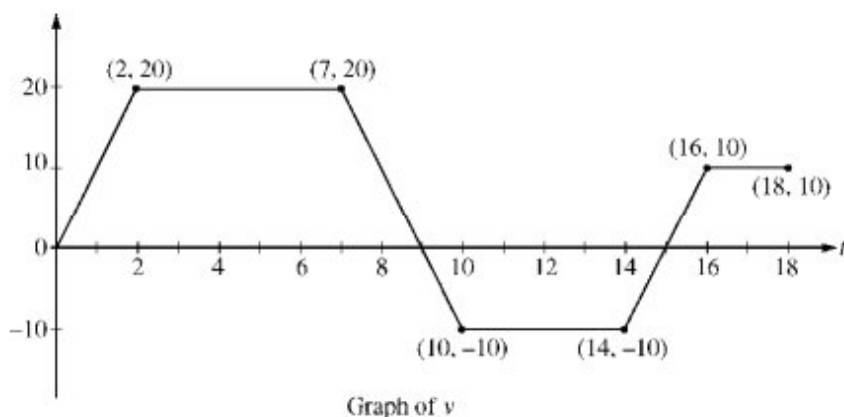
The only value in $[0, 6]$ to satisfy $S(t) = R(t)$ is $a = 5.117865$.

t	$Y(t)$
0	2500
a	2492.3694
6	2493.2766

The amount of sand is a minimum when $t = 5.117$ or 5.118 hours. The minimum value is 2492.369 cubic yards.

5-5b

85.



A squirrel starts at building A at time $t = 0$ and travels along a straight, horizontal wire connected to building B . For $0 \leq t \leq 18$, the squirrel's velocity is modeled by the piecewise-linear function defined by the graph above.

- At what times in the interval $0 < t < 18$, if any, does the squirrel change direction? Give a reason for your answer.
- At what time in the interval $0 \leq t \leq 18$ is the squirrel farthest from building A ? How far from building A is the squirrel at that time?
- Find the total distance the squirrel travels during the time interval $0 \leq t \leq 18$.
- Write expressions for the squirrel's acceleration $a(t)$, velocity $v(t)$, and distance $x(t)$ from building A that are valid for the time interval $7 < t < 10$.

Part A

1 point is earned for the t -values

1 point is earned for the explanation

The squirrel changes direction whenever its velocity changes sign. This occurs at $t = 9$ and $t = 15$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the t -values

1 point is earned for the explanation

The squirrel changes direction whenever its velocity changes sign. This occurs at $t = 9$ and $t = 15$.

Part B

1 point is earned for identifying candidates

5-5b

1 point is earned for answers

Velocity is 0 at $t = 0$, $t = 9$, and $t = 15$.

t	position at time t
0	0
9	$\frac{9+5}{2} \cdot 20 = 140$
15	$140 - \frac{6+4}{2} \cdot 10 = 90$
18	$90 + \frac{3+2}{2} \cdot 10 = 115$

The squirrel is farthest from building A at time $t = 9$;

its greatest distance from the building is 140.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for identifying candidates

1 point is earned for answers

Velocity is 0 at $t = 0$, $t = 9$, and $t = 15$.

t	position at time t
0	0
9	$\frac{9+5}{2} \cdot 20 = 140$
15	$140 - \frac{6+4}{2} \cdot 10 = 90$
18	$90 + \frac{3+2}{2} \cdot 10 = 115$

The squirrel is farthest from building A at time $t = 9$;

its greatest distance from the building is 140.

Part C

1 point is earned for the answer

5-5b

The total distance traveled is $\int_0^{18} |v(t)| dt = 140 + 50 + 25 = 215$.



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

The total distance traveled is $\int_0^{18} |v(t)| dt = 140 + 50 + 25 = 215$.

Part D

1 point is earned for $a(t)$

1 point is earned for $v(t)$

2 points are earned for $x(t)$

$$\text{For } 7 < t < 10, a(t) = \frac{20 - (-10)}{7 - 10} = -10$$

$$v(t) = 20 - 10(t - 7) = -10t + 90$$

$$x(7) = \frac{7+5}{2} \cdot 20 = 120$$

$$x(t) = x(7) + \int_7^t (-10u + 90) du$$

$$= 120 + (-5u^2 + 90u) \Big|_{u=7}^{u=t}$$

$$= -5t^2 + 90t - 265$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $a(t)$

1 point is earned for $v(t)$

2 points are earned for $x(t)$

5-5b

$$\text{For } 7 < t < 10, a(t) = \frac{20 - (-10)}{7 - 10} = -10$$

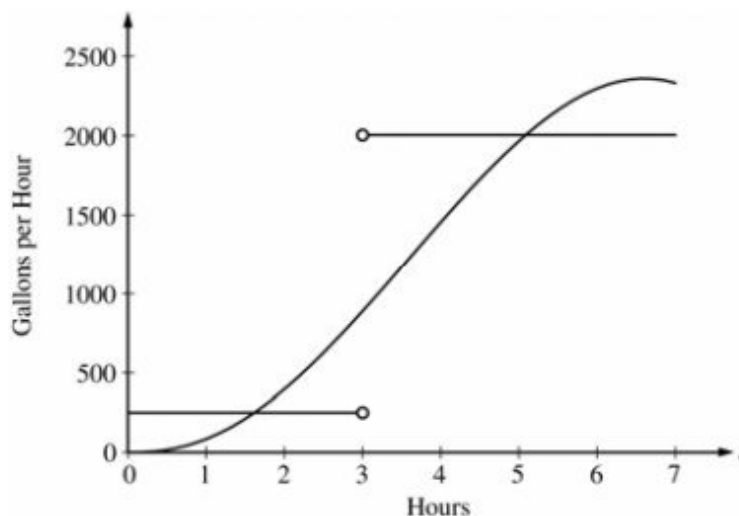
$$v(t) = 20 - 10(t - 7) = -10t + 90$$

$$x(7) = \frac{7+5}{2} \cdot 20 = 120$$

$$\begin{aligned} x(t) &= x(7) + \int_7^t (-10u + 90) du \\ &= 120 + (-5u^2 + 90u) \Big|_{u=7}^{u=t} \\ &= -5t^2 + 90t - 265 \end{aligned}$$

5-5b

86.



The amount of water in a storage tank, in gallons, is modeled by a continuous function on the time interval $0 \leq t \leq 7$, where t is measured in hours. In this model, rates are given as follows:

- i. The rate at which water enters the tank is $f(t) = 100t^2 \sin(\sqrt{t})$ gallons per hour for $0 \leq t \leq 7$.
- ii. The rate at which water leaves the tank is

$$g(t) = \begin{cases} 250 & \text{for } 0 \leq t < 3 \\ 2000 & \text{for } 3 < t \leq 7 \end{cases} \text{ gallons per hour.}$$

The graphs of f and g , which intersect at $t = 1.617$ and $t = 5.076$, are shown in the figure above. At time $t = 0$, the amount of water in the tank is 5000 gallons.

- (a) How many gallons of water enter the tank during the time interval $0 \leq t \leq 7$? Round your answer to the nearest gallon.
- (b) For $0 \leq t \leq 7$, find the time intervals during which the amount of water in the tank is decreasing. Give a reason for each answer.
- (c) For $0 \leq t \leq 7$, at what time t is the amount of water in the tank greatest? To the nearest gallon, compute the amount of water at this time. Justify your answer.

Part A

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^7 f(t) dt \approx 8264 \text{ gallons}$$

5-5b



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^7 f(t)dt \approx 8264 \text{ gallons}$$

Part B

1 point is earned for the intervals

1 point is earned for the reason

The amount of water in the tank is decreasing on the intervals $0 \leq t \leq 1.617$ and $3 \leq t \leq 5.076$ because $f(t) < g(t)$ for $0 \leq t < 1.617$ and $3 < t < 5.076$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the intervals

1 point is earned for the reason

The amount of water in the tank is decreasing on the intervals $0 \leq t \leq 1.617$ and $3 \leq t \leq 5.076$ because $f(t) < g(t)$ for $0 \leq t < 1.617$ and $3 < t < 5.076$.

Part C

1 point is earned if identifies $t = 3$ as a candidate

1 point is earned for the integrand

1 point is earned for the amount of water at $t = 3$

1 point is earned for the amount of water at $t = 7$

1 point is earned for the conclusion

5-5b

Since $f(t) - g(t)$ changes sign from positive to negative only at $t = 3$, the candidates for the absolute maximum are $t = 0, 3$, and 7 .

t (hours)	gallons of water
0	5000
3	$5000 + \int_0^3 f(t) dt - 250(3) = 5126.591$
7	$5126.591 + \int_3^7 f(t) dt - 2000(4) = 4513.807$

The amount of water in the tank is greatest at 3 hours. At that time, the amount of water in the tank, rounded to the nearest gallon, is 5127 gallons.



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned if identifies $t = 3$ as a candidate

1 point is earned for the integrand

1 point is earned for the amount of water at $t = 3$

1 point is earned for the amount of water at $t = 7$

1 point is earned for the conclusion

Since $f(t) - g(t)$ changes sign from positive to negative only at $t = 3$, the candidates for the absolute maximum are $t = 0, 3$, and 7 .

t (hours)	gallons of water
0	5000
3	$5000 + \int_0^3 f(t) dt - 250(3) = 5126.591$
7	$5126.591 + \int_3^7 f(t) dt - 2000(4) = 4513.807$

The amount of water in the tank is greatest at 3 hours. At that time, the amount of water in the tank, rounded to the nearest gallon, is 5127 gallons.

5-5b

87. Let $f(x) = 12 - x^2$ for $x \geq 0$ and $f(x) \geq 0$.

(a) The line tangent to the graph of f at the point $(k, f(k))$ intercepts the x -axis at $x = 4$. What is the value of k ?

(b) An isosceles triangle whose base is the interval from $(0,0)$ to $(c,0)$ has its vertex on the graph of f . For what value of c does the triangle have maximum area? Justify your answer.

Part A

1 point is earned for $f'(x) = -2x$

1 point is earned for finding slope of line through $(4,0)$ & $(k, f(k))$ **or** equates to slope of tangent line

1 point is earned for write equation of tangent line through $(k, f(k))$ or $(4,0)$ **or** uses other point to get a quadratic

1 point is earned for finding roots of quadratic

1 point is earned for eliminating one answer

$$f(x) = 12 - x^2; f'(x) = -2x$$

slope of tangent line at

$$(k, f(k)) = -2k$$

line through $(4,0)$ & $(k, f(k))$

$$\text{has slope } \frac{f(k) - 0}{k - 4} = \frac{12 - k^2}{k - 4}$$

$$\text{so } -2k = \frac{12 - k^2}{k - 4} \Rightarrow k^2 - 8k + 12 = 0$$

$k = 2$ or $k = 6$ but $f(6) = -24$ so 6 is not in the domain.

$$k = 2$$



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $f'(x) = -2x$

5-5b

1 point is earned for finding slope of line through $(4,0)$ & $(k,f(k))$ **or** equates to slope of tangent line

1 point is earned for write equation of tangent line through $(k,f(x))$ or $(4,0)$ **or** uses other point to get a quadratic

1 point is earned for finding roots of quadratic

1 point is earned for eliminating one answer

$$f(x) = 12 - x^2; f'(x) = -2x$$

slope of tangent line at

$$(k, f(k)) = -2k$$

line through $(4,0)$ & $(k, f(k))$

$$\text{has slope } \frac{f(k) - 0}{k - 4} = \frac{12 - k^2}{k - 4}$$

$$\text{so } -2k = \frac{12 - k^2}{k - 4} \Rightarrow k^2 - 8k + 12 = 0$$

$k = 2$ or $k = 6$ but $f(6) = -24$ so 6 is not in the domain.

$$\boxed{k = 2}$$

Part B

$$1 \text{ point is earned for } A = \frac{1}{2}C\left(12 - \frac{C^2}{4}\right)$$

$$1 \text{ point is earned for computing } \frac{dA}{dC} \text{ \& set } = 0$$

1 point is earned for answer

1 point is earned for justification with end points

5-5b

$$A = \frac{1}{2}c \cdot f\left(\frac{c}{2}\right) = \frac{1}{2}c \left(12 - \frac{c^2}{4}\right)$$

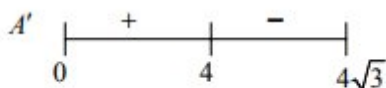
$$= 6c - \frac{c^3}{8} \text{ on } [0, 4\sqrt{3}]$$

$$\frac{dA}{dC} = 6 - \frac{3c^2}{8}; 6 - \frac{3c^2}{8} = 0 \text{ when } c = 4.$$

Candidate test

c	A
0	0
4	16 ← Max
$4\sqrt{3}$	0

First derivative



second derivative

$\frac{d^2A}{dc^2}\bigg|_{c=4} = -3 < 0$ so $c = 4$ gives a relative max. $c = 4$ is the only critical value in the domain interval, therefore maximum



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $A = \frac{1}{2}C(12 - \frac{C^2}{4})$

1 point is earned for computing $\frac{dA}{dC}$ & set = 0

1 point is earned for answer

1 point is earned for justification with end points

$$A = \frac{1}{2}c \cdot f\left(\frac{c}{2}\right) = \frac{1}{2}c \left(12 - \frac{c^2}{4}\right)$$

$$= 6c - \frac{c^3}{8} \text{ on } [0, 4\sqrt{3}]$$

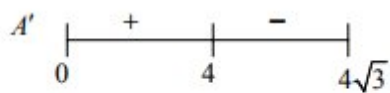
$$\frac{dA}{dC} = 6 - \frac{3c^2}{8}; 6 - \frac{3c^2}{8} = 0 \text{ when } c = 4.$$

5-5b

Candidate test

c	A
0	0
4	16 ← Max
$4\sqrt{3}$	0

First derivative



second derivative

$\frac{d^2 A}{dc^2} \Big|_{c=4} = -3 < 0$ so $c = 4$ gives a relative max. $c = 4$ is the only critical value in the domain interval, therefore maximum

5-5b

88. Water is pumped into an underground tank at a constant rate of 8 gallons per minute. Water leaks out of the tank at the rate of $\sqrt{t+1}$ gallons per minute, for $0 \leq t \leq 120$ minutes. At time $t = 0$, the tank contains 30 gallons of water.
- How many gallons of water leak out of the tank from time $t = 0$ to $t = 3$ minutes?
 - How many gallons of water are in the tank at time $t = 3$ minutes?
 - Write an expression for $A(t)$, the total number of gallons of water in the tank at time t .
 - At what time t , for $0 \leq t \leq 120$, is the amount of water in the tank a maximum? Justify your answer.

Part A

Method 1:

2 points are earned for the definite integral

1: limits

1: integrand

1 point is earned for the answer

- or -

Method 2:

1 point is earned for antiderivative with C

1 point is earned for solves for C using $L(0) = 0$

1 point is earned for the answer

Method 1: $\int_0^3 \sqrt{t+1} \, dt = \frac{2}{3}(t+1)^{3/2} \Big|_0^3 = \frac{14}{3}$

- or -

Method 2: $L(t)$ = gallons leaked in first t minutes

$$\frac{dL}{dt} = \sqrt{t+1}; \quad L(t) = \frac{2}{3}(t+1)^{3/2} + C$$

$$L(0) = 0; \quad C = -\frac{2}{3}$$

$$L(t) = \frac{2}{3}(t+1)^{3/2} - \frac{2}{3}; \quad L(3) = \frac{14}{3}$$

5-5b



0	1	2	3	4	5	6
---	---	---	---	---	---	---

The student response earns all of the following points:

Method 1:

2 points are earned for the definite integral

1: limits

1: integrand

1 point is earned for the answer

- or -

Method 2:

1 point is earned for antiderivative with C

1 point is earned for solves for C using $L(0) = 0$

1 point is earned for the answer

Method 1: $\int_0^3 \sqrt{t+1} \, dt = \frac{2}{3}(t+1)^{3/2} \Big|_0^3 = \frac{14}{3}$

- or -

Method 2: $L(t)$ = gallons leaked in first t minutes

$$\frac{dL}{dt} = \sqrt{t+1}; \quad L(t) = \frac{2}{3}(t+1)^{3/2} + C$$

$$L(0) = 0; \quad C = -\frac{2}{3}$$

$$L(t) = \frac{2}{3}(t+1)^{3/2} - \frac{2}{3}; \quad L(3) = \frac{14}{3}$$

Part B

1 point is earned for the answer

$$30 + 8 \cdot 3 - \frac{14}{3} = \frac{148}{3}$$

5-5b



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$30 + 8 \cdot 3 - \frac{14}{3} = \frac{148}{3}$$

Part C

Method 1:

1 point is earned for: $30 + 8t$

1 point is earned for: $-\int_0^t \sqrt{x+1} \, dx$

- or -

Method 2:

1 point earned for antiderivative with C

1 point is earned for the answer

Method 1:

$$\begin{aligned} A(t) &= 30 + \int_0^t (8 - \sqrt{x+1}) \, dx \\ &= 30 + 8t - \int_0^t \sqrt{x+1} \, dx \end{aligned}$$

- or -

Method 2:

5-5b

$$\frac{dA}{dt} = 8 - \sqrt{t+1}$$

$$A(t) = 8t - \frac{2}{3}(t+1)^{3/2} + C$$

$$30 = 8(0) - \frac{2}{3}(0+1)^{3/2} + C; \quad C = \frac{92}{3}$$

$$A(t) = 8t - \frac{2}{3}(t+1)^{3/2} + \frac{92}{3}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

Method 1:

1 point is earned for: $30 + 8t$

1 point is earned for: $-\int_0^t \sqrt{x+1} \, dx$

- or -

Method 2:

1 point earned for antiderivative with C

1 point is earned for the answer

Method 1:

$$\begin{aligned} A(t) &= 30 + \int_0^t (8 - \sqrt{x+1}) \, dx \\ &= 30 + 8t - \int_0^t \sqrt{x+1} \, dx \end{aligned}$$

- or -

Method 2:

5-5b

$$\frac{dA}{dt} = 8 - \sqrt{t+1}$$

$$A(t) = 8t - \frac{2}{3}(t+1)^{3/2} + C$$

$$30 = 8(0) - \frac{2}{3}(0+1)^{3/2} + C; \quad C = \frac{92}{3}$$

$$A(t) = 8t - \frac{2}{3}(t+1)^{3/2} + \frac{92}{3}$$

Part D

1 point is earned for set $A'(t) = 0$

1 point is earned for solves for t

1 point is earned for the justification

$$A'(t) = 8 - \sqrt{t+1} = 0 \text{ when } t = 63$$

$A'(t)$ is positive for $0 < t < 63$ and negative for $63 < t < 120$. Therefore there is a maximum at $t = 63$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for set $A'(t) = 0$

1 point is earned for solves for t

1 point is earned for the justification

$$A'(t) = 8 - \sqrt{t+1} = 0 \text{ when } t = 63$$

$A'(t)$ is positive for $0 < t < 63$ and negative for $63 < t < 120$. Therefore there is a maximum at $t = 63$.